

CHAPTER ONE

Concept of management and Function of the controller

Learning objectives:

After reading this chapter you should be able to:

- ❖ Define cost accounting and distinguish between cost and costing
- ❖ Distinguish between cost accounting, management accounting and financial accounting
- ❖ Identify planning and control responsibilities of middle and operating management levels
- ❖ Discuss the role of cost accounting in manufacturing organizations.
- ❖ Define and discuss responsibility accounting and controller's participation in planning and control.
- ❖ Describe the benefit of budgeting as a control tool.
- ❖ Appreciate the essentiality of organization chart to the development of a cost system and cost reports.

INTRODUCTION

Traditionally, *cost accounting* is considered as the technique and process of ascertaining costs of a given thing. In sixties, the definition of cost accounting was modified as 'the application of costing and cost accounting principles, methods and techniques to the science, art and practice of cost control and ascertainment of profitability of goods, or services'. It includes the presentation of information derived therefrom for the purpose of managerial decision making. It clearly emphasizes the importance of cost accountancy achieved during the period by using cost concepts in more and more areas and helping management to arrive at good business decisions. Today, the scope of cost accounting has enlarged to such an extent that it now refers to the collection and providing all sorts of information that assists the executives in fulfilling the organizational goals. Modern cost accounting is being termed as management accounting, since managers being the primary user of accounting information are increasingly using the data provided by the accounts, setting objectives and controlling the operations of the business.

FINANCIAL ACCOUNTING AND COST ACCOUNTING

In financial accounts, the monetary transactions of the business are recorded, classified and analyzed in an orderly manner, so as to prepare periodic results in the form of profit and loss account or income statement and balance sheet, indicating the financial position of the business at the end of that period. The financial accounting is guided by various

rules and regulations, some of which are mandatory. The system cannot normally deviate from the accepted accounting practices.

The object of financial accounting is to provide information mainly to outsiders such as shareholders, investors, government authorities, financial institutions, etc. The analysis and interpretation of financial data contained in the income statement and the balance sheet enable persons interested in the business to make meaningful judgment about the profitability, liquidity and solvency of the enterprise. Besides, income-tax, central excise, banks and insurance companies rely on the data contained in the financial statements. Cost accounting, on the other hand, deals with the ascertainment of the cost of product or service. It is a tool of management that provides detailed records and reports on the costs and expenses associated with the operations, mainly for internal control and decision making. Cost accounting basically relates to the utilization of resources, such as material, labor, machines, etc. and provides information like products cost, process cost, service or utility cost, inventory value, etc. so as to enable management taking important decisions like fixing price, choosing products, preparing quotations, releasing or withholding inventory, etc.

The objective of cost accounting is to provide information to internal managers for better planning and control of operations and taking timely decisions. In the early stages, cost accounting was considered as an extension of financial accounting. Cost records were maintained separately. Cost information and data were collected from financial books, since all monetary transactions are entered in the financial accounts only. After developing product cost or service cost and valuation of inventory, the costing profit and loss account is prepared.

The profit and loss figures so derived by the two sets of books i.e. financial accounts and cost accounts, would have to be reconciled, since some of the income and expenditure recorded in financial books do not enter into product cost, while some of the expenses are included in cost accounts on notional basis i.e. without having incurred actual expense. However, a system of integrated account has been developed subsequently, wherein cost and financial accounts are integrated avoiding maintaining two sets of books. The basic difference between financial and cost accounting may be summarized as follows:

<i>Financial Accounting</i>	<i>Cost Accounting</i>
<i>Accounting of monetary transactions of the business.</i>	<i>Accounting of product cost or service cost.</i>
<i>Consists of recording, classification and analysis of financial transactions.</i>	<i>Consists of developing product or service cost with element wise cost breakdown.</i>
<i>Leads to preparation of income statement and balance sheet at periodic interval.</i>	<i>Leads to development of product or service cost, indicating profitability of each product or service as and when required.</i>
<i>Aims at external reporting to the shareholders,</i>	<i>Aims at internal reporting both routine as well as</i>

<i>investors, Government authorities and other outside parties.</i>	<i>special reporting to managers for internal control and decision making.</i>
<i>The accounting systems are mandatory and structured as per legal and other requirements.</i>	<i>The system is much less structured and is not mandatory, except those covered by cost audit required u/s 233-B of the Companies Act, 1956</i>
<i>Subject to verification by external auditor.</i>	<i>Cost audit is not compulsory but selective to some specific industries/products.</i>

DEFINITION AND SCOPE OR COST ACCOUNTANCY

The terminology of cost accountancy published by the Institute of Cost and Management Accountants, London defines cost accountancy as “the application of costing and cost accounting principles, methods and techniques to the science, art and practice of cost control and the ascertainment of profitability. It includes the presentation of information derived therefrom for the managerial decision-making.”

On analysis of the above definition, the following features of cost accountancy become evident:

- “Cost accountancy” is used in the broadest sense when compared to “cost accounting” and “costing”. This is so because cost accountancy is concerned with the formulation of principles, methods and techniques to be applied for ascertaining cost and profit.
- Having ascertained ‘cost’ and ‘profit’, cost accountancy is concerned with presentation of information to management. To enable management to carry out its functions, reports must be promptly made available at the right time, to the right person and in a proper form.
- The information so provided is to serve the purpose of managerial decision making such as introducing a new line of product, replacement of manual labor by machines, make or buy, decisions, etc.

SCOPE OF COST ACCOUNTANCY

The scope of any subject refers to the various areas of study included in that subject. As regards the scope of cost accountancy is concerned, it has vast scope. The following topics fall under the purview of cost accountancy: (1) Costing, (2) Cost Accounting, (3) Cost Control Techniques, (4) Budgeting and (5) Cost Audit.

1. Costing

The terminology of ICMA, London, defines costing as “the technique and process of ascertaining the cost.” The above definition is very significant in as much as it carries the main theme of cost accountancy. This definition emphasizes two important aspects, viz.

- The technique and process of costing:*** The technique of costing involves two distinct steps, namely, (i) collection and classification of costs according to

various elements and (ii) allocation and apportionment of the expenses which cannot be directly charged to production. As a process, costing is concerned with the routine ascertainment of cost with a formal procedure.

- b) **Ascertainment of cost:** It involves three steps, viz., (i) collection and analysis of expenses, (ii) measurement of production at different stages and (iii) linking up of production with the expenses. To achieve the first step, costing has developed different systems such as Historical, Estimated and Standard Cost. For achieving the second step, costing has developed different methods such as single or output costing, Job costing, contract costing, etc. Finally, for achieving the last step costing has developed important techniques such as *Absorption Costing*, *Marginal Costing* and *Standard Costing*.

2. Cost Accounting

Kohler in his dictionary for Accountants defines cost accounting as “that branch of accounting dealing with the classification, recording, allocation, summarization and reporting of current and prospective costs.” Mr. Wheldon defines cost accounting as “the classifying, recording and appropriate allocation of expenditure for the determination of the costs of products or services, the relation of these costs to sales values, and the ascertainment of profitability.”

The above definitions reveal the following aspects of cost accounting:

- a) *Cost classification:* This refers to grouping of like items of cost into a common group.
- b) *Cost recording:* This refers to posting of cost transactions into the various ledger maintained under cost accounting system.
- c) *Cost allocation:* This refers to allotment of costs to various products or departments.
- d) *Cost determination or cost finding:* This refers to the determination of the cost of goods or services by informal procedure, i.e., procedures that do not carry on the regular process of cost accounting on a continuous basis.
- e) *Cost reporting:* This refers to furnishing of cost data on a regular basis so as to meet the requirements of management.

3) Cost Control

According to Kohler, cost control represents the employment of management devices in the performance of any necessary operation so that pre-established objectives of quality, quantity and time may be attained at the lowest possible outlay for goods and services. The terminology published by ICMA, London, defines cost control as “The guidance and regulation by executive action of the cost of operating an undertaking.” According to this definition, cost control aims at guiding the actuals towards the lines of target and regulates the actuals if they deviate from the targets. This guidance and regulation is done

by the executive who is responsible for causing the deviation. This process will become clear by enumerating the steps involved in any cost control technique.

- a) Fixation of targets in terms of cost and production performance.
- b) Ascertaining the actual cost and production performance.
- c) Comparison of actuals with the targets.
- d) Analysing the variance by causes and the person responsible for it.
- e) Taking remedial steps to set right unfavourable variations.

Cost control is exercised through a variety of techniques such as inventory control, quality control, budgetary control, standard costing, etc. The advantages of cost control are as follows:

- a) It helps in utilising the resources to the full extent.
- b) It helps in reduction of prices which are benefited by customers.
- c) It helps in competing successfully in the market.
- d) It increases the profit earning capacity of the business.
- e) It increases the goodwill of the business.

4) Budgeting

Mr. Heiser in his book *Budgeting–Principles and Practice*, defines budget as “an overall blue print of a comprehensive plan of operations and actions expressed in financial terms. According to him budgeting process involves the preparation of a budget and its fullest use not only as a device for planning and co-ordinating but also for control.

5) Cost Audit

The terminology of ICMA, London, defines cost audit, as “the verification of the correctness of cost accounts and of the adherence to the cost accounting plan.

NATURE OF COST ACCOUNTING

The nature of cost accounting can be brought out under the following headings:

1. **Cost accounting is a branch of knowledge:** Though considered as a branch of financial accounts, cost accounting is one of the important branch of knowledge, i.e., a discipline by itself. It is an organised body of knowledge consisting of its own principles, concepts and conventions. These principles and rules of course vary from industry to industry.
2. **Cost accounting is a science :** Cost accounting is a science as it is a body of systematic knowledge relating to not only cost accounting but relating to a wide variety of subjects such as law, office practice and procedure, data processing, production and material control, etc. It is necessary for a cost accountant to have intimate knowledge of all these field of study in order to carry on his day-to-day activities. But it is to be admitted that it is not a perfect science as in the case of natural science.
3. **Cost accounting is an art:** Cost accounting is an art in the sense it requires the ability and skill on the part of cost accountant in applying the principles, methods

and techniques of cost accountancy to various management problems. These problems include the ascertainment of cost, control of costs, ascertainment of profitability, etc.

4. **Cost accounting is a profession:** In recent years cost accounting has become one of the important professions which has become more challenging. This view is evident from two facts. First, the setting up of various professional bodies such as National Association of Accountants (NAA) in USA. The Institute of Cost and Management Accountants in UK, the Institute of Cost and Works Accountants in India and such other professional bodies both in developed and developing countries have increased the growing awareness of costing profession among the people. Secondly, a large number of students have enrolled in these institutes to obtain costing degrees and memberships for earning their livelihood.

PURPOSES OR OBJECTS OF COST ACCOUNTS

Costing serves number of purposes among which the following are considered to be most important.

1. **Ascertainment of cost:** This was considered to be the primary objective of cost accounting in the initial stages of its development. However, in modern times this has assumed the secondary objective of cost accounting. Cost ascertainment involves the collection and classification of expenses at the first instance. Those items of expenses which are capable of charging directly to the products manufactured are allocated. Then the other expenses which are not capable of direct allocation are apportioned on some suitable basis. Thus the cost of production of goods manufactured is ascertained. In this process, cost accounts involve maintenance of different books to record various elements of cost. Cost of production is ascertained by using any of the costing technique such as historical costing, marginal costing, etc.
2. **Cost control:** At one time cost control was considered as secondary objective of cost accounts. But in modern times it constitutes the primary purpose because of its utmost importance in all business undertakings. Cost control is exercised at different stages in a factory, viz., acquisition of materials, recruiting and deployment of labour force, during the production process and so on. As such we have material cost control, labour cost control, production control, quality control and so on. However, control over cost is exercised through the techniques of budgetary control and standard costing. The control techniques enable the management in knowing the operating efficiency of a business.
3. **Determination of selling price:** Every business organisation aims at maximizing profit. Total cost of production constitutes the basis on which selling price is fixed by adding a margin of profit. Cost accounting furnishes both the total cost of

production as well as cost incurred at each and every stage of production. No doubt other factors are taken into consideration before fixing price such as market conditions the area of distribution, volume of sales, etc. But cost plays the dominating role in price fixation.

4. **Frequent preparation of accounts and other reports:** The management of every business constantly rely upon the reports on cost data in order to know the level of efficiency relating to purchase, production, sales and operating results. Financial accounts provide information only at the end of the year because closing stock value is available only at the end of the year. But cost accounts provide the value of closing stock at frequent intervals by adopting a “continuous stock verification” system. Using the value of closing stock it is possible to prepare final accounts and know the operating results of the business.
5. **To provide a basis for operating policy:** Cost data to a great extent helps in formulating the policies of a business and in decision-making. As every alternative decision involves investment of capital outlay, costs play an important role in decision-making.

Therefore availability of cost data is a must for all levels of management. Some of the decisions which are based on cost are (a) make or buy decision, (b) manufacturing by mechanization or automation, (c) whether to close or continue operations in spite of losses.

ORGANISATION OF COST ACCOUNTING DEPARTMENT

The organization of cost accounting department depends upon the size of the concern. Whatever may be the structure of cost accounting department in a factory, it is established to serve the following purposes: (i) To compile cost data in order to meet the statutory requirement wherever applicable, (ii) To provide necessary cost data to management to carry out its functions efficiently, and (iii) To ensure efficiency and economy in the functioning of cost accounting department. To achieve the above purposes cost accounting department usually performs the following functions:

- 1) Designing and installation of appropriate method of costing.
- 2) Accumulation of cost data by process, department and product.
- 3) Analysis of such cost by elements of cost.
- 4) Estimation of cost of production.
- 5) Reporting of cost information to all levels of management.
- 6) Advising management in relation to investment based on cost information.

In a small and medium-sized concern, the cost accounting department may be set up as a section of financial accounting system. The cost accountant who is in charge of cost accounting department may be authorized to report to the chief accountant. In large-sized concern, a separate cost accounting department is established under the supervision of a full-fledged cost accountant. The cost accounting department is equipped with sufficient

staff each to look after different facets of cost accounting function. While important functions such as budgeting, cost analysis, etc. are performed by cost accountant, cost recording, cost reporting and such other functions are performed by cost clerks.

The cost accounting department may be organized either on the principle of centralization or decentralization. Under centralized system, the functions of cost accounting departments relating to all firms belonging to same industry is performed at a common central place.

The extent of centralization of cost accounting department depends upon the following factors :

- a) The philosophy of management regarding divisional responsibility.
- b) The ready availability of cost data from each firm.
- c) Size of the firm.
- d) The area of operations of every firm.
- e) The economy involved in centralization process.

Advantages of Centralized Cost Accounting Department

1. It facilitates full utilization of services of costing staff.
2. It permits mechanization of accounting which is not possible under decentralization system.
3. It reduces paper work and economies stationery costs.
4. It facilitates prompt reporting.

Under decentralization system a separate cost accounting department is set up for each and every firm under the supervision of a competent cost accountant. This system has certain advantages.

- a) It tends to increase the initiative of the cost accountants of every firm as the responsibility to control lies in their hands.
- b) It eliminates duplication of recording and reporting.
- c) It increases the speed of functioning of cost accounting department.

Relationship of Cost Department to Other Departments

1. **Cost accounting department and production department:** It can well be said that production department and cost department are interwoven together as the former cannot function efficiently without the existence of the latter. The production process is concerned with the utilization of materials, money and human resources. The cost department helps in estimating the material cost, labour cost and other expenses for manufacturing a product. It also helps in controlling these costs so as to minimize the cost of production. In fact, the main objective of cost accounting system is to reduce the cost of production of goods or services manufactured and rendered by business units. The other areas where cost accounting department is helpful in manufacturing process are : (a) Engineering department which is concerned with designing a product, (b) Research and development department which is concerned with development of a new product,

(c) Production planning and control department which ensures completion of production within the time schedule, and (d) Quality control section, which ensures quality of products. All these departments heavily rely on cost accounting department because costs to be incurred in these departments have a direct impact over the functioning of these departments.

2. **Cost accounting department and personnel department:** The personnel department which is concerned with proper recruitment, selection, training, time-keeping, fixation of wage rate and preparation of payroll, will work with close co-ordination of cost department. Each and every function performed by personnel department is again influenced by additional cost to be incurred on such function as for example promotion of employees leads to incurrence of additional wages.
3. **Cost accounting department and finance department:** The finance department is concerned with receiving and disbursement of cash. The allocation of investments on fixed and working capital entirely depends upon the cost reports submitted to it by the cost department. Judicious utilization of available capital is possible only when priority is given for more important areas of investment. This is facilitated by furnishing prompt report by cost department.
4. **Cost accounting department and purchase department:** In majority of firms purchase of raw materials at right quantity, of right quality, from right supplier, at a right time not only ensures ready supply but also at a reasonably low price. Any purchase of substandard quality of materials will lead to dissatisfaction among customers and consequently it leads to loss of orders. Further, purchase of materials at high rate will increase the cost of production tremendously. Delay in getting supplies of materials lead to delay in executing customers' orders. In these respects cost accounting department can assist purchase department to ensure efficient purchasing. Cost department also help in reducing (a) waste of materials, (b) the risk from theft, and (c) excessive investment in inventories.
5. **Cost accounting department and marketing department:** Marketing department relies on cost information in order to (a) estimate future product cost for fixing the selling price, (b) in knowing the expenses incurred in marketing the products so that if the amount exceeds the target, control measures can be taken to reduce such expenses, (c) to consider alternative selling methods and promotional measures, and (d) to make further investment in warehousing and distribution process. Cost department provides all such information as is required by the marketing department for its efficient functioning.
6. **Cost accounting department and financial accounting department:** The existence of cost accounting department makes the financial accounting department a complete organization by furnishing additional cost data to the chief accountant. Cost department enable financial accounts department in carrying out the latter's function furnishing necessary data in respect of the following :

- a) Supply of material cost, labour cost and expenses to facilitate preparation of manufacturing account.
- b) Provision of the value of closing stock frequently to facilitate the preparation of interim final accounts.
- c) Assist financial accounting department in matters relating to taxation, insurance and in solving legal matters.
- d) Enables financial accounting department to settle the bills by duly approving them.
- e) Helps financial accounting department in budgeting.

ADVANTAGES OF COST ACCOUNTING

A good costing system serves the needs of large sections of people. The advantages of cost accounting are discussed below.

Advantages of Cost Accounting to Management

1. **Fixation of responsibility:** Whenever a cost center is established, it implies establishing a kind of relationship between superior and subordinates. Thus responsibilities are fixed on every individual who is concerned with incurrence of cost.
2. **Measures economic performance:** By applying cost control techniques such as budgetary control and standard costing it helps in assisting the performance of business.
3. **Fixation of price:** By providing cost data it helps management to fix the selling price in advance. Hence, quotations can be supplied to prospective customers to secure orders.
4. **Aids in decision-making:** It helps management in making suitable decisions such as make or buys, replaces manual labour by machines, shut down or continue operations based on cost reports.
5. **Helps in the preparation of interim final accounts:** By the process of continuous stock taking it enables to know the value of closing stock of materials at any time. This facilitates preparation of final accounts wherever desired.
6. **Helps in minimizing wastages and losses:** Cost accounting system enables to locate the losses relating to materials, idle time and underutilization of plant and machinery.
7. **Facilitates comparison:** It facilitates cost comparison in respect of jobs, process, and departments and also between two periods. This reveals the efficiency or otherwise of each job process or department.
8. **Assists in increasing profitability:** Costing reports provide information about profitable or unprofitable areas of operation. The management can discontinue that product line or that department which are responsible for incurring losses. Thereby only profitable line of activities alone is retained.

9. **Reconciliation with financial accounts:** A well maintained cost accounting system facilitates reconciliation with financial accounts to check the arithmetical accuracy of both the systems.
10. **It guides future production policy:** Cost data help management in determining future production policy. Any expansion or contraction of production for the future is based on past cost data.

LIMITATIONS OF COST ACCOUNTING

1. **It is expensive:** The system of cost accounting involves additional expenditure to be incurred in installing and maintaining it. However, before installing it, care must be taken to ensure that the benefits derived are more than the investment made on this system of accounting.
2. **The system is more complex:** As the cost accounting system involves number of steps in ascertaining cost such as collection and classification of expenses, allocation and apportionment of expenses, it is considered to be complicated system of accounts. Moreover the system makes use of several documents and forms in preparing the reports. This will tend to delay in the preparation of accounts.
3. **Inapplicability of same costing method and technique:** All business enterprises cannot make use of a single method and technique of costing. It all depends upon the nature of business and type of product manufactured by it. If a wrong technique and method is used, it misleads the results of business.
4. **Not suitable for small scale units:** A cost accounting system is applicable only to a large-sized business but not to a small-sized one. Hence, there is limitation to its application to all types of business.
5. **Lack of accuracy:** The accuracy of cost accounts get distorted owing to the use of notional cost such as standard cost, estimated cost, etc.
6. **It lacks social accounting:** Cost accounting fails to take into account the social obligation of the business. In other words, social accounting is outside the purview of cost accounts.

Self-test questions

- A. True or False? Give reason in brief.
- 1) In responsibility accounting, basic method of control is the same as used in the budgetary control and standard costing.
 - 2) Organizing is that force which guides management in achieving objectives by comparing performance with policies and decisions.
 - 3) Middle management group consists of /represented by/division manger, vice-presidents and the executives in charge of marketing.
 - 4) Cost accounting is a branch of financial accounting.
 - 5) The main purpose of cost accounting is to maximize profit.

B. Choose the answer form the following given alternatives.

- 1) One of the following is true :
 - a) Cost accounting helps mangers in choosing among alternatives.
 - b) Accountability stands for power to command others.
 - c) Authority arises usually from lower management groups.
 - d) Controlling is establishment of work framework through which the required activities are to be performed.
 - e) All
 - f) None
- 2) Which of the following is not the responsibility of cost department?
 - a) Record manufacturing and non-manufacturing costs.
 - b) Analyze all costs.
 - c) Issue significant control reports.
 - d) All but C
 - e) All
 - f) None
- 3) Management's systematic effort to achieve objectives by comparing performance to plan is called
 - a) Controlling
 - b) Planning
 - c) Budgeting
 - d) Organizing
 - e) All
 - f) None
- 4) All of the following are the advantages of cost accounting *except*,
 - a) It facilitates use of specialized cost reduction techniques.
 - b) It helps to form cost centres and responsibility centres to exercise control.
 - c) It provides the base for taking the best decision and give outright solution of the problem.
 - d) It helps to formulate operating policies like whether to make or buy from out side sources.
 - e) None
- 5) Construction of detail operating programs for all phases of operation is called
 - a) Budgeting

- b) Planning
 - c) Controlling
 - d) Organizing
 - e) None
- 6) controller is the executive manager responsible for a company's accounting function and performs all of the following functions except;
- a) Coordinate management's participation in planning and control phases of attaining objectives.
 - b) Coordinate management's participation in determining the effectiveness of policies.
 - c) Coordinate management's participation in creating organizational structure and procedures.
 - d) All
 - e) None
- 7) One of the following statements is not true cost accounting and management accounting comparison.
- a) Scope of cost accounting is limited to providing cost information for managerial uses.
 - b) Scope of cost accounting is broader than that of management accounting.
 - c) Evolution of cost accounting is mainly due to the limitation of financial accounting.
 - d) The main emphasis of management accounting is on planning and controlling.
 - e) None
- 8) One of the following is not the management accounting task:
- a) Supplying information for planning and decision making
 - b) Performance evaluation/to see whether assets are used efficiently.
 - c) Looking at the reasonableness of the cost incurred.
 - d) Cost determination.
 - e) All
 - f) None
- 9) The primary objective of cost accounting is:
- a) Guide to business policy.
 - b) Control of cost

- c) Determination of selling price
 - d) Ascertainment of cost.
- 10) _____ is defined as reduction in firm's equity, other than form withdrawal of capital for which no compensating value has been received.
- a) Expense
 - b) Loss
 - c) Cost
 - d) Depreciation

B.Fill in the blank spaces.

- 1) _____ is a process of determining desirable and possible objectives, and of deploying resources to match then objectives.
- 2) _____ is the force that binds the organization together and the power to command others to perform or not to perform certain activities.
- 3) The management function that brings the many functional units of an enterprise into a well-conceived structure is known as _____.
- 4) _____ is the executive manager responsible for a company's accounting function.
- 5) The _____ department interviews, screens, and select employees for various job classification and maintains the wage rates and method of remuneration for each employee.

C. Objective questions

- 1) What is cost accounting? Discuss briefly its objectives and advantages.
- 2) Why has the budget cited as the most essential tool in cost accounting.
- 3) In what manner does the controller exercise control over the activities of other member of management?
- 4) State and explain the main differences between cost accounting and management accounting in your own words.
- 5) What are the functions of cost accountants in an industrial organization?
- 6) Numerous non-accounting departments require cost data and must also feed basic data to the cost department. Discuss.

CHAPTER TWO

Cost Concepts and the Cost Accounting Information System

2.1. Introduction

This chapter explains several widely recognized cost concept and terms. Various cost concept and terms are useful in many contexts, including decision making in all areas of the value chain. They help managers decide such issues as, how much should we spend for research and development? What is the effect of product design changes on manufacturing costs? Should we replace some production assembly with a robot? Should we spend more of the marketing budget on sales promotion coupons and less on advertising? Should we distribute from a central warehouse or from regional warehouse?

2.2. COST CONCEPT AND COST OBJECT

The dictionary meaning of cost is “a loss or sacrifice”, or “an amount paid or required in payment for a purchase or for the production or upkeep of something, often measured in terms of effort or time expended”. *C I M A Terminology* defines cost as ‘resources sacrificed or forgone to achieve a specific objective’. Cost is generally measured in monetary terms.

Cost is the amount of expenditure (actual or notional) incurred on or attributable to, a specified thing or activity. Thus, material cost of a product will mean the expenses incurred in procuring, storing and using materials in the product. Similarly, labour cost will represent that part of payment made to the workmen for time spent on the product during its manufacture.

Again, the term ‘cost’ can hardly be meaningful without using a suffix or a prefix. The cost is always ascertained with reference to some object, such as, material, labour, direct, indirect, fixed, variable, job, process, etc. Thus, each suffix or prefix implies certain attribute which will explain its nature and limitations.

Cost object is defined by Charles T. Horngren as ‘any activity for which a separate measurement of cost is desired’. It may be an activity, or operation in which resources, like materials, labour, etc. are consumed. The cost object may be a product or service, a project or a department, or even a program like eradication of illiteracy. Again, the same cost may pertain to more than one cost objects simultaneously. For example, material cost may be a part of product cost as well as production department cost.

Cost Management System

A *cost management system* (CMS) is a collection of tools and techniques that identifies how management's decisions affect costs. The primary purposes of a cost management system are:

- ☞ to provide cost information for strategic management decisions,
- ☞ to provide cost information for operational control, and
- ☞ measure of inventory value and cost of goods sold for financial reporting.

2.3. Cost Accounting Information Systems

Cost accounting is that part of the cost management system that measures costs for the purposes of management decision making and financial reporting.

The cost accounting information system accumulates and classifies expenditure according to the elements of costs, and then, the accumulated expenditure is allocated and apportioned to cost objects i.e. cost centres and cost units. We should, therefore, know what are cost elements, cost centres and cost units.

2.3.1. Elements of Cost

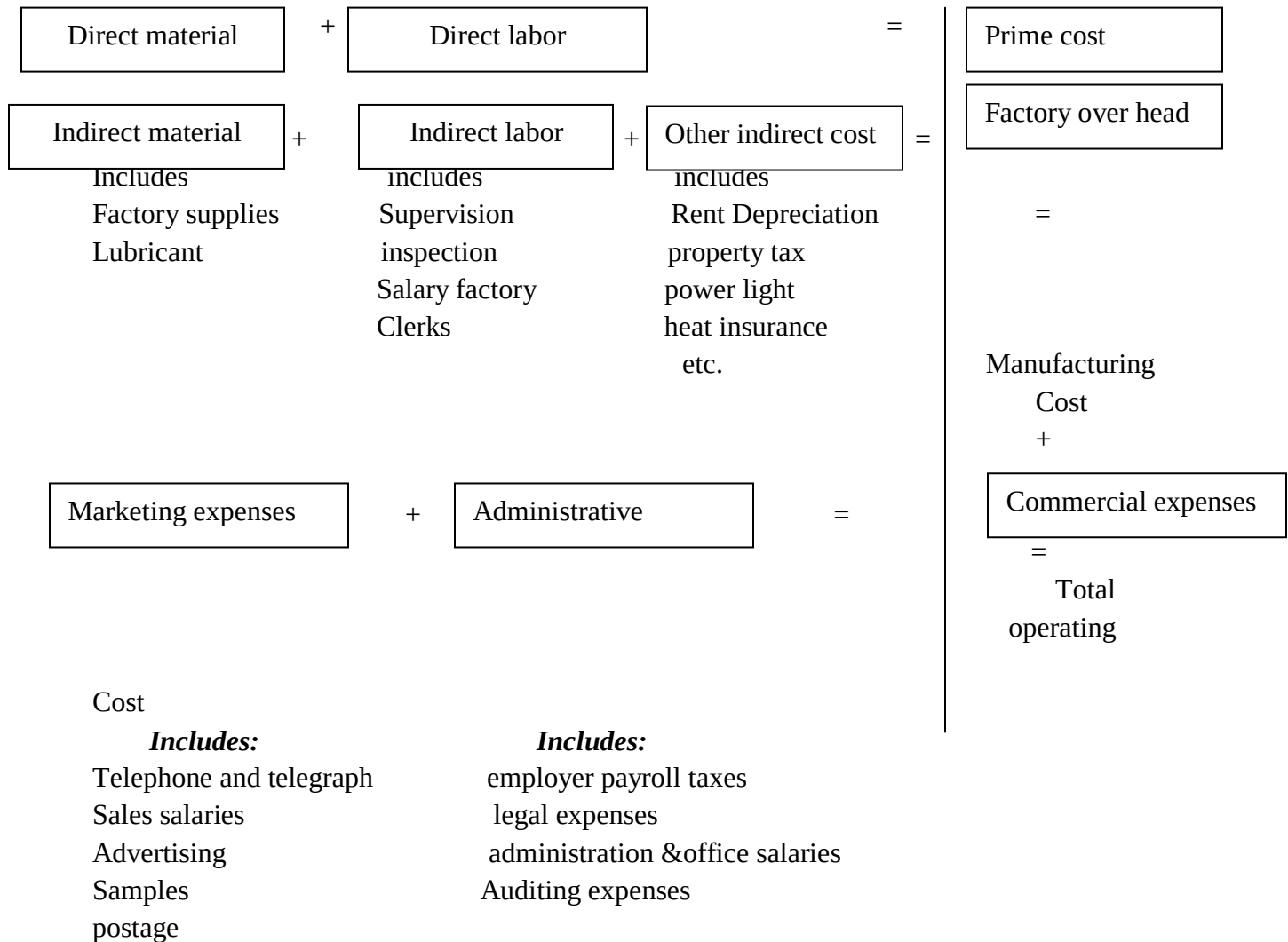
For the purpose of identification, accounting and control, breakup of cost into its elements is essential. Elements are related to the process of manufacture i.e. the conversion of raw materials into finished products. Costs are normally broken down into **three** basic elements, namely, **material, labour and expense**. *Material cost* includes all materials consumed in the process of manufacture up to the primary packing. *Labour cost* includes all remuneration paid to the staff and workmen for conversion of raw materials into finished products. *Expenses* consist of the cost of utilities and services used for the conversion process including notional cost for the use of owned assets.

Each of the cost elements can be further divided into direct and indirect cost. Direct costs are those which can be identified with or related to the product or services, so much so that an increase or decrease of a unit of product or service will affect the cost proportionately.

Indirect cost, on the other hand, cannot be identified or traced to a given cost object in economical way and are related to the expenses incurred for maintaining facilities for such production or services. Thus, the elements of cost may be summarized as follows –
(a) Direct materials and indirect materials, (b) Direct wages and indirect wages, (c) Direct expense and indirect expense

The aggregate of direct materials cost, direct wages and direct expense is called Prime cost, while indirect materials cost, indirect wages and indirect expenses are collectively called overhead cost.

Overheads are classified into production overheads i.e. indirect costs relating to manufacturing activities, administration overheads i.e. costs relating to formulating the policy, directing the organization and controlling operations and selling and distribution overheads i.e. indirect costs relating to the activity of creating and stimulating demand and securing orders as well as operations relating to distribution of goods from factory warehouse to customers. Factory cost, cost of production and cost of sales are arrived at by adding respective overheads to prime cost, factory cost and cost of production as indicated in the chart below :-



<i>Production cost = Prime cost + factory overhead</i>

$$\text{OR}$$
$$= \text{Direct materials} + \text{Conversion cost}$$

$$\text{Conversion Cost} = \text{Direct labor cost} + \text{Factory overhead cost}$$

Prime cost is the sum of direct materials cost and direct labor cost. Conversion cost is the sum of direct labor cost and manufacturing overhead cost.

The term conversion cost is used to describe direct labor and manufacturing overhead because these costs are incurred to convert materials into the finished product.

Total cost = Prime cost + Overheads (admin, selling, distribution cost)

$$\text{OR}$$
$$= \text{Production cost} + \text{period cost (administrative, selling, distribution and finance cost)}$$

Period cost is treated as expenses and matched against sales for calculating profit, e.g. office rental.

2.3.2. Cost Centre

Cost center is a location where costs are incurred and may or may not be attributed to cost units. It can be defined as a location, person or item of equipment (or group of them) in respect of which costs may be ascertained and related to cost units for the purposes of cost control. It is the smallest segment of activity or area of responsibility for which costs are accumulated. Thus cost centers can be of **two kinds**, viz.

- a) **Impersonal cost center** consisting of a location or item of equipment (or group of these) such as machine shop, and
- b) **Personal cost center** consisting of a person or a group of persons such as factory manager, sales manager, etc.

From functional point of view, cost centers may be the following two types:

- a) **Production cost centers.** - are those cost centers where actual production work takes place. **Examples are**
 - Weaving department in textile factory
 - Melting shop in a steel mill
 - Cone crushing shop in a sugar mill, Etc.
- b) **Service cost center.** There are those cost centers which are ancillary to and render services to production cost centers, so that manufacturing activities can take place. In a biscuit manufacturing company, making, baking and packing are production cost centers while personnel, purchase, stores, canteen, accounts are service cost centers.

Example: power house

Stores department
Repair shop, etc.

The main purpose of cost center is twofold:–

- i. **Recovery of cost:** Costs are collected, classified and accumulated in respect of a location, person or an item of equipment and then the costs are distributed over the products for recovery of incurred cost, and
- ii. **Cost control:** Cost centers assist in making a person responsible for the control of expenditure incurred by the cost center. Manager of each cost center shall control costs incurred in his area of responsibility.

The size of the cost center depends on the activity and operation, and feasibility of cost control. Sub-cost centers are created if the size of the cost centers becomes too big from control point of view.

3.2.3. Cost Unit

While cost centers assist in ascertaining costs by location, person, equipment, operation or process, cost unit is a unit of product, service or a combination of them in relation to which costs are ascertained or expressed.

The selection of suitable cost unit depends upon several factors, such as, nature of business, process of information, requirements of costing system, etc. but usually relates to the natural unit of the product or service. For example, in steel and cement industry, the cost unit is ‘tone’, while in transportation services, the unit may be passenger-kilometer or tone-km, etc. It may be noted that while the former is a single cost unit, the latter is a composite unit, i.e. a combination of two units. A few examples of cost units are given below:–

<i>Industry or product</i>	<i>Cost unit</i>
Automobile	Number
Biscuit	Kilogram
Bread	Thousand loaves
Breweries	Barrel
Chemical kilogram	Liter, gallon,
Gas meter	Cubic foot or cubic
Hospital	Patient day

Hotel
room, etc.

Guest-day, guest

Power and electricity

Kilowatt-hour

Transport

Passenger kilometer,

Cost Versus Expenses and loss:

Expense is defined as an expired cost resulting from a productive usage of an asset. An expense is map portion of the revenue earning potential of an asset w/c has been consumed in the generation of the revenue.

Loss: Is a reduction in firm's equity other than from withdrawals of capital for which no compensating value has been received. It is an expired cost resulting from the decline in the service potential of an asset that generated no benefit to the firm.

Ex. Obsolescence or destruction of stock by fire

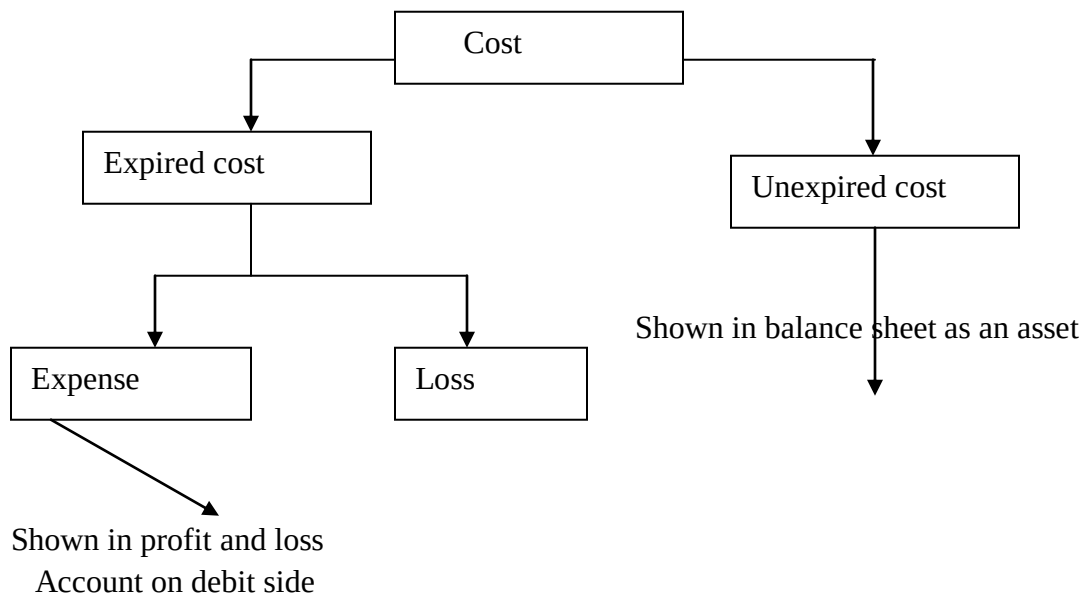


Fig 2.1 Relation of cost, expense and loss

2.4. ACCOUNTING FOR LOSSES

There is always a chance of occurrence of losses during the manufacturing process and even in the finished goods go down. Such losses are classified as normal loss and abnormal loss. Occurrence of **Normal loss** is always expected. It is unavoidable loss and is inherent in the manufacturing process or its chances of happening are more likely than not. For example; while transporting petrol, it is normal that a little quantity will be evaporated. **Abnormal loss** is an unexpected loss. Measures are always taken to avoid abnormal losses. For example; security system is installed to secure loss by theft from go

down of finished goods. Safety measures are undertaken during manufacturing process against any loss of breakage of the output. Special care is taken during transportation of petrol to avoid its leakage.

Normal and abnormal losses are treated differently in the financial accounting and in the cost accounting as well. Following is the self-explanatory chart that will help in understanding the difference between accounting treatment of the two different types of losses.

Treatment in Cost Accounting	
Normal loss	Abnormal loss
1. Charged to FOH account. 2. Overall per unit cost increases 3. No impact on individual job cost.	1. Charged to specific WIP account.

2.5. COST CLASSIFICATION

Cost classification refers to the process of grouping costs according to their common characteristics, such as nature of expense, function, variability, controllability and normality. Cost classification can be done on the basis of time, their relation with the product and accounting period. Cost classification is also made for planning and control and decision making. Thus, classification is essential for identifying costs with cost centres or cost units for the purpose of determination and control of cost:

- a) **By nature of expenses:** Costs can be classified into material, labour and expenses as explained earlier.
- b) **By function:** Costs are classified, as explained earlier, into production or manufacturing cost, administration cost, selling and distribution cost, research and development cost.

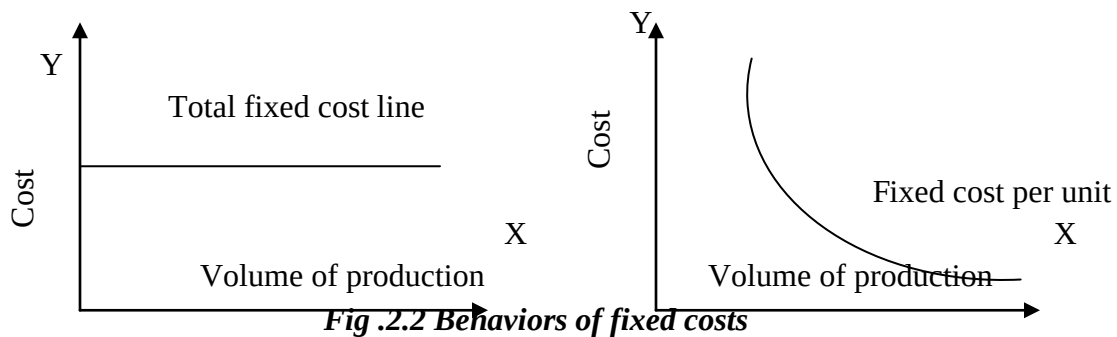
Production cost begins with the process of supplying material labour and services and ends with primary packing of the finished product. **Administration cost** is the aggregate of the costs of formulating the policy, directing the organization and controlling the operations of an undertaking, which is not related directly to production, selling, distribution, research and development activity or function. **Selling cost** refers to the expenditure incurred in promoting sales and retaining customers. **Distribution cost** begins with the process of making the packed product available for dispatch and ends with making the reconditioned returned empty package available for reuse. **Research and development cost**

relates to the costs of researching for new or improved products, new application of materials, or new or improved methods, processes, system or services, and also the cost of implementation of the decision including the commencement of commercial production of that product or by that process or method.

Pre-production cost refers to the part of development cost incurred in making trial production run preliminary to formal production, either in a new or a running factory. In a running factory, this cost often represents research and development cost also. Pre-production costs are normally considered as deferred revenue expenditure and are charged to the cost of future production.

- c) **By variability:** Costs are classified into fixed, variable and semi-fixed / semi-variable costs according to their tendency to vary with the volume of output.

Fixed costs are tend to remain unaffected by the variation or change in the volume of output, such as supervisory salary, rent, taxes, etc. These costs remain constant in total amount over a wide range of activity for a specified period of time. These costs do not increase or decrease when the volume of production change. But fixed cost “per unit “decreases when volume of prod increases and vice versa.



Fixed cost can be further classified as:

Committed – are those costs that are incurred in maintaining physical facilities & managerial setup. Once the decision to incur them has been made, they are unavoidable invariant in the short run.

E.g. Depreciation of machinery & equipment

Discretionary- are those costs which can be avoided by management decision e.g. advertising, research & development, salaries of low level managers ,etc., because these cost may be avoided or reduced in the short run.

The characteristics of fixed cost

Special features that make them different from other types of costs. These are:

- ☞ Fixed total amount within a relevant output range
- ☞ Increase or decrease in per unit when level of production changes
- ☞ Apportioned to department on some arbitrary basis
- ☞ Controlled mostly by top level management.

Variable costs tend to vary in direct proportion to the volume of output. Thus, in general, variable costs show the following characteristics:

- ☞ Variability of the total amount in direct proportion to the volume of output.
- ☞ Fixed amount per unit in the face of changing volume
- ☞ Easy and reasonably accurate & apportionment to department.
- ☞ Such costs can be controlled by functional management groups

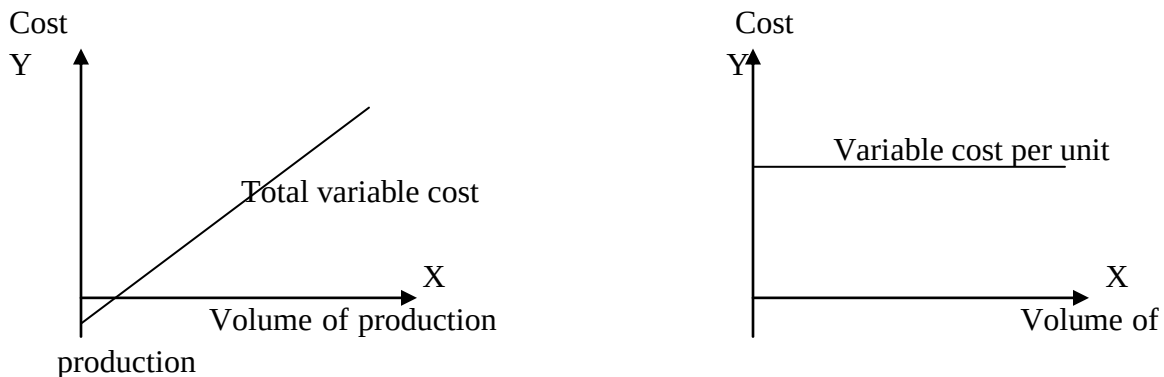


Fig: 2:3 Behaviors of variable costs

d) By controllability: Costs can be classified under controllable cost and uncontrollable cost.

Controllable cost can be influenced by the action of a specified member of an undertaking. **Uncontrollable cost** cannot be influenced by the action of a specified member of an undertaking.

e) By normality: Costs can be divided into normal cost and abnormal cost.

Normal cost refers to the cost, at a given level of output in the conditions in which that level of output is normally attained. Abnormal cost is a cost which is not normally incurred at a given level of output in the conditions in which that level of output is normally attained.

f) On the basis of time: Costs may be classified into historical or actual cost and predetermined or future cost.

Historical cost relates to the usual method of determining actual cost of operation based on actual expenses incurred during the period. Such evaluation of costs takes longer time, till the accounts are closed and finalized, and figures are ready for use in cost calculations.

Predetermined cost as the name signifies is prepared in advance before the actual operation starts on the basis of specifications and historical cost data of the earlier period and all factors affecting cost. Predetermined cost is the cost determined in advance and may be either estimated or standard. **Estimated cost** is prepared before accepting an order for submitting price quotation. It is also used for comparing actual performance. **Standard cost** is scientifically predetermined cost of a product or service applicable during a specific period of immediate future under current or anticipated operating conditions. The method consists of setting standards for each elements of cost, comparing actual cost incurred with the standard cost, evaluating the variance from standard cost and finding reasons for such variance, so that remedial steps can be taken promptly to check inefficient performances.

- g) **In relation to the product:** Costs may be classified into direct and indirect costs.

Direct costs are those which are incurred for a particular cost unit and can be conveniently linked with that cost unit. Direct costs are termed as product cost.

Indirect costs are those which are incurred for a number of cost units and also include costs which though incurred for a particular cost unit are not linked with the cost unit. Since such costs are incurred over a period and the benefit is mostly derived within the same period, they are called period costs.

- h) **Common and Joint Costs**

Common costs: costs of facilities or services employed by two or more operations. **Joint costs:** occur when the production of one product may be possible only if one or more other products are manufactured at the same time.

- i) **Capital expenditure and Revenue expenditure**

Capital expenditure: intend to benefit future periods and is recorded as an asset.

Revenue expenditure: benefit the current period and is recorded as an expense. Ultimately the asset (capital expenditure) will flow in to expense stream as it is consumed or when it loses its usefulness.

- j) **Cost analysis for decision making:** Here costs are classified under **relevant costs** (e.g. additional fixed cost, incremental cost, opportunity cost) and **irrelevant costs** (e.g. sunk cost, committed costs, etc.)

Opportunity Cost

Opportunity cost is the potential benefit that is given up when one alternative is selected over another. To illustrate this important concept, consider the following examples:

Example 1 Vicki has a part-time job that pays \$200 per week while attending college. She would like to spend a week at the beach during spring break, and her employer has agreed to give her the time off, but without pay. The \$200 in lost wages would be an opportunity cost of taking the week off to be at the beach.

Example 2 Suppose that Neiman Marcus is considering investing a large sum of money in land that may be a site for a future store. Rather than invest the funds in land, the company could invest the funds in high-grade securities. If the land is acquired, the opportunity cost is the investment income that could have been realized by purchasing the securities instead.

Example 3 Steve is employed by a company that pays him a salary of \$38,000 per year. He is thinking about leaving the company and returning to school. Since returning to school would require that he give up his \$38,000 salary, the forgone salary would be an opportunity cost of seeking further education.

Sunk Cost

A **sunk cost** is a cost *that has already been incurred* and that cannot be changed by any decision made now or in the future. Because sunk costs cannot be changed by any decision, they are not differential costs. And because only differential costs are relevant in a decision, sunk costs can and should be ignored.

To illustrate a sunk cost, assume that a company paid \$50,000 several years ago for a special-purpose machine. The machine was used to make a product that is now obsolete and is no longer being sold. Even though in hindsight purchasing the machine may have been unwise, the \$50,000 cost has already been incurred and cannot be undone. And it would be folly to continue making the obsolete product in a misguided attempt to “recover” the original cost of the machine. In short, the \$50,000 originally paid for the machine is a sunk cost that should be ignored in current decisions.

2.6. Producing department Vs Service department cost

This classification is a classification of costs in relation to manufacturing departments. The departments of a factory generally fall in to two categories:

- a) Producing department
 - b) Service department
- a) **Producing department** is also known as operating department. A producing department is one in which manual and machine operation are performed directly up on any part of the product manufactured in many cases producing departments are further subdivided in to cost pools.
- b) **Service department** is one that is not directly engaged in production but render a particular type of service for the benefit of other service department as well as the producing department

Examples:

- ✓ Repair and maintenance department
- ✓ Receiving and inspection department

- ✓ Data processing, time keeping
- ✓ Store room, cafeteria, payroll, budgeting, Plant protection, etc.

2.7. The manufacturing cost Accounting cycle

The manufacturing process and the physical arrangement of a factory are the skeleton upon which the cost system and cost accumulation procedures for the manufacturing function are built.

Since cost accounts are an expansion of general accounts, they should, as a basic accounting procedure be related to them. This relationship leads to the concept of the tie-in of cost accounts with general accounts as illustrated below.

Accounts describing manufacturing operation are material, payroll, factory overhead control, work-in-process, finished goods, and cost of goods sold. Each fiscal period, these accounts recognize and measure the flow of costs from the acquisition of materials, through factory operations, to the cost of product sold.

Work in Process Inventory Account		Work in Process Inventory Account	
Balance 12/31/x3: <u>\$10,000</u>	Used during 20x4:	Balance 12/31/x3: \$ 2,000	Completed during 20x4: <u>\$30,000</u>
Total direct materials purchased during 20x4: <u>20,000</u>	<u>\$25,000</u>	Direct materials used during 20x4: <u>25,000</u>	
Balance 12/31/x4: <u>\$5,000</u>		Direct labor 20x4: 12,000	
		Manufacturing overhead 20x4: 6,000	
		Balance 12/31/x4 <u>\$15,000</u>	

$$\text{Beginning Balance} + \text{DM purchase} - \text{Ending DM Balance} = \text{DM used}$$

Factory Payroll Account		Work in Process Inventory Account	
Direct labor earned during 20x4:	20x4:	Balance 12/31x3:	Completed
	<u>\$12,000</u>		<u>\$ 2,000\ during 20x4:</u>
20x4:			<u>\$30,000</u>
<u>\$12,000</u>		Direct materials used during 20x4:	25,000
		Direct labor 20x4:	
Balance 12/31/x4:			<u>12,000</u>
	<u>\$ 0</u>	Manufacturing overhead 20x4:	6,000
		Balance 12/31/x4	<u>\$15,000</u>

Manufacturing Overhead Control Account		Work in Process Inventory Account	
Total manufacturing overhead incurred during 20x4:	20x4:	Balance 12/31/x3:	Completed
	<u>\$ 6,000</u>	<u>\$2,000</u>	<u>during 20x4:</u>
20x4:			<u>\$30,000</u>
<u>\$ 6,000</u>		Direct materials used during 20x4:	25,000
		Direct labor 20x4:	12,000
Balance 12/31/03: <u>\$ 0</u>		Manufacturing overhead 20x4:	
			<u>6,000</u>
		Balance 12/31/x4	<u>\$15,000</u>

Work in Process Inventory Account		Finished Goods Inventory Account	
Balance 12/31/x3: \$2,000	Completed during 20x4: <u>\$30,000</u>	Balance 12/31/x3: \$6,000	Sold during 20x4: <u>\$24,000</u>
Direct materials used during 20x4: 25,000		Completed during 20x4: 30,000	
Direct labor 20x4: 12,000			
Manufacturing overhead 20x4: 6,000		Balance 12/31/x4: <u>\$12,000</u>	
Balance 12/31/x4 \$15,000			

Finished Goods Inventory Account		Cost of Goods Sold Account
Balance 12/31/x3: \$6,000	Sold during 20x4: \$24,000	Sold during 20x4: \$24,000
Completed during 20x4: 30,000		
Balance 12/31/x4: \$12,000		

2.8. Reporting the Results of Operation

The balance sheet

The balance sheet or statement of financial position, of a manufacturing company is similar to that of a merchandising company. However, their inventory accounts differ. A merchandising company has only one class of inventory—goods purchased from suppliers for resale to customers.

In contrast, manufacturing companies have three classes of inventories—raw materials, work in process, and finished goods. Raw materials are the materials that are used to make a product. Work in process consists of units of product that are only partially complete and will require further work before they are ready for sale to a customer. Finished goods consist of completed units of product that have not yet been sold to customers. Ordinarily, the sum total of these three categories of inventories is the only amount shown on the balance sheet in external reports.

The Income Statement

At first glance, the income statements of merchandising and manufacturing companies are very similar. The only apparent difference is in the labels of some of the entries in the computation of the cost of goods sold. The computation of cost of goods sold relies on the following basic equation for inventory accounts:

Basic Equation for Inventory Accounts

$$\text{Beginning balance} + \text{Additions to inventory} = \text{Ending balance} + \text{Withdrawals from inventory}$$

The beginning inventory consists of any units that are in the inventory at the beginning of the period. Additions are made to the inventory during the period. The sum of the beginning balance and the additions to the account is the total amount of inventory available. During the period, withdrawals are made from inventory. The ending balance is whatever is left at the end of the period after the withdrawals.

The cost of goods sold for a manufacturing company is determined as follows:

Cost of Goods Sold in a Manufacturing Company

$$\text{Beginning finished goods inventory} + \text{Cost of goods manufactured} = \text{Ending finished goods inventory} + \text{Cost of goods sold}$$

Or

$$\text{Cost of goods sold} = \text{Beginning finished goods inventory} + \text{Cost of goods manufactured} - \text{Ending finished goods inventory}$$

To determine the cost of goods sold in a merchandising company, we only need to know the beginning and ending balances in the Merchandise Inventory account and the purchases. Total purchases can be easily determined in a merchandising company by simply adding together all purchases from suppliers.

To determine the cost of goods sold in a manufacturing company, we need to know the cost of goods manufactured and the beginning and ending balances in the Finished Goods inventory account. The cost of goods manufactured consists of the manufacturing costs associated with goods that were finished during the period.

EXHIBIT 2-2

Schedule of Cost of Goods Manufactured

Direct materials:

Beginning raw materials inventory -----	\$xx
Add: Purchases of raw materials -----	xx
Raw materials available for use -----	xx
Deduct: Ending raw materials inventory -----	<u>xx</u>
Raw materials used in production -----	<u>\$xx</u>

Direct materials

Direct labor ----- \$xx

Direct labor

Manufacturing overhead:

Insurance, factory -----	\$xx
Indirect labor -----	xx
Machine rental -----	xx
Utilities, factory -----	xx
Supplies -----	xx
Depreciation, factory -----	xx
Property taxes, factory -----	<u>xx</u>
Total manufacturing overhead cost -----	<u>\$xx</u>

Manufacturing overhead

Illustration

BARAF manufacturing enterprise begins a new fiscal with the following financial position

BARAF manufacturing company**Balance sheet****Jan 1, 2008****Asset**

Current Asset:

Cash..... \$183,000

Marketable securities 76,000

Account receivable (net) 313,100

Inventories:

Finished goods 68,700

Work in process 234,300

Material 135,300 438,300Prepaid expenses..... 15,800

Total Current Assets 1,026,200

Property, plant and equipment:

Land 41,500

Building 580,600

Machinery and equipment 1,643,000

Accumulated depreciation..... (1,010,700) 1,212,900Total property, plant and equipment..... 1,254,400Total Assets 2,280,600

Liabilities

Current Liabilities:

Accounts payable.....	\$553,000
Estimated income tax payable	35,700
Due on long term debt.....	20,000
Total current liabilities	608,700
Long-term debt	<u>204,400</u>
Total liability	813,100

Stockholders' Equity

Common stock	528,000
Retained earnings	939,500
Total stockholders' equity.....	<u>1,467,500</u>

Total	liabilities	and	stockholders'	equity
.....			<u>2,280,600</u>	

During the month of January, the company completed the following transactions:

- a) Materials purchased and received on account for \$ 100,000

Materials	100,000
Accounts payable	100,000

- b) Materials requisitioned in the month amount \$80,000 and \$12,000 for production and indirect factory use respectively.

Work in process.....	80,000
Factory Overhead Control	12,000
Materials	92,000

- c) Total gross payroll, \$160,000. The payroll is paid to employees after 7% FICA tax and 12% federal income tax are deducted.

Payroll.....	160,000
Income tax payable	19,200
FICA tax payable	11,200
Cash	129,600

- d) The distribution of payroll was: Direct Labor (65%), indirect factory labor (15%) marketing salary (13%) and administrative salaries (7%).

Work in Process.....	104,000
Factory Overhead Control.....	24,000
Marketing Expense Control.....	20,800
Administrative Expense Control.....	11,200
Payroll	160,000

- e) Additional 10% is recorded for the employer's payroll taxes: FICA TAX 7%, Federal unemployment insurance tax 1% and state unemployment insurance tax 2%.

Factory Overhead Control.....	12,800
Marketing Expense Control.....	2,080
Administrative Expense Control.....	1,120
FICA tax.....	11,200
Federal unemployment insurance tax	1,600
State unemployment insurance tax	3,200

- f) Factory Overhead consisting \$8,500 and 1,200 for depreciation and prepaid insurance respectively is recorded

Factory Overhead control	9,700
Accumulated depreciation	8,500
Prepaid Expense	1,200

- g) General overhead cost (not itemized) amounts \$26,340. 70% of these expenses were paid in cash and the balance is credited to Account Payable account

Factory Overhead control	26,340
Cash.....	18,438
Accounts payable	7,902

- h) \$205,000 is collected from credit customers in settling their obligations.

Cash.....	205,000
Accounts receivable	205,000

- i) Account payable \$ 227,000 and Estimated income tax \$35,700 is paid

Account payable	227,000	
Estimated income tax	35,700	
Cash		262, 700

- j) The balance of Factory Overhead control account is transferred to the work in process account

Work in process.....	84,840	
Factory Overhead control	84,840	

- k) Work completed and transferred to the finished goods amounts \$320,000

Finished Goods.....	320,000	
Work in process	320,000	

- l) Sales, \$384,000. 40% paid in cash and the balance is charged to Accounts Receivable account. The cost of goods sold was 75% of sales.

Cash.....	153,600	
Accounts receivable	230,400	
Sales		384,000

Cost of Goods sold	288,000	
Finished Goods		288,000

- m) Provision for income tax, \$26,000

Provision for income tax	26,000	
Estimated income tax payable	26,000	

BARAF manufacturing company
Cost of Goods Sold statement
For the month ended Jan 31, 2008

Direct Materials:

Materials inventory, Jan 1, 2008	135,300	
Purchases	100,000	
Materials available for use	235,300	
Less: indirect materials issued	12,000	
Materials inventory, Jan 31, 2008	<u>143,300</u>	155,300
Direct material consumed		80,000
Direct labor		96,000
Factory overhead:		
Indirect materials	12,000	
Indirect labor	24,000	
Payroll taxes	12,800	

Depreciation	8,500	
Insurance	1,200	
General factory overhead	<u>26,340</u>	<u>84,840</u>
Total manufacturing cost		260,040
Add: work in process inventory as of Jan1, 2008		<u>234,300</u>
		494,300
Less: work in process as of Jan 31, 2008		<u>174,340</u>
Cost of goods manufactured		320,000
Add: finished goods inventory as of Jan 1, 2008		<u>68,700</u>
Cost of goods available for sale		388,700
Less: finished goods inventory as of Jan 31, 2008		<u>100,700</u>
Cost of goods sold		<u>288,000</u>

BARAF manufacturing company

Income statement

For the month ended Jan 31, 2008

Sales	384,000	
Less: CGS	<u>(288,000)</u>	
Gross profit		96,000
Less: commercial expenses		
Marketing expenses	22,880	
Administrative expenses	<u>12,320</u>	<u>35,200</u>
Income from operation		60,800
Less provision for income tax		<u>26,000</u>
Net income		<u>34,800</u>

Calculating Unit Cost- Manufacturing

The cost per unit of a product is found by adding all its cost and dividing it by the number of units produced.

In manufacturing, for example, it will include:

Materials	\$1,000
Labor	\$2,000
Overhead	<u>\$5,000</u>
Total Cost	<u>\$8,000</u>

Number of Units Produced 10,000

Unit cost =total cost / number of units produced

$$\text{Unit Cost} = (10,000 / \$8,000) = \underline{\$1.2}$$

Self-Test Questions

A. **True or False.** Indicate whether each of the following statement is true or false. Give reasons in brief.

- 1) Variable cost per unit remain unchanged when out put is increased or decreased.
- 2) Loss means expired cost while expense means expired cost as well as unexpired cost.
- 3) Abnormal cost is uncontrollable
- 4) Conversion cost is the aggregate of direct labour and manufacturing overhead.
- 5) Nails used in furniture manufacture are an indirect material cost.
- 6) A producing department is one in which manual and machine operations are performed indirectly upon any part of the product manufactured.
- 7) Commercial expenses are the aggregate of marketing expenses and administrative expenses.
- 8) Direct materials + direct labour + factory overhead + commercial expenses = total operating costs.
- 9) Job order costing applies where work is under taken to customers' special requirements.
- 10) The basis for the classification of cost as direct and indirect is the identifiability of costs to cost units or process or cost centres.

B. Fill in the blank spaces

- 1) Aggregate of all direct costs is known as _____.
- 2) Factory cost plus office and administration overhead is equals to _____.
- 3) _____ is a unit of product, service or time in relation to which costs may be ascertained.
- 4) Indirect materials plus indirect labour plus _____ = overhead.
- 5) _____ are future costs which are ascertained in advance of production on the basis of a specification of all the factors affecting cost.

C. multiple choice questions

- 1) Warehousing cost is :
 - a) production overhead
 - b) selling overhead
 - c) administration overhead
 - d) distribution overhead
- 2) Telephone bill falls in the category of
 - a) fixed cost
 - b) semi-variable cost
 - c) variable cost
 - d) stepped cost
- 3) Allotment of whole items of overhead to cost centres is known as:
 - a) allocation
 - b) apportionment
 - c) classification
 - d) absorption
- 4) The cost that form a part of production and incurred normally on expected line at a given level of output is:
 - a) indirect cost
 - b) controllable cost
 - c) uncontrollable cost
 - d) abnormal cost
 - e) normal cost
- 5) Service department is one that is not directly engaged in production but render a particular type of services for the benefit of other service department as well as the producing department. Based on this definition, one the following can not be examples for service department.
 - a) Repair and maintenance department
 - b) Data processing department
 - c) Storeroom
 - d) Cafeteria
 - e) Budgeting department
 - f) None
- 6) _____ are those cost centres where actual production work takes place.

- a) Personal cost centres
 - b) Service cost centres
 - c) Production cost centres
 - d) Impersonal cost centres
- 7) Identify wrong statement(s) from the followings. (*There may be more than one answer*)
- A. variable cost per unit varies with increase or decrease in the volume of output
 - B. conversion cost is equal to direct wages plus factory overhead
 - C. fixed cost per unit remains fixed
 - D. All costs are controllable.
 - E. An item of cost that is direct for one business may be indirect for another
- 8) ----- are those costs that are incurred in maintaining physical facilities, managerial setup and unavoidable once the decision to incur them has been made.
- A. Period costs
 - B. Pre-determined
 - C. Committed costs
 - D. Discretionary costs
 - E. Historical costs
- 9)) If Dm represents direct materials; DL represents direct labor ;PC represents prime cost ;FOH stands for factory overhead ; IM represents indirect materials ; IL represents indirect labors ; M represents manufacturing costs ; CE stands for commercial expenses ; Top represents total operating costs ,which one is wrong about the relationship of these operating costs ?
- A. $PC + FOH = M$
 - B. $DM + DL = Pc$
 - C. $TOP - CE = PC + FOH$
 - D. $M = TOP + CE$
- 10) Based on their traceability with cost unit or jobs or processes, costs can be classified as----- &----- costs.
- A. Fixed and variable
 - B. Product costs &period costs
 - C. Direct & indirect
 - D. Controllable & non-controllable
 - E. Normal & abnormal

D. Objective questions

- 1) What is classification and why is it so important to business organization?

- 2) Give a realistic example of each of the following
 - a) Direct materials
 - b) Direct labour
 - c) Indirect material
 - d) Indirect labour
 - e) Indirect expenses
- 3) What are prime costs? And identify the components of prime costs.
- 4) Define committed and discretionary costs and compare their nature with the general characteristics listed under fixed costs.
- 5) Explain why a variable cost is said to be variable and why a fixed cost is said to be fixed.

CHAPTER THREE

JOB-ORDER COSTING

Chapter Learning Objectives

After you have studied this chapter, you should be able to:

- ❖ Describe the differences between job-order costing and process costing and identify the types of firms that would use each.
- ❖ Identify and set up the basic source documents used in a job order costing system.
- ❖ Identify the accounts used to capture costs in a job- order costing system.
- ❖ Compute the predetermined overhead rate and use the rate to assign over head to unit or services produced.
- ❖ Describe the cost flows associated with job-order costing

3.1 Introduction

Cost accounting is a sub-field of accounting that measures, records, reports, and interprets information about costs. In any type of firm, therefore, Cost accounting system focuses on cost determination. For manufacturing firms, cost accounting system will systematically determine the three Manufacturing cost elements, cost of direct materials, cost of direct labor and cost of factory (Manufacturing) overheads incurred to produce an item.

The knowledge of determining cost of a product is a pre-requisite to the use of cost data for different purposes. A product costing system accumulates the costs incurred in a production process and assigns those costs to the organization's final products or services. Product costs are needed for a variety of purposes in both financial accounting as well as managerial accounting.

Use in Financial Accounting: In financial Accounting, product costs are needed to value inventory on the balance sheet and to compute cost of goods sold expenses on the income statement. Under the Generally Accepted accounting Principles (GAAP),

inventory is valued at its cost until it is sold then the cost of the inventory becomes an expense on the period in which it is sold.

Use in Managerial Accounting: In managerial accounting, product costs are needed to help managers with planning and to provide them with data for decision making. Decisions about product prices, the mix of products to be produced, and the quantity of output to be manufactured are among those for which product cost information is needed.

The cost- benefit (contexts approach) is of relevance in designing and choosing costing systems. The detailed accounting procedures used in product costing systems depend on the type of industry involved.

3.2 Types of Product –costing systems

There are two main types of product costing systems. Companies select a method that best matches the flow of work in their business. These methods are used to allocate all production costs: direct materials, direct labor and factory overhead. Each costing system gathers and reports on the same information. The method used depends on the needs of the business.

These product costing systems are:

- 1) Job-order costing system
- 2) Process costing system

1) Job-order costing (or simply job costing)

Job-costing is used when different types of products, jobs, or batches are produced or it is the method of allocating costs of products that are readily identified by individual units or batches, each of which requires varying degree of attention and skill. The cost unit (object) is the jobs, the work order or contract; the records will show the cost of each. Industries that commonly use job- order method include construction, furniture, manufacturing of tailor made or unique goods, ship builders, and etc.

The cost accounting procedures are designed to assign costs to each job. The costs assigned to each job are averaged over the units of production in the job to obtain average

cost per unit. For example, suppose that Mega printing enterprise worked on two printing jobs during Miazia 2000, and the following costs were incurred:

	Job A 27 (1,000 Campaign Posters)	Job B 39 (100 wedding invitations)
Direct Materials	Br100	Br36
Direct labor	250	40
Factory overhead	<u>150</u>	<u>24</u>
Total manufacturing cost	Br <u>500</u>	Br <u>100</u>

The cost per campaign poster is Br 0.50 (Br 500 divided by 1000 posters) and the cost per wedding invitation is Br 1.00 (Birr 100 divided by 100 invitations)

2 Processes costing

Process costing is used by companies that produce large numbers of identical products and/or when units are not distinguishable from one another during one or more manufacturing processes. This costing method accumulates all the production costs for a large number of units of output, and then these costs are averaged over all of the units produced. The cost object is the department, processing center or the production line.

The following conditions may also exist in process costing:

- 1) The product of one process becomes the material (input) of the next process.
- 2) Different products or even by-products are produced by the same process.

The process costing method is applicable to industries such as flour mills, breweries, chemical Industries, oil refining, and many others.

For example, Moha millennium plant at Hawassa produced 400,000 bottles of soft drink specifically Pepsi cola during sane, 2000. The following manufacturing costs were incurred in sane.

Direct Materials	Birr 150,000
Direct labor	200,000
Factory overhead	<u>150, 000</u>
Total manufacturing cost	Birr <u>500,000</u>

The cost per bottle is birr 1.25(total manufacturing cost of birr 500,000 divided by 400,000 bottles produced)

These costing systems have got two major purposes that they satisfy simultaneously. These purposes are identified as:

- 1) planning and controlling purpose; and
- 2) Product costing purpose

The planning and control purpose focuses on cost data provided through cost accounting system for planning, decision making, and performance evaluation. Managers need to know future cost data in setting their goals and objectives. They want to know how much cost is incurred to produce an item and also they need to know how work is performed by comparing the actual costs with the budgeted ones and then identify, analyze and take actions on the deviations. Variance analyses and performance evaluation will be discussed in detail in cost and management accounting II.

The product costing purpose focuses on the inventory valuation, determination of periodic net income and income tax reporting. Accountants need to determine, thus, total cost of products and costs of individual unit of product in order to determine cost of inventories and cost of goods sold. In this chapter and in the coming two chapters, how this purpose is achieved through the job-order costing system and the process costing system will be discussed in detail.

Activity 3.1

- Distinguish the differences between job-order costing and process costing
- List at least three organizations which use either job or process costing system to accumulate their cost.
- Discuss the major purposes of product costing systems

3.3. Accumulating costs in job- order costing system

Main features of job-costing

- ◆ Production is undertaken against customer's orders.
- ◆ Each job has its own characteristics and satisfies the requirement to the customer.
- ◆ Duration of job is normally short. However, a large order may extend beyond one year.
- ◆ Identity of each order is maintained through out the manufacturing process.
- ◆ Only prime costs elements are traceable and overheads are apportioned to each job on some suitable basis.

The job-order costing system is used when products are identifiable lots or groups, or when the products are manufactured according to customer specifications. This costing system is, therefore, used in those manufacturing firms where many different products, jobs or batches of production, each receiving varying degrees of attentions and skills are produced each period.

For example, a printing enterpriser may receive an order to print 4,000 copies of a university graduation bulletin. This bulletin is prepared as directed by the graduation committee; it differs in form and content from the printing work done for other customers.

In job-order costing, each job is an accounting unit to which material, labor, and factory overhead costs are assigned by means of job-order numbers. Therefore, the costs of individual jobs must be kept separate. This is accomplished by the use of a job-order cost sheet, or job-cost records. On the job-order cost sheet, all the three manufacturing cost elements –cost of direct materials, cost of direct labor, and cost of factory overhead attributable to the job are recorded. The job sheet also provides the summary of total and unit cost inputs and out puts. A typical job cost sheet provides information about on the date products are started and completed, job number, number of units completed and description.

Job – cost sheets differ in form, content, and arrangement in each business. In each firm, the upper section provides space for the job number, the name of the customer, a description of the items to be produced, the quantity, the date started, the date completed, and the lower section furnishes summarized cost information. The job cost sheets are the source documents to record the required journal entries and post to the general ledger accounts. In short, job- cost sheet is a subsidiary ledger maintained to keep track of the manufacturing costs assigned to each job and it is controlled by work in process control account. The following is a sample job – order cost sheet

XYZ Company
Job Cost Record /Sheet/

Job Number: AMU -1000

Description: Armed chair

Date completed December 29, 2007

Customer Name: AMU

Number of Units completed: 1,000

Date Started: December 2, 2007

Direct Material				
Date	Requisition Number	Quantity	Unit price	Cost
DD MM,YY	DM 25	3000	25	\$75,000

Direct Labor				
Date	Time card Number	Hours	Rate	Cost
DD,MM,YY	DL 25	2000	50	\$100,000

Factory Over head				
Date	Activity Base	Quantity	Application rate	Cost
DD,MM,YY	Labor hours	6000	2.20	\$ 13,200

Cost summary	
Cost Item	Amount
Total direct material	\$ 75,000
Total direct Labor	\$ 100,000
Total factory over head	\$13,200
Total Cost	\$188,200
Unit Cost	\$188.20

NB: - All figures are assumed

\$188.20= \$188,200 ÷ 1,000

DD- Stands for day

1,000= Number of units completed

MM- stands for month

YY – Stands for year

Activity 3.2

- Describe the features of the three manufacturing cost elements.
- Identify and discuss about the main features of job- order costing.
- In job-order costing products are manufactured according to customer specifications, why? Support your discussion with your own example.
- Why job-order cost sheet is called “the source document”?
- List down the components of job-cost record

3.3.1 Accounting for materials

In manufacturing enterprise, it is common practice to record all materials and supplies in one control account, Materials. Procedures and events that affect that materials account involve:

- 1) Purchase of materials
- 2) Issuance of materials for factory use or for production.

1/ Recording the purchase of materials

The accounting techniques used for the purchase of materials vary little from thus studied in general accounting. The account debited is Materials inventory now used in place of purchases, and the credit is made to Accounts payable or cash as materials are received. Each purchase is also entered on an individual material ledger card or material records. These subsidiary records maintain a continuous record of additions to and reductions from stock. At a minimum these records would contain columns for quantity received, issuance for jobs and balance. There is a separate subsidiary materials record for each type of materials in the subsidiary record. The following illustrative example (Transactions 1-9) considers the events that took place in December, 2007 in XYZ CO.

Transaction 1:

Purchase of materials (direct and indirect) of \$ 89,000 on account. When materials are purchased, the approved invoice is recorded as:

Materials inventory	89,000
Accounts payable	89,000

After the above journal entry is passed, the asset materials inventory and the liability, Accounts payable accounts are increased. In case when materials are returned to the vendor, the following journal entry is maintained:

Accounting payable	***
Materials	***

An entry for the returned materials is also made on the appropriate Materials ledger card or materials records

2 Recording the Issuance of materials:

When production is to be initiated, production order is prepared and approved by appropriate official(s). The production order authorizes the manufacture of a product and

contains a description of a product to be produced, the quantity required, estimated date of completion, job order number, date of production order and the authorization signature. Once the production is issued, a material requisition form would be prepared. This source document authorizes the transfer of material from warehouse to production department to start production. The materials requisition form should contain specific description of materials needed and distinguish between direct and indirect materials.

The materials requisition form uses as:

- ❖ A means of controlling the flow of materials into production since the store room clerk issues materials based on the receipt of properly authorized materials requisition form.
- ❖ A source document to pass journal entries in the accounting records. The following is an example of materials requisition form:

XYZ Company Materials requisition form Materials requisition number ----- M-132 Job- number to be charged-----Amu-1000 Date -----December 9, 2007 <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 10px;"> <thead> <tr> <th style="width: 20%;">Quantity</th> <th style="width: 20%;">Description</th> <th style="width: 15%;">Unit cost</th> <th style="width: 15%;">Total cost</th> <th style="width: 30%;">classification</th> </tr> </thead> <tbody> <tr> <td>60 Board feet</td> <td>Wooden board</td> <td>\$ 1,350</td> <td>\$ 81,000</td> <td>direct</td> </tr> <tr> <td>10 tubes</td> <td>Glue</td> <td>400</td> <td>4000</td> <td>indirect</td> </tr> </tbody> </table>					Quantity	Description	Unit cost	Total cost	classification	60 Board feet	Wooden board	\$ 1,350	\$ 81,000	direct	10 tubes	Glue	400	4000	indirect
Quantity	Description	Unit cost	Total cost	classification															
60 Board feet	Wooden board	\$ 1,350	\$ 81,000	direct															
10 tubes	Glue	400	4000	indirect															

Transaction 2:

Materials sent to manufacturing plant floor or materials requisitioned (issued) for production, direct materials \$ 81, 000 and indirect materials \$ 4000. The flow of materials from the storeroom to factory results, in a transfer of materials from materials inventory account to the work in process or factory (manufacturing) overhead account. Each requisition results in a debit to Work in process or Factory overhead and a credit to materials inventory. Work in process account will be charged for direct materials costs and the factory overhead control account will be charged for all indirect materials. Materials requisitions are Summarized and recorded as follows:

Work in process	81,000
-----------------	--------

Factory overhead control	4000
Materials inventory	85, 000

When materials originally requisitioned for a job are not used and returned to the store room accompanied by a returned materials report, materials inventory is debited and work in process is credited as shown below:

Materials	***
Work in process	***

This requires entries on the materials ledger card or materials record as well as on the job- order cost sheet. The effect of this transaction on the materials inventory account is shown below. Assume that materials costing \$ 100,000 were available on December 1, 2007 as beginning inventory.

Materials inventory	
Balance (inventory) \$100,000	(2) Direct materials requisitioned \$81, 000
(1) Purchase \$ 89,000	(2) indirect Materials requisitioned \$4, 000
Balance \$ <u>104,000</u>	

3.32 Accounting for labor

The labor cost is accumulated on each job in the same way as material cost. The entire payroll procedure may be divided into two distinct phases:

- 1) Collection of payroll data, computation of earnings, calculation of payroll taxes, and the payment of wage or salaries.
- 2) Distribution and allocation of labor costs to jobs, departments, and other cost classifications.

To compute the labor of a given order, the time spent on each job during a day must be recorded on each worker's time ticket. The time tickets in turn, are priced in the payroll department to permit computation of employee's gross earnings. The job time ticket or simply time tickets are priced using the employee's rate of payment. Time tickets for the various jobs worked on each day are sorted, priced, and summarized; at regular intervals usually at the end of the payroll period. The labor time and labor cost for each job are entered in the space provided on the job order cost sheets. Posting to the cost sheets constitutes the distribution of the direct labor costs. A summary of the time ticket data facilitates journalizing, general ledger posting, and preparation of reports.

Transition 3:

Assume Xyz Company met two payrolls during the month of December \$ 12,000 of direct labor and \$ 1,800 of indirect labor on the 15th and \$ 15,000 of direct labor and \$2,200 of indirect labor on the 31st. Assume further that the company with holds 15 percent (15%) for income tax and 6.5 Percent (6.5%) for FICA (Federal Insurance Contribution ACT) tax. The appropriate journal entries would be:

December 15th		
Payroll	13,800	
Employees income tax payable		2,070
FICA tax payable		897
Accrued payroll		10,833
<i>For the incurrence of labor cost (liability)</i>		
◆ Payroll: Direct labor cost + indirect labor cost		
➤ \$ 12,000 + \$ 1800 = \$ <u>13,800</u>		
◆ Employees income tax payable:		
➤ \$ 13,800 X 15% = \$ <u>2,070</u>		
◆ FLCA tax payable :		
➤ \$ 13,800 X 6.5% = \$ <u>897</u>		
◆ Accrued payroll :		
➤ \$ 13,800 - \$ 2,070 - \$ 897 = \$ <u>10,833</u>		
December 31st		
Payroll	17,200	
Employees income tax payable		2,580
FICA tax payable		1,118
Accrued payroll		13,502

In the above two journal entries (December 15th and December 31st) the employer is assumed to act as a collection agent for the federal Inland Revenue authority or some other concerned body to collect taxes. When the amounts collated are transferred to the concerned body, the employees' income tax payable and FICA tax payable accounts are debited and cash will be credited for the amount collected and transferred.

Transaction 4:

At each payroll date or payment date, the entry to record the payment to the workers would be recoded as follows:

December 15th		
Accrued payroll	10,833	
Cash		10,833
<i>(For the payment of labor cost or liability)</i>		

December 31st	
Accrued payroll	13,502
Cash	13,502
<i>(For the payment of labor cost or liability)</i>	

Transaction 5:

As employee, employers are expected to pay FICA tax and in addition the employers' should have to contribute for state unemployment tax and federal unemployment tax from which the employees are exempted. The employer's payroll taxes are commonly recorded at the end of each month. Assume that the employer is expected to contribute a 6.5% FICA tax, a 2.7% state unemployment tax. The payroll tax for both direct and indirect factory labor are treated as factory overhead.

The journal entry for this event would be:

Factory overhead control	3,069	
FICA tax payable		2,015
State unemployment tax payable		837
Federal unemployment tax payable		217
◆ \$ 2,015 = \$31,000 x 6.5%		
◆ \$ 837 = \$31,000 x 2.7%		
◆ \$ 217 = \$31,000 x 0.70%		
<i>(\$31,000 = payrolls at 15th and 31st of December)</i>		

Payroll is the labor clearing account kept in the records as a convenience, pending analysis of the labor time tickets and distribution of the labor costs to the proper accounts. Distribution entries are usually recorded on a weekly basis, so that labor cost remains current on the job order cost sheets and available to operating management

The accrued payroll account represents the liability for net pay, i.e., gross pay less withholding. The amounts withheld for employees income tax payable, federal insurance contribution act (FICA) tax, and other deductions are credited to the appropriate liability accounts. A summary of information from source documents (labor time tickets) is used to determine the amount of direct and indirect labor costs to distribute and the individual amounts of direct labor cost to charge to each job. The direct labor costs are charged (distributed) to work in process account and individual jobs (i.e. job cost sheets). Indirect

labor costs are charged to the factory overhead or manufacturing overhead account because these costs cannot easily be identified with any particular job.

Transaction 6:

The following month end summary entry would be made in order to distribute direct and indirect labor costs to their respective accounts:

Work-in process	27000
Factory overhead control	4000
Payroll	31000

The payroll account and employer payroll taxes accounts may also include amounts applicable to marketing and administrative personnel accounts. Such costs would be charged to marketing and administrative expense accounts.

Activity 3.3

- What is the purpose of job order cost card? Identify the kinds of information recorded on that document.
- Describe the purpose of labor time ticket.

3.3.3 The accounting for factory overhead

Factory overhead or manufacturing overhead cost must be considered along with direct material costs and direct labor costs in determining unit cost of products. Factory overhead includes expenses such as rent, power, heat, light, insurance, depreciation on machinery, equipment and other factory facilities. Although such costs are not directly traceable to specific jobs, they are part of the basic inputs to perform a job. The assignment of factory overhead costs to jobs completed or units produced is often a difficult task. This is because:

1. Factory overhead cost is an indirect cost which cannot be traceable to a product or job in a straightforward and reasonably exact manner.
2. Factory overhead cost consists of many unlike items involving both variable and fixed costs.
 - Variable cost items: Costs which vary with the quantity of inputs manufactured.
 - Fixed costs: Costs which remain constant regardless of the amount of production
3. The actual Factory overhead cost is unknown until the end of a fiscal period.

Given these and other related problems, the only acceptable way to assign Factory overhead cost to a particular job is to do so through an allocation process. The allocation process is done by computing an application rate. This application rate is termed as predetermined factory overhead rate. This rate is based on estimates of overhead cost and application base. The application base is any activity base which is common to all jobs or products that the company manufactures, or common to all services that the company renders. In selecting the application base care should be taken because product costs can be easily distorted if inappropriate base is selected. The best application base is the one that is a cost driver. A cost driver is a measure of an activity such as machine hours operated, number of beds occupied in a hospital, number of hours flown by an airplane, number of miles driven by an automobile, direct labor hours worked, direct labor cost incurred etc. which is a causal factor for the incurrence of factory overhead costs. If a company made only one product, it could allocate overhead cost to units of products simply by dividing total overhead cost by the number of units produced and allocating that per unit amount to each unit. However, such a calculation is inappropriate for a job-order costing because units made are not uniform. Therefore, application of overhead is inescapable.

3.3.3.1 Applied factory overhead

Factory overhead is entered on the job-order cost sheets on the basis of predetermined factory overhead rate based on an appropriate allocation base such as direct labor hours, direct labor cost and etc. In principle, the accountant determines a causal relationship between two factors and uses this relationship as the means of charging factory overhead to jobs. The predetermined overhead rate is computed by dividing the estimated units of selected cost driver into the estimated factory overhead cost for the period.

$\text{Pre-determined Overhead rate (POR)} = \frac{\text{Estimated factory overhead costs}}{\text{Estimated units of cost driver}}$

For example: Suppose that XYZ company's direct labor hours for the month were estimated to be 7,000 hours and factory overhead was estimated to be \$15,400. Therefore, pre-determined overhead rate will be:

$$\text{POR} = \frac{\$15,400}{7,000} = \$2.20 \text{ per direct labor hour worked}$$

7,000 labor hours

This estimate leads to the assumption that for each hour of direct labor worked there is \$ 2.20 of factory overhead to be allocated.

To determine the predetermined overhead rate the basic steps to be followed are:

1. Select an appropriate activity base (cost driver).
 - ❖ Here the activity base selected is direct labor hours
2. Estimate (budget) the total factory overhead costs for the period.
 - ❖ The factory overhead cost for the period is estimated to be \$15,400
3. Estimate (budget) the total units of activity base (cost driver) for the period.
 - ❖ The direct labor hours to be worked for the period are estimated to be 7000 hours.
4. Divide the budgeted factory overhead costs by the budgeted total units in the activity base (cost driver) i.e., $\$15,400 / 7000 \text{ hours} = \$ \underline{\underline{2.20}}$

Transaction 7:

After the predetermined rate is computed, the factory overhead cost is applied to work in process based on the actual units of the activity base (application base) i.e. the rate multiplied by the actual units of application base of the period. Now, XYZ Co. can use the POR to apply or allocate the factory overhead to specific jobs. It would do this by multiplying the POR by the number of direct labor hours actually operated on the jobs. Suppose XYZ Co. has 6000 direct labor hours actually worked during a period, therefore, the applied factory overhead for the period would be \$ 13,200 ($\$2.20 \times 6000 \text{ hours}$).

The journal entry is, therefore, to debit the working process inventory account and credit the applied factory overhead as shown below:

Work-in-process	13,200
Applied factory overhead	13,200

It is common practice to use an applied factory overhead account because it keeps applied costs and actual costs in separate accounts. Some companies do not use the applied factory overhead account and post the credit directly to factory overhead control by the following journal entry:

Work-in process

Factory overhead control

This entry eliminates the transfer of applied expenses to actual expenses. The applied factory overhead account is closed to the actual factory overhead control account at the end of the accounting period by entry shown below:

Applied factory overhead	13,200
Factory overhead control	13,200

After the applied factory overhead account is closed to the actual factory overhead account at the end of the operation period, the balance in the applied factory overhead account will be zero.

3.3.3.2 Actual factory over head

While job order cost sheets receive factory overhead on the basis of a predetermined factory overhead rate, actual factory overhead (consisting of indirect materials, indirect labor, payroll taxes, invoices received for overhead items, and monthly adjusting entries for expired costs such as depreciation and insurance) is charged to factory overhead control.

Assume depreciation for the period is \$682 and an insurance amounting \$516 is expired during the period, therefore, the required journal entry to be maintained by XYZ Co. would be:

Factory overhead control	1,198
Accumulated depreciation	682
Prepaid insurance	516

For XYZ Co. \$ 4000 of indirect materials was requisitioned from storeroom, \$ 4,000 of indirect labor was used, and payroll taxes on factory labor totaled \$ 3,069. Other factory expenses comprised of depreciation \$ 682 and insurance \$ 516 were expired during the period. The factory overhead control account would reflect these facts as shown below:

Factory overhead control	
Indirect materials \$ 4000	Overhead applied
54	

Indirect labor	4000	to Work in process \$ 13, 200
Payroll-taxes.....	3, 069	
Depreciation - Machinery	682	
Insurance expires.....	516	

Balance \$ 933

The balance of \$ 933 in the factory overhead control account indicates that overhead applied to jobs exceeded the actual expenses; stated differently, overhead was over applied.

Transaction 8:

Completion and transfer of jobs to finished goods storeroom.

After charging all the three manufacturing cost elements and the job is completed, the cost should be transferred to the finished goods inventory account. When finished goods are transferred from factory floor to the finished goods storeroom, the asset finished goods control is increased and the asset work-in –process control is decreased. The total cost of each job is calculated in the subsidiary ledger as each job is completed. During the period XYZ Co. has completed four individual jobs with \$ 95,000. The appropriate journal entry for this event would be:

Finished Goods	95,000
Work – in – process	95, 000

In job- order costing system, the finished goods inventory ledger is a controlling account for all jobs. For each job separate card should be prepared and it serves as a subsidiary ledger.

Transaction 9:

To complete the cost accounting cycle, two journal entries are necessary. These are:

1. To record the sales of goods (delivery of completed jobs to customers) and
2. To record the cost of producing the goods (jobs) which are delivered to customers.

By the end of the period, units costing \$ 85,000 was shipped to customers at total sales volume of \$ 110,000 on account.

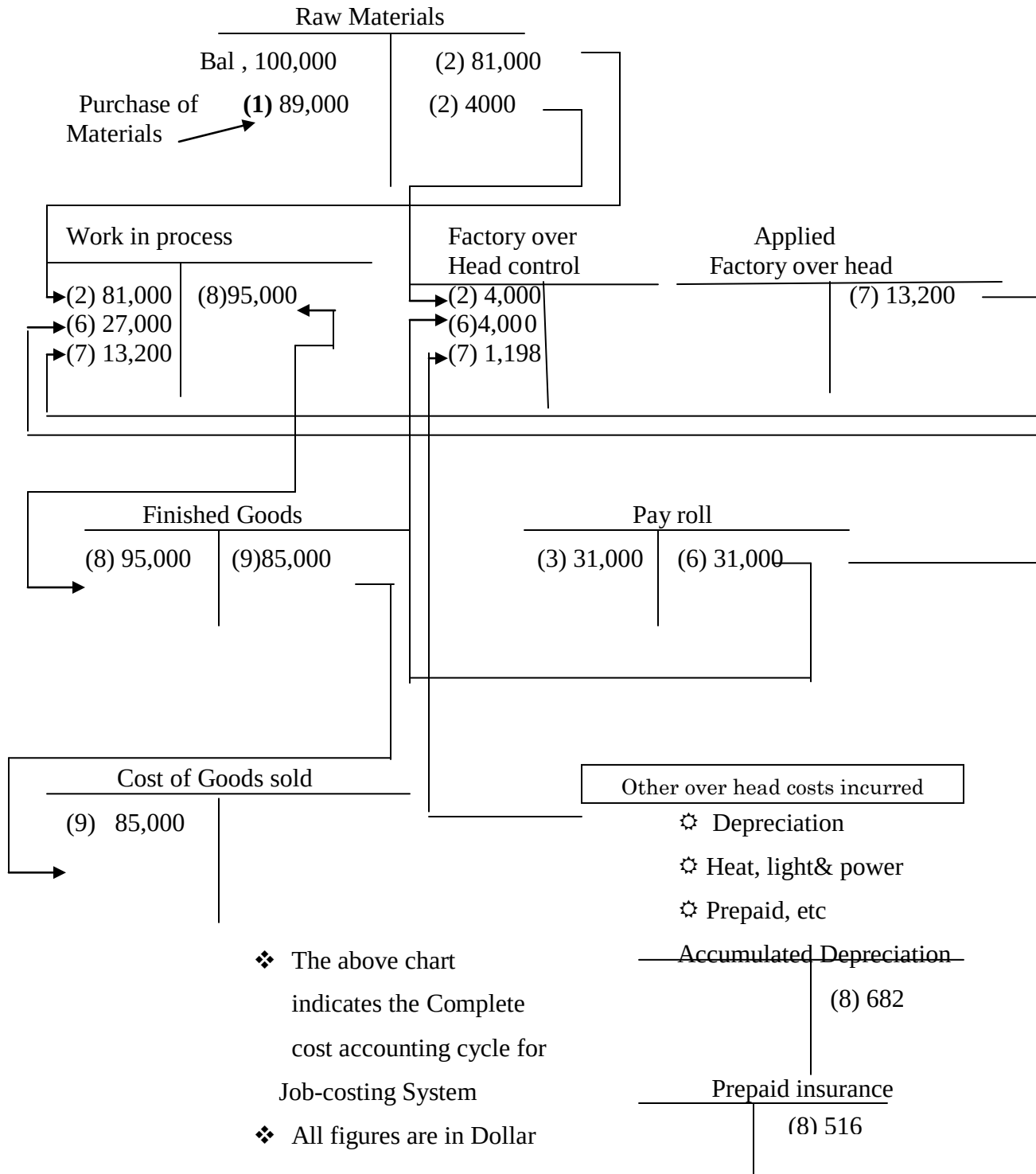
The appropriate journal entries are:

Accounts receivable	110,000
---------------------	---------

Sales 110, 000
To record sales of completed goods to customers

Cost of goods sold 85, 000
 Finished goods 85,000
To record cost of goods sold to customs

Flow of Manufacturing Costs in Job – order costing system



Activity 3.4

- Explain the benefits of using a predetermined overhead rate instead of the actual overhead rate.
- Describe the role of a manufacturing overhead allocation base in job costing.
- Give two reasons for under or over application of indirect costs at the end of an accounting period.
- Describe the cost flow in a job order costing system for a manufacturing Company.

3.3.4 Problems of over- head Application and Period –end adjustments**3.3.4.1 Problems of overhead Application**

Budgeted (predetermined) indirect cost rates have the advantage of being **timelier** than actual indirect cost rates. With predetermined rates, indirect cost can be assigned to individual jobs on an ongoing basis rather than waiting until the end of the accounting period when actual cost will be known. However, the disadvantage of predetermined rates is that they will probably be inaccurate, having been made up to 12 months before actual costs are incurred. Besides, the following may contribute immensely for the difference between the actual rate and the predetermined overhead rate:

- 1) The use of single or dual application rate, and
- 2) The use of plant –wide or departmental overhead application rate.

1) The use of single or dual application rate

It is obvious that manufacturing overhead costs can be grouped in to variable and fixed cost depending on the behavior of costs. The presence of fixed manufacturing overhead costs is a major reason for difficulties in overhead application. So as to minimize the difficulties faced and for other purposes, firms may develop different application rates to apply manufacturing overhead costs to production. Moreover, the cost driver may differ for variable and fixed manufacturing overhead cost. The basic procedure will remain the same, except that the individual overhead cost would be gathered into one cost pool or the other account depending on its behavioral pattern. The two different application rates

would be used to put overhead cost into work in process inventory. These are the variable overhead application rate and the fixed overhead application rate.

2. The use of plant wide or departmental overhead application base

A firm may have many departments. The question of overhead application is whether to use single, plant wide application rate or to use overhead application rate computed (developed) for each department involved in the production process. Sometimes the assumption that a single application rate can serve may not warrant. In such cases firms will use different overhead application rates for each department instead of a single, plant wide rate.

A departmental overhead application rate is used to apply overhead costs of a single department. Each department will have its own rate, and might use different application base. The basic procedures of applying overhead costs remain the same as in single rate method except that the relevant cost drivers selected in each department governs the amount of overhead applied. Besides, over applied or under applied overhead cost is completed for individual department. There are two conditions which may encourage firms to use departmental overhead application rates. These are:

1. Different departments' Overhead costs may be associated with different measure of activity cost driver. Some departments may be capital intensive and the cost driver tends to be total machine hours operated, others may be, labor intensive and hence the overhead cost driver tends to be direct labor hour worked or direct labor cost. In such situations using single rate developed on single measure of activity will not suffice.
2. Some departments may have different rate of usage as compared to other with regard to overhead items. The single rate method obse cures this important difference.

Illustration:

<u>Department</u>	<u>Budgeted over head (a)</u>	<u>Budgeted labor house(b)</u>	<u>Overhead rate (a/b)</u>
Assembly	\$ 240,000	50,000	$\$240,000 \div 50,000\text{hrs} = \$4.80 / \text{DLH}$
Finishing	<u>120,000</u>	<u>50,000</u>	$\$120,00 \div 50,000 \text{ hrs} = \$2.40 / \text{DLH}$

Total \$360,000 100,000 hrs \$360,000 /100,000hrs =3.60/ DLH

DLH: Direct labor hours.

Assume that job 101 and job 102 were worked in both assembly and finishing departments. The following data are on hours used by each job in each department:

<u>Job</u>	<u>Assembly hours</u>	<u>Finishing hours</u>	<u>Total hours</u>
101	300	50	350
102	150	200	350

Solution:

Overhead applied to each job would be:-

➤ **Using single rate method:**

Job 101: \$3.60 x350 hrs = \$1,260

Job 102: \$3.60 x 350 hrs = \$1,260

As illustrated above, jobs with differing times in different departments but same total time have the same overhead charge when a single rate is used.

➤ **Using departmental rate :**

Job101: From Assembly department, \$ 4.80x300 hrs =\$1.440

From finishing department, \$2.40 x 50 hrs = 120

Total cost allocated to job101= \$ 1,560

Job 102: From Assembly department t = \$4.80x150 hrs = \$ 720

From finishing department = \$ 2.40 x 200 hrs = 480

Total cost allocated to Job102 = \$ 1,200

The central point of the above example is that misleading results might follow the use of a single plant wide application rate.

3.3.4.2 The Disposition of under applied or over applied Factory overhead (or period end adjustments)

When factory overhead cost incurred exceeds the factory overhead cost applied, the overhead is said to be **under applied**. Conversely, when the manufacturing (factory) overhead cost applied exceeds the factory overhead cost incurred, the overhead is said to be **over applied**. It is rare that the overhead applied will equal the actual overhead in any

given period. Total applied overhead is the predetermined overhead rate multiplied by actual activity (cost driver). This formula gives us some insight. Thus, if the total budgeted activity equals total actual activity and total budgeted overhead equals total actual overheads, there will be no over applied or under applied amount. However, it is not likely that people making the predictions will hundred (100%) percent perfect. Therefore, it is necessary to make adjustments for under applied or over applied factory overhead at the end of the operating period.

In short:

- ✚ Under applied indirect costs occur when the applied amount of indirect costs in an accounting period is less than the actual (incurred) amount in the period.
- ✚ Over applied indirect costs occur when the applied amount of indirect cost in an accounting period exceeds the actual (incurred) amount in that period.

Under or Over applied = Actual factory Factory overhead overhead incurred	–	Applied factory overhead
---	---	---

Assume the following actual data for XYZ Company in 2005:

Factory overhead control	Applied factory overhead
Balance Dec.31, 2005 \$ 330,000	Balance Dec.31, 2005 \$ 250,000

Based on the above assumption, the XYZ Company's the under or over applied Factory overhead will be computed as follows:

$$\begin{aligned}
 &\text{Under or over applied} = \text{Actual Factory} - \text{Applied Factory} \\
 &\text{Factory overhead} \quad \quad \quad \text{overhead} \quad \quad \quad \text{overhead} \\
 &\Rightarrow \$ 330,000 - \$ 250,000 = \underline{\underline{80,000}} \text{ under applied}
 \end{aligned}$$

Actual Factory overhead: The sum of the actual costs in all the individual Overhead categories

(Such as indirect materials, indirect labor, power and rent) for
the
whole accounting period.

Applied Factory overhead: Is the record of the factory overhead applied to individual jobs on

the basis of the budgeted rate multiplied by the actual direct labor hours

There are two main approaches to disposing of this \$ 80,000 under applied of factory overhead in XYZ's costing system.

1. Adjusted allocation rate
2. Proration approach.

1. Adjusted Allocation rate approach

The adjusted allocation approach, in effect, restates all entries in the general ledger by using actual cost rate rather than budgeted cost rates. First, the actual indirect cost rate is calculated at the end of each period. Then, every job to which indirect costs were allocated (applied) during the period has its amount recalculated using the actual indirect cost rate (rather than the budgeted indirect cost rate). Finally, end - of - period closing entries are made. The result is that every single job cost record as well as the ending stock and cost of goods sold accounts accurately represents actual indirect cost incurred. This approach increases the Accuracy of each individual product cost amount and the accuracy of the end-of-year account balances for stocks and cost of goods sold. This increase in accuracy is an important benefit. After the fact analysis of individual product profitability can provide managers with useful insights for future decisions about product pricing and about which products to emphasize.

2. Proration Approach

Proration is the term we use to refer to spreading of under or over applied overhead among ending stocks (inventory) and cost of goods sold. As we have seen before XYZ Company's factory overhead is applied on the basis of direct labor hours. Materials are not applied any overhead costs. It is not until materials are put into work in process that production commences. Only the ending work in process and finished goods stocks will have an applied factory overhead component. Thus, it is only two ending stock amounts and cost of goods sold for which end of- period proration is an issue. Assume the following actual results for XYZ Company in 2005:

	End of year balances (Before proration)	Factory overhead Applied component of Year end balances (Before proration)
Work in process	\$12,500	\$3,250
Finished Goods	18,750	6,250
Cost of goods sold	<u>593,750</u>	<u>240,500</u>
	<u>\$ 625,000</u>	<u>\$ 250,000</u>

There are three methods for prorating the under applied of \$ 80,000 factory overhead at the end of 2005.

Method 1

This proration approach is based on the total amount of indirect cost applied in the ending balances of work in process, Finished Goods and cost of goods sold. The process of prorating is shown below.

Account balance	Account balance	Indirect cost allocated Component	Percentage	Proration	Account (after proration)
(1)	(2)	(3)	(4)	(5)	(6) =
(2) + (5)					
	Work in Process	\$ 12,500	\$ 3,250	1.3% = 0.013	
	\$80,000x0.013=1,040	13,540			
	Finished Goods	18,750	6,250	2.5%= 0.025	80,000x0.025= 2000
	20,750				
	Cost of Goods Sold	<u>593,750</u>	<u>240,500</u>	<u>96.2%= 0.962</u>	<u>80,000x0.962 =76,960</u>
	<u>670,710</u>				
	<u>\$ 625,000</u>	<u>\$250,000</u>	<u>(100%)</u>	<u>1.00</u>	<u>\$80,000</u>
	<u>705,000</u>				<u>\$</u>

Percentage Computation

\$ 3,250 /\$ 250,000	= 1.3%
\$6,250/\$250.000	=2.5%
\$ 240, 500/\$250,000	= 96.20%

If XYZ Company had chosen to prorate under applied overhead using method 1, the following journal entry would have been made:

Work in process	1,040
Finished Goods	2000
Cost of goods Sold	76,960
Applied Factory overhead	250,000
Factory overhead Control	330,000

This journal entry results in the 2005 ending balances for work in process, Finished Goods and cost of Goods sold being restated to what they would have been had actual cost rates rather than budgeted cost rates been used.

Method 2

Proration is based on total ending balances (before proration) in work in process, Finished Goods and cost of Goods sold. In our XYZ company example, the \$ 80,000 under applied overhead is prorated over the three pertinent accounts in proportion to their ending balances (before proration) in column 2 in the following table, resulting in the ending balances (after proration) in column 5.

Accounts balance (Before proration)	Proration of \$ 80,000 FOH	Account
balance	Under applied	(after
Peroration)		
(1)	(2)	(3)
(2) + (4)	(4)	(5) =
Working in process	\$12,500	2% = 0.02
\$14,100	\$80,000x0.02=	\$1,600
Finished Goods	18,750	3% = 0.03
21,150	80,000x0.03 =	2,400
Cost of Goods Sold	<u>593,750</u>	<u>95% = 0.95</u>
<u>669,750</u>	80,000x0.95 =	<u>76,000</u>
	<u>\$ 625,000</u>	<u>100% 1.00</u>
		<u>\$ 80,000</u>
<u>\$705,000</u>		

Percentage Calculation:

$$\text{\$ } 12,500 / \text{\$ } 650,000 \times 100 = 2\%$$

$$\text{\$ } 18,750/\text{\$ } 650,000 \times 100 = 3\%$$

$$\text{\$ } 593,750/\text{\$ } 650,000 \times 1000 = 95\%$$

The journal entry for this proration would be:

Work in process	1,600
Finished Goods	2,400
Cost of Goods Sold	76,000
Applied Factory overhead	250,000
Factory overhead control	330,000

NB. Note that if factory overhead had been over applied, the work in process, Finished Goods and cost of Goods sold Accounts would be decreased (credited) instead of increased (debited).

Method 3

Proration is based on year end write - off to cost of Goods sold. Here the total under applied overhead is included in this year's (2005) cost of Goods sold as shown below:

Cost of Goods Sold.	80,000
Applied Factory overhead	250,000
Factory overhead Control	330,000

XYZ's two factory overhead accounts (actual factory overhead and Applied factory overhead) are closed out with all the difference between them now included in the cost of Goods sold after proration is \$593,750 before portion + \$ 80,000 under applied overhead amount = \$673,750.

Choice among Approaches

Choice among the approaches should be guided by how the resultant information to be used. If managers wish to develop the most accurate record of individual job costs for profitability analysis propose, the adjusted allocation rate approach is preferred. The proration approach does not make any adjustment to individual job cost records.

If the Purpose is confined to reporting the most accurate stock and cost of Goods sold figures, either the adjusted allocation rate or the method1 proration should be used. Both give the ending balances of work in process, Finished Goods and cost of Goods Sold that would have been reported had an actual indirect cost rate been used. Method 2 is frequently justified as being a lower cost way of approximating the results from method

1. The implicit assumption in method 2 is that the ratio of factory overhead costs applied to total manufacturing costs is similar in work in process, Finished Goods and cost of Goods sold. Many Companies use method 3 for its simplicity.

Activity 3.5

- Distinguish between single rate and dual rate and plant wide or departmental rate overhead application rate.
- Describe the alternative ways to prorate end -of -period adjustments for under or over applied factory overhead.

Summary

Product costing is the process of accumulating costs of a production process and assigning them to the firm's products. Product costs are needed for three major purposes: (1) to value inventory and cost of goods sold in financial accounting; (2) to provide managerial accounting information to managers for planning, cost control and decision making; and (3) to provide cost data to various organizations outside the firm, such as governmental agencies or insurance companies. Information about the costs of producing goods and services is needed in manufacturing companies, service industry firms, and nonprofit organizations.

Two types of product-costing systems are used depending on the type of product manufactured. . Process costing works well whenever, relatively homogeneous products pass through a series of processes and receive similar amounts of manufacturing costs. A process costing system traces the costs to the processes. Job-order costing, the topic of this chapter is used by firms that engage in either job-shop or batch production operations. Such firms produce relatively small numbers of dissimilar products, such as feature films and custom furniture. Job cost system is used when many different products are produced each period, products are manufactured to order, the unique nature of each

order requires tracing or allocating costs to each job, and maintaining cost records for each job.

In job-order costing system the costs of direct material, direct labor, and factory (manufacturing) overhead are first entered into the work-in process inventory account. When goods are completed, the accumulated manufacturing costs are transferred from work-in-process inventory to Finished-goods inventory. Finally, these product costs are transferred from Finished-goods inventory to Cost of goods sold when sales occur. Direct material and direct labor are traced easily to specific batches of production called job orders. In contrast, factory overhead is an indirect cost with respect to job orders or units of product. Therefore, overhead is applied to production jobs using a predetermined overhead rate, which is based on estimates of factory overhead and the level of some cost driver (or activity base). The most commonly used volume based cost drivers are direct labor hours, direct-labor cost and machine hours. Since these estimates will seldom be completely accurate, the amount of overhead applied during an accounting period to work-in-process inventory will usually differ from the actual costs incurred for overhead items. The difference between actual overhead and applied overhead called over applied or under applied overhead may be closed out into cost of goods sold or prorated among work-in- process Inventory, Finished –goods inventory, and Cost of Goods sold.

The accuracy of product costs often can be increased by the use of departmental overhead rates instead of a single plant wide overhead rate.

Job-order costing methods also are used in a variety of service industry firms and nonprofit organizations. Accumulating costs of projects, contracts, cases, programs, or missions provides important information to managers in such organizations as hospitals, law firms, and government agencies.

Self-Check Exercise

Part I: Say “True” if the statement is correct and “False” if the statement is incorrect.

Enter a T or an F in the blank to indicate whether the statement is true or false.

- ___ 1. A company producing many different kinds of furniture would probably use a job-order cost system.
- ___ 2. The predetermined overhead rate is computed using estimates of overhead cost and the amount of the allocation base.
- ___ 3. Actual manufacturing overhead costs are charged directly to the Work in Process account as the costs are incurred.
- ___ 4. Selling and administrative expenses are charged to the Manufacturing Overhead account.
- ___ 5. If more overhead is applied to Work in Process than is actually incurred, then overhead cost will be over applied.
- ___ 6. A debit balance in the Manufacturing Overhead account at the end of a period would mean that overhead was under applied for the period.
- ___ 7. Any balance in the Work in Process account at the end of a period should be closed to Cost of Goods Sold.

Part II: Multiple Choices

Choose and Circle the best answer

- 1. Original records that support accounting transactions and journal entries are called
 - a. Cost Pools
 - b. Job Sheets
 - c. Source Documents
 - d. Cost Drivers

2. An organization would record which of the following to account for over applied manufacturing overhead?
 - a. A debit to Cost of Goods Sold
 - b. A debit to Work-in-Process
 - c. A credit to Revenue
 - d. A credit to Cost of Goods Sold
3. The following statement is TRUE about Managerial and Cost Accounting:
 - a. Managerial Accounting data is subject to Generally Accepted Accounting Principles.
 - b. Managerial Accounting data is often distributed to company creditors and other investors.
 - c. Managerial Accounting data only focuses on financial accounting data that can be objectively measured and audited.
 - d. Managerial Accounting data may be focused on future activities of the firm, as well as statistical forecasts.
4. In a job-order costing system, the basic document for accumulating costs for a specific job is:
 - a. the materials requisition form;
 - b. the job cost sheet;
 - c. the Work in Process inventory account;
 - d. the labor time ticket.
5. Suppose \$20,000 of raw materials are withdrawn from the storeroom to be used in production. Of this amount, \$15,000 consists of direct materials and \$5,000 consists of indirect materials. What account or accounts will be debited?
 - a. Work in Process \$15,000 and Raw Materials \$5,000;
 - b. Raw Materials \$15,000 and Manufacturing Overhead \$5,000;
 - c. Manufacturing Overhead \$15,000 and Work in Process \$5,000;
 - d. Work in Process \$15,000 and Manufacturing Overhead \$5,000
6. A typical job-cost record would provide information about all of the following items related to an order except:

- a. the cost of direct materials used.
 - b. administrative costs.
 - c. direct labor costs incurred.
 - d. applied manufacturing overhead.
 - e. direct labor hours worked.
7. Last year, a company reported estimated overhead, \$100,000; actual overhead, \$90,000; and applied overhead, \$92,000. The company's overhead cost for the year would be:
- a. under applied, \$10,000;
 - b. under applied, \$8,000;
 - c. over applied, \$2,000;
 - d. over applied, \$10,000.
8. On January 1, Habib Company's Work in Process account had a balance of \$18,000. During the year, direct materials costing \$35,000 were placed into production. Direct labor cost for the year was \$60,000. The predetermined overhead rate for the year was set at 150% of direct labor cost. Actual overhead costs for the year totaled \$92,000. Jobs costing \$190,000 to manufacture according to their job cost sheets were completed during the year. On December 31, the balance in the Work in Process inventory account would be:
- a. \$13,000;
 - b. \$18,000;
 - c. \$15,000;
 - d. \$8,000.
9. If overhead is over applied for a period, it means that:
- a. the predetermined overhead rate used to apply overhead cost to Work in Process was too low;
 - b. the company incurred more overhead cost than it charged to Work in Process;
 - c. too much cost has been assigned to jobs;
 - d. none of these.

10. ABC Company had actual overhead cost of Birr 1,210,000 and a predetermined overhead rate of Birr 4.00 per machine hour. ABC worked 290,000 machine hours during the period. At the end of the period, ABC's overhead was:
- Birr 50,000 over applied
 - Birr 50,000 under applied
 - Birr 60,000 over applied
 - Birr 60,000 under applied
11. Assume that, at the end of the year, the balance in the factory overhead account represents over applied overhead for the year. When this account balance is closed to the cost of goods sold account, the Company's net income will:
- Increase
 - Be unaffected
 - Decrease
11. Which of the following equations is used to calculate the cost of goods sold during the period?
- Beginning finished goods + cost of goods manufactured + ending finished goods.
 - Beginning finished goods - ending finished goods.
 - Beginning finished goods + cost of goods manufactured.
 - Beginning finished goods + cost of goods manufactured - ending finished goods.
 - Beginning finished goods + ending finished goods - cost of goods manufactured.
12. Cox Industries began June with a finished-goods inventory of \$13,000. The finished-goods inventory at the end of June was \$10,000 and the cost of goods sold during the month was \$20,000. The cost of goods manufactured during June was:
- \$7,000.
 - \$17,000.
 - \$20,000.
 - \$27,000.
 - some other amount.
13. A custom-home builder would likely utilize:
- job-order costing.
 - process costing.
 - mass customization.
 - process budgeting.

- e. joint costing.
14. A manufacturing firm produces goods in accordance with customer specifications, commencing production upon receipt of a purchase order. To accumulate the cost of each order, the company would use a:
- job-cost record.
 - cost allocation document.
 - production log.
 - overhead sheet.
 - manufacturing cost record.
15. Which of the following is the correct method to calculate a predetermined overhead rate?
- Budgeted total manufacturing cost $\div$ budgeted amount of cost driver.
 - Budgeted overhead cost $\div$ budgeted amount of cost driver.
 - Budgeted amount of cost driver $\div$ budgeted overhead cost.
 - Actual overhead cost $\div$ budgeted amount of cost driver.
 - Actual overhead cost $\div$ actual amount of cost driver.

Part III: Work out

1. TATA Company has established the following budget for 1999:

	<u>Assembling</u>	<u>Welding</u>
Factory overhead	\$175,000	\$225,000
Direct labor cost	\$300,000	\$375,000
Direct labor hours	25,000	37,500
Machine hours	13,000	37,500

TATA Company uses predetermined rates to apply factory overhead. These rates are based on Machine hours in Assembling and on the direct labor cost in Welding.

Required: Compute the predetermined factory overhead rate for each department

2. During May, the job cost sheet for Job no. 55 showed the following:

	<u>Assembling</u>	<u>Welding</u>
Direct material cost	\$1,000	\$4,000
Direct labor cost	\$3,000	\$1,250
Direct labor hours	250	125

Machine hours

120

75

Required: Using the overhead rates computed in Q1; compute the total cost of Job No.55

3. The following selected account balances are taken from the books of Paradise Company as of January 1 of the most recent year:

<i>Cash</i>	<i>Work in Process</i>	<i>Accounts Payable</i>	<i>Sales</i>
12,000	40,000	75,000	
<i>Accounts Receivable</i>	<i>Finished Goods</i>	<i>Salaries and Wages Payable</i>	<i>Cost of Goods Sold</i>
48,000	100,000	12,000	
<i>Prepaid Insurance</i>	<i>Accumulated Depreciation</i>		
8,000	120,000		
<i>Raw Materials</i>	<i>Manufacturing Overhead</i>		
30,000			

The following data relate to the activities of Paradise Company during the year:

1. Raw materials purchased on account, \$150,000.
2. Raw materials issued to production, \$145,000 (all direct materials).
3. Advertising cost incurred for the year, \$50,000 (credit accounts payable).
4. Utilities cost incurred for the factory, \$35,000 (credit accounts payable).

5. Salaries and wages costs incurred: direct labor, \$250,000 (30,000 hours); indirect labor, \$75,000; selling and administrative, \$140,000.
6. Depreciation recorded for the year, \$20,000, of which 75% related to the factory and 25% related to selling and administrative functions.
7. Other factory overhead costs incurred for the year, \$30,000 (credit accounts payable).
8. Other selling and administrative expenses incurred for the year, \$25,000 (credit accounts payable).
9. Prepaid insurance of \$4,000 expired during the year; all of this is related to the factory.
10. The company applies overhead on the basis of direct labor-hours at \$5.50 per hour.
11. The cost of goods manufactured for the year totaled \$550,000.
12. Goods that cost \$540,000 according to their job cost sheets were sold on account for \$800,000.
13. Collections on account from customers during the year totaled \$790,000.
14. Cash disbursed during the year: on accounts payable, \$300,000; for salaries and wages, \$460,000.

Required:

- a. Post the above entries directly to Paradise Company's T-accounts on the previous page. Key your entries with the numbers 1-14.
 - b. Compute the ending balance in each T-account.
 - c. Is overhead under applied or over applied for the year? Close the balance to Cost of Goods Sold. (Key the entry as #15.)
 - d. Prepare an income statement for the year
4. The Comfort furniture company uses Job order cost system to Manufacture home and office furniture. The operation period of the company is from Harmed 1 to sene 30. As of Hamle 1, 1999 E.C, its records showed:

Materials	Birr 140,000
Work in process	270,000

The Work in process inventory consisted Job All:

Direct Materials	Birr 129,600
Direct labor	91,800
Applied Factory over head	<u>48,600</u>
Total	<u>Birr 270,000</u>

The company Applies overhead to jobs using a predetermined overhead rate based on machine hours. At the beginning of the year the company estimated that it would work 33,000 machine hours, and incurs Birr 330,000 in factory overhead cost. During the current fiscal period the company had started job A 12.

The following transactions were recorded for the year:

1. Materials were purchased during the fiscal year at a total cost of birr 295,000 On account.
2. During the year, the cost of materials requisitioned for production on the two jobs amounted birr 290,000, of this amount the direct materials costs traceable to Job All and Job A12 are Birr 95,000 and Birr 160,000 respectively. The remainder is cost of indirect materials used up during the year on both jobs.
3. Total wages and salaries of manufacturing employees and administrative salaries for the year amounted Birr 284,000 and Birr 155,000. From the total administrative salaries, 60% is marketing expense and 40% is administrative personnel salary expense. The wages and salaries traceable to Job All and Job A12 are Birr 94,000 and Birr 145,000. The remainder is considered as indirect costs. Assume the company with holds 12% for income tax and 4% for pension contribution.
4. The accrued payroll, pension contribution and income tax are paid to the employees and concerned parties
5. All the employee costs or payroll costs are distributed to the respective accounts.
6. The following factory overhead costs were incurred during the year.

Utilities (light, power, telephone)	Birr 50,000
Rent on Factory building	120,000
Property taxes	45,000
Insurance (expired)	<u>7,500</u>
Total	<u>Birr 222,500</u>

7. Depreciation for the year was Birr 127,000 of which Birr 80,000 is related to factory operation and birr 47,000 is related to office equipment.
8. Factory overhead was applied to job. The actual level of activity for the year on both jobs was 34,000 machine hours (11,900 machine hours on Job All and 22,100 machine hours on Job A12).
9. During the year Job All was completed and Job A12 was still in process.
10. $\frac{3}{4}$ of Job All was shipped to customers at 155% of cost on account

Required:

- a) Determine the Predetermined Factory overhead rate.
- b) Record all the summarized transactions in a general journal form.
- c) Post to T- accounts and determine the end balances of the work in process inventory and finished goods inventory.
- d) Prorate the over or under applied factory overhead among Finished goods, cost of goods sold, and works in process in proportion of their account balances and pass the appropriate journal entry.
- e) Prepare statement of cost of goods manufactured and cost of goods sold.
- f) Prepare income statement if all under or over applied factory overhead is directly closed to cost of goods sold

CHAPTER FOUR

Process costing

Chapter learning objectives

After studying this chapter, you should be able to:

- ❖ List and explain the similarities and important differences between job-order and process costing.
- ❖ Describe when process-costing system is appropriate.
- ❖ Define equivalent units and explain their role in process costing.
- ❖ Describe the concept of timing of lost units and their accounting treatment.
- ❖ Explain and prepare the cost of production report
- ❖ Distinguish the accounting procedures of lost units in the originating department and in subsequent department.
- ❖ Describe the possible effects of additions of materials in subsequent departments.

4.1 Introduction

Cost accounting system should be designed to fit the nature of manufacturing operation. The job-order costing system is designed to trace costs to a job, while a process costing system traces the costs to the processes, instead of the job. A process costing system is used where there is a mass production of like units which usually pass in a continuous fashion through a series of uniform production stages called process or operations. In process costing system, it is necessary to establish specific process costing centers which are usually called processing departments. The basic approach to process costing is to accumulate costs in a particular operation or processing department for the entire period (a month, a quarter, a year) and then divide the total accumulated cost by the total number of units completed (produced) during the Period to determine the unit cost of the product. This unit cost is an average unit cost for all units produced since all units are the same and indistinguishable. Firms which employ a process costing system is petroleum refineries, textile factories, flourmills, paint factories and etc.

The main focus of this chapter is the manner in which costs flow through the perpetual inventory accounts in process costing system. The chapter includes 2 sections.

The first section introduces the concepts, characteristics and procedures associated with process cost accumulation method and the cost of production report for the originating (first) department. Section two emphasizes on the issues of cost of production report for the second and third departments, additions of materials to subsequent departments, and timing of lost units.

Section 1

4.2 Nature (Characteristics) and procedures of process costing

The characteristics of process costing are:

- 1) A cost of production report is used to collect, summarize, and compute total and unit costs.
- 2) Production is accumulated and reported by departments or processes.
- 3) Costs are posted to departmental work in process accounts.
- 4) Production in process at the end of a period is restated in terms of completed units.
- 5) Total cost charged by a department is divided by total production units of the department to determine the unit cost for a specific period.
- 6) Costs of completed units of a department are transferred to the next processing department in order to arrive at the total costs of the finished products during a period.

Procedures of process costing

The procedures of process costing are designed to:

- 1) Accumulate materials, labor, and factory overhead costs by departments.
- 2) Determine a unit cost for each department.
- 3) Transfer costs from one department to the next and to finished goods.
- 4) Assign costs to the inventory of work still in process

Costing by departments

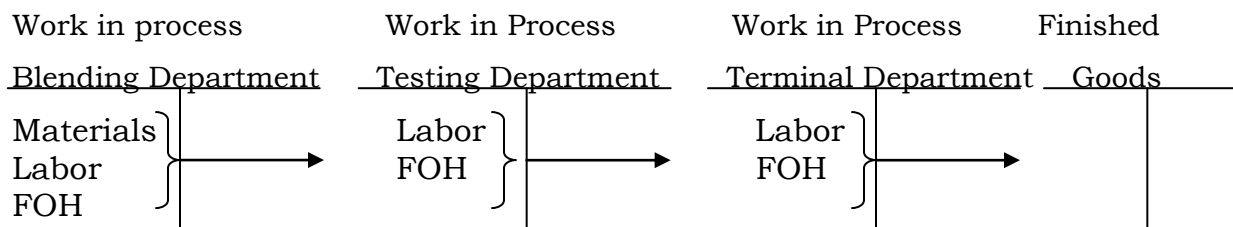
The nature of manufacturing operations in firms using process or Job-order cost procedures usually takes production tasks from different departments. For that matter, whatever the cost procedure is applied, each department is responsible to perform a specific operation or process towards the completion of production. In processing, each department may consume (charge) a certain amount of materials, labor and factory overhead for the accomplishment of its tasks. For example, have a look at the diagram below (sequential product flow), it is clearly depicted that just after a Blending department has completed the starting phase of work on the product, units are transferred to Testing department after which they may go to Terminal department for completion and transfer to finished goods store room. When units are transferred from one department to the other the associated cost incurred in each department will also be transferred along with the physical flow of products. Finally, the cost of completed units is determined by dividing the total cost of a period by the total units produced during the same period. Departmental total and unit costs are determined by the use of the cost of production report.

4.2.1 Product Flow Methods

A product can flow through a factory in many ways. Three product flow formats associated with process costing; sequential, parallel, and selective are discussed below:

a) Sequential product flow

Sequential processing is a method of processing manufacturing in which units pass from one process to another in a sequential pattern and each unit is processed using the same series of steps as illustrated below:



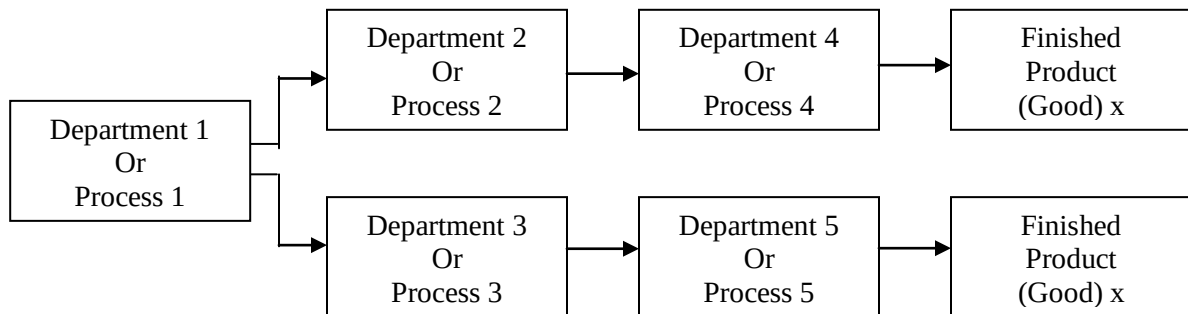
FOH= Factory overhead

Materials are placed into production in the Blending department, and labor and factory overhead are added. When work is finished in Blending department, it moves to the Testing department. The second process and any succeeding processes may add more materials or simply work on the partially completed input from the preceding processes (department), adding only labor and factory overhead, as in this example. After the product has been processed by the Terminal department, it is a completed product and becomes a part of finished goods inventory.

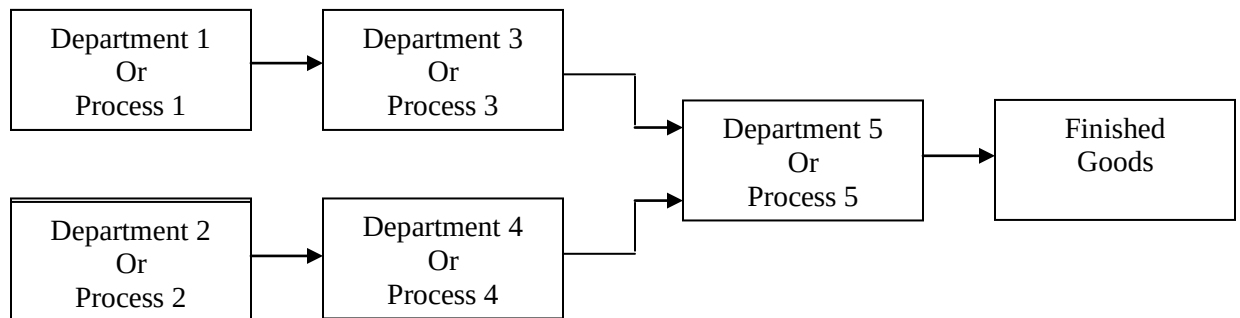
b) Parallel Product Flow

Two examples of parallel processing (product flow) are:

- i. Two different products emerge from a common process, and then each flows through its own separate series of processes for completion.



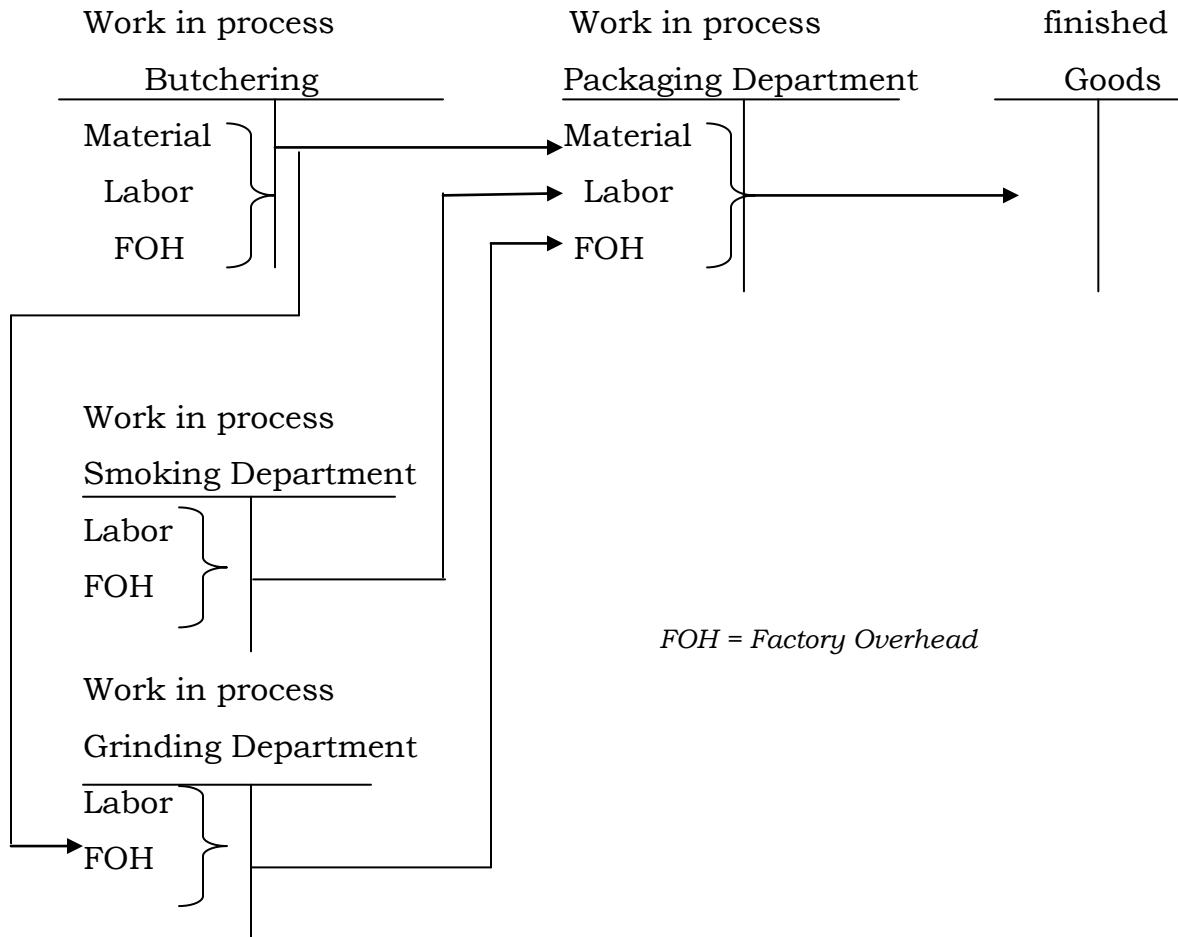
- ii. The products flow through their separate series of processes and then eventually flow through a common department.



NB As in the previous illustration, materials may be added in subsequent processes.

C) Selective Product Flow

In this method, the product moves to different departments within the plant, depending up on the desired final product.



For example in meat processing, after the initial butchering process, some of the product goes directly to the packaging department and then to finished goods inventory; some goes to the smoking department and then to the packaging department and finally to finished goods inventory, and some goes to the grinding department, then to the packaging department, and lastly to finished goods inventory.

4.3 Procedures for Materials, Labor and Factory overhead Costs

In process costing, materials, labor and factory overhead costs are accumulated in the usual accounts, using normal cost accounting procedures. Costs are then analyzed by department or processes and charged to departments by appropriate journal entries. In process costing, materials are issued only to the process in originating department; subsequent departments add labor and factory overhead. If materials are needed in a

department other than the first, they are charged to the department performing the specific operation.

For control purposes materials need not always be priced individually or by requisition forms. The cost of materials used can be determined at the end of the production period through inventory difference procedures, i.e., adding purchases to beginning inventory and then deducting ending inventory. Or consumption reports which state the cost of materials or quantity of materials put into process by various departments can be used.

Journal entries in process-costing systems are basically like those made in job-costing systems with respect to direct materials and conversion cost. The main difference is that, in process costing, there is often more than one work-in-process account, one for each process. As direct materials and direct labor are used in production department one (1), these costs are added to the work in process inventory account for department one (1). Overhead is applied using a predetermined overhead rate. The predetermined overhead rate is computed in the same way in job-order and process costing.

The appropriate journal entry to be maintained when materials are charged for production in Blending department during a period is:

Work in process-Blending Department	***
Materials	***

In process costing, labor costs are identified by and charged to departments, and thus eliminating the detailed clerical work of accumulating labor costs by jobs. Summary of labor charges are made to departments through an entry which distributes the direct manufacturing payroll as shown below:

Work in process-Blending Department	***
Work in process-Testing department	***
Work in process-Terminal Department	***
Payroll	***

Factory overhead incurred in process costing as well as in job order costing should be accumulated in factory overhead subsidiary ledger for producing and service departments. This procedure is consistent with requirements for responsibility accounting and responsibility reporting.

In process costing, either the predetermined overhead or the actual overhead can be charged to products. However, charging actual overhead to products is feasible when production remains comparatively stable from period to period, since factory overhead

will then remain about the same from one month to the next. Predetermined overhead rates for producing departments should be used if;

1. Production is not stable
2. Factory overhead, especially fixed overhead is a significant cost.

The use of predetermined rates is highly recommended for improving cost and facilitating cost analysis. Prior to charging factory overhead to departments via their respective work in process accounts, expenses must be accumulated in a factory overhead control account.

As expenses are incurred, the appropriate entry to be maintained is:

Factory overhead control	***	
Accounts payable		***
Accumulated depreciation-machinery		***
Prepaid insurance		***
Materials		***
Payroll		***

As the end of the period, the predetermined rates are multiplied by the respective actual activity base (e.g. Direct labor hours) for each producing department, the entry charging these expenses to work in process is as follows:

Work in process-Blending Department	***
Work in process-Testing Department	***
Work in process-Terminal Department	***
Factory overhead control	***

Activity 4.1

- List down at least four characteristics and the procedures of process costing.
- Describe what costing by department means.
- Identify and discuss the situations when either actual or the predetermined overhead rate is appropriate for charging indirect costs to processes.

4.4 The cost of production report

The key document in a typical process-costing system is the departmental production report, prepared for each production department at the end of every accounting period. It shows all costs chargeable to a department. This report replaces the job-cost record, which is used to accumulate costs by job in a job-order costing system. The departmental production report summarizes the flow of production quantities through departments, and

it shows the amount of production cost transferred out of the department's work in process inventory account during the period. It is not only the source for summary journal entries at the end of the month, but also a most convenient vehicle for presenting and disposing of costs accumulated during the month.

A cost of production report shows:

1. Total and unit costs transferred to it from a preceding department.
2. Materials, labor and factory overhead added by the department.
3. Unit costs added by the department.
4. Total and unit costs accumulated to the end of operations in the department.
5. The cost of the beginning and ending work in process inventories, and
6. Cost transferred to a succeeding department or to a finished goods store room.

It is customary to divide the cost section of the report into two parts: One is showing costs for which the department is accountable, including departmental and cumulative total and unit costs, the other showing the disposition of these costs. A quantity schedule showing the total number of units for which a department is accountable and the disposition made of these units is also part of each department's cost of production report. Information on this schedule, adjusted for equivalent production unit, is used to determine the unit costs added by a department, the costing of the ending work in process inventory, and the cost to be transferred out of the department.

A cost of production report determines periodic total and unit costs. As controlling demands detailed presentation of cost data, showing total figures by itself mean very little. Therefore in most instances, the total cost is broken down by cost element for each department responsible for the costs incurred. Furthermore, detailed departmental figures are needed because of the various completion stages of the work in process inventories. Either in the cost of production report itself or in the supporting schedule, each item of material used by a department is listed; every labor operation is shown separately; factory overhead components are noted individually; and a unit cost is derived for each item.

Illustration 4.1

ABC Company manufactures a single product in three producing departments namely, Blending, Testing and Terminal. The originating (Blending) department put 50,000 units in process during the month of January, 2007. Of which 45,000 units are completed and transferred to the next department .4, 000 units are still in process and 1,000 units are lost during production. Materials, labor and factory overhead added by the department are amounted \$24,500, \$29,140 and \$28,200 respectively. In addition, the ending work in process for the department is fully complete with respect to materials and $\frac{1}{2}$ complete with respect to labor and factory overhead.

Required: prepare the cost of production report for Blending department.

Solution***Supportive computation:***

Equivalent unit of production: Equivalent production represents the production of a process in terms of completed units, work in process at the end of an accounting period are converted into equivalent units. This is done by the following formula

$$\text{Equivalent completed units} = \left(\frac{\text{No of units of work in process}}{\text{work in process}} \right) \times \left(\frac{\text{Degree of completion in percentage or fraction}}{\text{percentage or fraction}} \right)$$

Equivalent production:

$$\text{Materials} = 45,000 + 4,000 = 49,000 \text{ units}$$

$$\text{Labor and factory overhead} = 45,000 + 4000(1/2) = 47,000 \text{ units.}$$

NB: All materials needed for the completion of production are added at the beginning of the process in the originating department. Therefore, the equivalent production for materials is 49,000 units.

All the units completed and transferred to the next department are fully complete with respect to materials and conversion costs in this department.

Unit costs: unit costs are determined for each cost component based on the respective computed production units.

$$\text{Materials} = \$24,500 \div 49,000 = \$0.50 \text{ per unit}$$

$$\text{Labor} = \$29,140 \div 47,000 = \$0.62 \text{ per unit}$$

$$\text{Factory overhead} = \$28,200 \div 47,000 = \$0.60 \text{ per unit}$$

ABC Company
Blending Department
Cost of production report
For the month of January, 2007

Quantity schedule:

Units started in process		<u>50,000</u>
Units transferred to the next department	45,000	
Units still in process (all materials, ½ labor and factory overhead)	4,000	
Units lost in process	<u>1,000</u>	<u>50,000</u>

Cost charged to the department:

Cost added by the department	<u>Total cost</u>	<u>unit cost</u>
Materials	\$24,500	\$0.50
Labor	29,140	0.62
Factory overhead	<u>28,200</u>	<u>0.60</u>
Total cost to be accounted for	<u>\$81,840</u>	<u>1.72</u>

Cost accounted for as follows:

Transferred to next department (45,000 x \$ 1.72)		\$77,400
Work in process-ending inventory:		
Materials (4,000 x \$ 0.50)	\$2,000	
Labor (4,000 x ½ x \$ 0.62)	1,240	
Factory overhead (4,000 x ½ x 0.60)	<u>1,200</u>	<u>4,440</u>
Total cost accounted for		<u>\$81,840</u>

NB: figures in bracket represent equivalent units in work in process inventory for each cost components and unit costs of the respective cost components.

Equivalent units of production: In many manufacturing processes all materials are issued at the start of production but all conversion costs are generally added to the processes uniformly through time. Therefore, the 4,000 units still in process for Blending department of ABC Company have all the materials needed for their completion but of labor and factory overhead are charged only 50% or (1/2). And we can infer that, the total amount of labor and factory overhead charged into units still in process can be converted (stated) into the amount of completed output units that could be made with the quantity of input. For example, if 50 units of a product in ending work in process inventory are 70% complete with respect to conversion costs there are 35 (70%×50 units) equivalent units of production for conversion costs. This restatement of uncompleted work in terms of completed units is called computation of equivalent units of production. That is, if all the conversion costs input in 50units inventory were used to make complete output units, the company would be able to make 35 units of output. Therefore, in light with the above

example we can say that (in terms of equivalent units of production) the balances of labor and factory overhead added to work in process inventory of the Blending department are sufficient to complete 2,000 units. Equivalent units are computed separately for each cost categories of materials, labor and factory overhead.

Unit cost:

Departmental cost of production report indicated the cost of units as they leave each department. These individual unit costs are accumulated into a completed unit cost for the period. Unit costs are determined for each cost components based on the respective computed equivalent production units.

Disposition of Departmental costs:

In the departmental cost report, the section titled **“cost charged to the department”** shows a total cost charged to the production under the custody the department (\$81,840). The section titled **“cost Accounted for as follows”** shows the disposition of this cost as transferred to the next department (\$77,400) and work in process (\$4,440). The work in process figure must be broken down into its component parts of materials, labor and factory overhead

Lost Units:

Continuous processing leads to the possibility of waste shrinkage and other factors which cause loss or spoilage of production units.

Units lost in the first department:

Lost units reduce the number of units over which total costs can be spread, causing an increase in unit costs. The 1,000 units lost in the Blending department increase the unit cost of materials, labor and factory overhead. Had these units not been lost, the equivalent production figure would be 50,000 units for materials and 48,000 for labor and factory overhead. The unit cost for materials would be \$0.49 instead of \$0.50; labor \$0.607 instead of \$0.62; and factory overhead \$0.588 instead of \$0.60. In the first department, the only effect of losing units is an increase in the unit cost of the remaining good units.

Activity 4.2

- What is the purpose of a departmental production report as compared to job cost record?
- List down the elements of a departmental production report and discuss their purposes?
- How is unit cost calculated in a process cost system?
- How the cost of lost units is accounted in the first department?

Section 2**Cost of production report for second (Testing) Department:**

The Blending Department transferred 45,000 units to the Testing Department, where labor and factor overhead were added before the units were transferred to the Terminal (third) department. Costs incurred by the Testing department resulted in additional departmental as well as cumulative unit costs.

The cost of production report of the Testing department differs from that of the Blending department in several respects. Several additional calculations are made, for which space has been provided on the report. The additional information deals with:

1. costs received from the preceding department,
2. an adjustment of the preceding department's unit cost because of lost units, and
3. cost received from the preceding department to be included in the cost of the ending work in process inventory.

The Testing department is responsible for labor and factory overhead used as well as for the cost of units received from the Blending department.

The journal entry that has to be maintained by Testing department when processes are received from Blending department is shown below:

Work in process- Testing department	77,400	
Work in process –blending department		77,400

Illustration: 4.2

Assume that the quantity schedule for the Testing department shows that the 45,000 units received from the Bending department were accounted for as follows: 40,000 units sent to the Terminal department, 3,000 units still in process, and 2,000 units lost. An analysis of work indicated that units in process are one – third (1/3) complete as to labor and factory overhead. Labor and factory overhead charged to the department are \$37,310 and \$32,800 respectively.

Required: prepare the cost of production report for testing department:

ABC Company
Testing Department
Cost of production Report
For the month of January 2007

Quantity schedule:

Units received from preceding department		<u>45,000</u>
Units transferred to the next department	40,000	
Units still in process (1/3 labor and factory overhead)	3000	
Units lost in process	<u>2,000</u>	<u>45,000</u>

Cost charged to the department:

Cast from preceding department	<u>Total cost</u>	<u>unit cost</u>
Transferred in during the month (45,000 units)	<u>\$77,400</u>	<u>\$1.72</u>
Cost added by department:		
Labor	\$37,310	\$0.91
Factory overhead	<u>32,800</u>	<u>0.80</u>
Total cost added	\$70,110	\$1.71
Adjustment for lost units	-	0.08
Total cost to be accounted for: -	<u>\$147,510</u>	<u>\$3.51</u>

Cost accounted for as follows

Transferred to next department (40,000x\$3.51)		\$140,400
Work in process- ending inventory:		
Adjusted cost from preceding department (3,000x (\$1.72+\$0.08))	\$5,400	
Labor (3,000x1/3 x \$ 0.91)	910	
Factory overhead (3,000x1/3 x \$ 0.80)	<u>800</u>	<u>7,110</u>
Total cost accounted for		<u>\$147,510</u>

Additional computations:**Equivalent production:**

Labor and factory overhead = 40,000+ 3,000 x 1/3 = 41,000 units

Unit costs:

Labor= \$37,310 ÷ 41, 000=\$0.91 part unit

Factory overhead = $\$32,800 \div 41,000 = \0.80 per unit

Adjustment for lost units: this is part of the additional computation and computed after the following discussion.

Units lost in departments subsequent to the first department

The Blending department's unit cost was \$ 1.72 when 45,000 units were transferred to the Testing department. However, because 2,000 of these 45,000 units were lost during processing in Testing department, the \$1.72 unit cost figure no longer applies and must be adjusted. The total cost of the units transferred remains at \$ 77,400 but 43,000 units must now absorb this total cost, causing an increase of \$0.08 in the cost per unit due to the loss of 2,000 units in the Testing department.

The lost unit cost can be computed by one of two methods:

Method 1: Here we determine a unit cost of work done in the preceding department and subtract the preceding department's old unit cost figure from the adjusted unit cost figure.

The difference between the two figures is the additional cost due to lost units.

New unit cost = $\frac{\$77,400}{43,000} = \1.80 per unit

Old unit cost = $\frac{\$77,400}{45,000} = \1.72 per unit

$\therefore$ Additional cost due to lost unit = $\$1.80 - \$1.72 = \underline{\$0.08}$ per unit

Method 2: Here we determine the lost units' share of total cost and this cost will be allocated over the remaining good units. Total cost previously absorbed by the units lost is \$3,440 (2,000 units x \$ 1.72) must now be absorbed by the remaining good units.

$\therefore$ The additional cost to be picked up by each remaining good unit is \$ 0.08
($\$3,440 \div 43,000$ units)

NB. The lost unit cost adjustment must be entered in the cost production report on the adjustment for lost units line.

In short the adjustment for lost units is computed as follows under the two methods:

Method no.1: $\$77,400 \div 43,000 = \1.80 ; $\$1.80 - \$1.72 = \underline{\$0.08}$ per unit

Method no.2: 2000 units x \$ 1.72 = \$3,440; $\$3,440 \div 43,000 = \underline{\$0.08}$ per unit

4.4.1 Timing of lost units

Lost units may occur at the beginning, during, or at the end of a process. For purposes of practicability and simplicity, it is ordinarily assumed that units lost at the beginning or during process were never put in process. The cost of units lost is spread over the units completed and units still in process.

When units are lost or are identified as lost at the end of a process, the cost of lost units is charged to completed units only. Assume that the 2,000 units lost by the Testing department found at the final inspection and all lost units are normal lost units, their cost would be charged only to the 40,000 finished units. The cost of production report for this case would be the following.

**ABC Company
Testing Department
Cost of Production Report
For the month January, 2007**

Quantity schedule:

Units received from preceding department		<u>45,000</u>
Units transferred to next department	40,000	
Units still in process (1/3 labor and factory overhead)	3,000	
Units lost in process (at end of process)	<u>2,000</u>	<u>45,000</u>

Cost charged to the department:

Cost from preceding department	<u>Total cost</u>	<u>unit cost</u>
Transferred in during the month (45,000 units)	<u>\$77,400</u>	<u>\$1.72</u>
Cost added by department		
Labor	\$37,310	\$0.87
Factory overhead	<u>32,800</u>	<u>0.76</u>
Total cost added	<u>\$70,110</u>	<u>\$1.63</u>
Total cost to be accounted for	<u>\$147,510</u>	<u>\$3.35</u>

Cost accounted for as follows:

Transferred to next department (40,000 units x (\$3.35 + \$ 0.1675))		\$140,720
Work in process ending inventory:		
From preceding department (3,000 x \$1.72)	\$5,160	
Labor (3,000x1/3 x \$0.87)	870	
Factory overhead (3,000x1/3x\$0.76)	<u>760</u>	<u>6,790</u>
Total cost accounted for		<u>\$147,510</u>

Additional computations:**Equivalent production:**

Labor and factory overhead = 40,000 + 3,000 x 1/3 + 2,000 lost Units = 43,000 units

Unit costs:

$$\text{Labor} = \$ 37,310 \div 43,000 = \underline{\$0.87} \text{ per unit}$$

$$\text{Factory overhead} = \$ 32,800 \div 43,000 = \underline{\$ 0.76} \text{ per unit}$$

For similarity and to absorb the error in determination of fractions, the cost to be assigned to units transferred out is computed by subtracting the ending work in process balance from total cost accounted to figure i.e., $\underline{\$140,720} = \$ 147,510 - \$6,790$

Normal versus Abnormal loss of Units

Normal loss: units might be lost through evaporation, shrinkage, substandard yields, spoiled work, or inefficient equipment. Such loss is quite expected under normal conditions. In many instances the nature of operations makes certain losses normal or unavoidable, because they are considered within normal tolerance limits for human and machine errors. The cost of these normally lost units does not appear as a separate item of cost but is spread over the remaining good units.

Abnormal loss: This type of loss consists of loss due to carelessness, machine breakdown, accident, use of defective materials, etc. A different situation is created by abnormal or avoidable spoilage or losses that are not expected to arise in normal efficient operating conditions. The cost of such abnormal spoilage or losses is charged either to factory overhead, or directly to the current period expenses account and reported as a separate item in the cost of goods sold statement. The following is the appropriate journal entry to charge cost of abnormal lost units to Factory overhead account.

Factory overhead control	6,700
Work in process- Testing department	6, 700

The cost of production report would show the abnormal spoilage or losses as follows:

Transferred to next department (40,000x \$3.35) -----	\$134,020*
Transferred to factory overhead (40,000 units x	
\$0.1675) or (2,000 lost units x \$ 3.35)	6,700
* <u>\$134,020</u> = \$147,510 - \$6,790 ending inventory - \$6,700.	

If the lost units were only partially complete, equivalent units of production calculation should consider their stage of completion when lost or spoiled, and the costing of the

abnormal loss should be weighted accordingly. If one part of the loss is normal and another part abnormal, each portion must be treated in accordance.

Activity 4.3

- Explain the primary differences between the cost of production report of the originating department and the second department.
- Identify and distinguish the effects of lost units in the first department and in subsequent departments.
- Differentiate between normal and abnormal loss.

Cost of production report for third (Terminal) department

Total and unit cost figures were derived by using procedures discussed for the cost of production report of the second (Testing) department. The work completed is transferred to the finished goods store room; thus, the title **“Transferred to finished goods store room”** is used in place of the title **“Transferred to next department”**.

The journal entry to be maintained by Terminal department for costs transferred from testing department would be:

Work in process terminal department	140,400
Work in process testing department	140,400

Illustration 4.3

Among the 40,000 units transferred from the Testing department, 35,000 units are transferred to finished goods store room; 4,000 units are still in process and 1,000 units are lost during production. Ending work in process balance is $\frac{1}{4}$ complete as to labor and factory overhead. Besides, the labor and factory overhead balances charged to the process in this department are \$ 32,400 and \$19,800 respectively.

Required: - prepare a cost of production report for Terminal department

ABC Company
Terminal department
Cost of production report
For the month of January, 2007

Quantity schedule:

Units received from preceding department		<u>40,000</u>
Units transferred to finished goods storeroom	35,000	
Units still in process (1/4 labor and factory overhead)	4,000	
Units lost in process	<u>1,000</u>	<u>40,000</u>

Cost charged to the department:

Cost from preceding department	<u>Total cost</u>	<u>unit cost</u>
Transferred in during the month (40,000 units)	\$140,400	\$3.51
Cost added by department:		
Labor	32,400	\$0.90
Factory overhead	<u>19,800</u>	<u>0.55</u>
Total cost added	\$52,200	\$1.45
Adjustment for lost units	<u>-</u>	<u>0.09</u>
Total cost to be accounted for	<u>\$192,600</u>	<u>\$5.05</u>

Cost accounted for as follows:

Transferred to finished goods storeroom (35,000x\$5.05)	\$176,750
Work in process – ending inventory:	
Adjusted cost from preceding department (4,000x (\$3.51+\$0.09))	\$14,400
Labor (4,000 x ¼ x \$0.90)	900
Factory overhead (4000x1/4 x \$0.55)	<u>550</u>
Total cost accounted for	<u>\$192,600</u>

Additional computations:*Equivalent production:*

Labor and factory overhead = 35,000 + 4000 x ¼ = 36,000 units

Unit costs:

Labor = \$32,400 ÷ 36,000 = \$0.90 per unit

Factory overhead = \$19,800 ÷ 36,000 = \$0.55 per unit

Adjustment for lost units:

Method no.1: \$140,400 ÷ 39,000 = \$3.60; \$3.60 - \$3.51 = \$0.09 per unit

Method no. 2 = 1,000 units x \$ 3.51 = \$3,510 = \$3,510 ÷ 39,000 = \$0.09 per unit

The entry to transfer finished units to the finished goods store room would be:

Finished Goods	176,750
Work in process - Terminal department	176,750

4.5 Additions of Materials in Subsequent Departments

In numerous industries, all materials needed for the product are put in process in the first department. However, additional materials might be required in subsequent departments in order to complete the units. The addition of such materials has two possible effects on units and costs in process.

- 1) The additional materials increase the unit cost, since these materials become a part of the product manufactured, but do not increase the number of final units. For example, in an automobile assembly plant additional parts, these materials are needed to give the product certain specified quantities, characteristics, or completeness.
- 2) The added materials increase the number of units and also cause a change in unit cost. In processing chemical, water is often added to a mixture, causing an increase in the number of units and a spreading of costs over a greater number of units.

4.5.1 Increase in units cost due to addition of materials

In the simplest case, added materials such as parts of an automobile do not increase the number of units but increase total cost and unit costs. A materials unit cost must be computed for the department, and a materials cost must be included in the work in process inventory.

Illustration 4.4

Assume the additional materials costing \$ 17,020 are placed in process and charged to the Terminal department. Assume further that the materials in work in process are sufficient to complete 2,000 of the 4,000 units; that is, units are 50% complete as to materials cost. The effect of the additional materials cost is shown in the cost report be below.

ABC COMPANY
Terminal Department
Cost of production Report
For the month of January, 2007

Quantity Schedule;

Units received from preceding department		<u>40,000</u>
Units transferred to finished goods store room	35,000	
Units still in process (½ materials, ¼ labor and factory overhead)	4,000	
Units lost in process	<u>1,000</u>	<u>40,000</u>

Cost charged to the Department:

Cost from preceding department	<u>Total cost</u>	<u>unit cost</u>
Transferred in during the month (40,000 units)	\$140,400	\$3.51
Cost added by department:		
Materials	\$17,020	\$0.46
Labor	32,800	0.90
Factory overhead	<u>19,800</u>	<u>0.55</u>
Total cost added	\$69,220	\$1.91
Adjustment for lost units	<u>-</u>	<u>0.09</u>
Total cost to be accounted for	<u>\$209,620</u>	<u>\$5.51</u>

Cost accounted for as follows:

Transferred to finished goods storeroom (35,000x\$5.51)		\$192,850
Work in process ending inventory		
Adjusted cost from preceding department (4,000x\$3.60)	\$14,400	
Materials (4,000 x 1/2 x \$0.46)	920	
Labor (4,000 x ¼ x \$ 0.90)	900	
Factory overhead (4000 x ¼ x \$0.55)	<u>550</u>	<u>16,770</u>
Total cost accounted for		<u>\$209,620</u>

Additional computations:

Equivalent production:

$$\text{Materials} = 35,000 + 4000 \div 2 = \underline{37,000} \text{ units}$$

$$\text{Labor and factory overhead} = 35,000 + 4000 \div 4 = \underline{36,000} \text{ units}$$

Unit costs:

$$\text{Materials} = \$ 17,020 \div 37,000 = \underline{\$0.46} \text{ per unit}$$

$$\text{Labor} = \$32,400 \div 36,000 = \underline{\$0.90} \text{ per unit}$$

$$\text{Factory overhead} = \$19,800 \div 36,000 = \underline{\$0.55} \text{ per unit}$$

Adjustment for lost units:

$$\text{Method No. 1} = \$140,400 \div 39,000 = \$3.60; \$3.60 - \$3.51 = \underline{\$0.09} \text{ per unit}$$

$$\text{Method No. 2} = 1,000 \text{ units} \times \$3.51 = \$3,510, \$3,510 \div 39,000 = \underline{\$0.09} \text{ per unit}$$

The only differences in the two cost reports (the report for the third department(Terminal) and the above report) are the \$17,020 materials cost charged to the department and \$0.46 materials unit cost (\$17,020 . 37,000). The additional materials cost is also reflected in the total cost to be accounted for, and in the ending work in process inventory.

4.5.2 Increase in units and change in unit cost Due to Addition of Materials

When additional materials result in additional units different computations are necessary. The greater numbers of units cause a decrease in unit cost which necessitates an adjustment of the preceding department's unit cost; since the increased number of units will absorb the same total cost transferred from preceding department.

Illustration 4.5

Assume Terminal department costs for labor and factory overhead of \$32,400 and \$19,800 respectively, an additional materials cost of \$17,020 and an increase of 8,000 units as the result of added materials.

Required: - Prepare the cost of production report for Terminal Department

ABC COMPANY
Terminal Department
Cost of Production Report
For the Month of January, 2007

Quantity Schedule:

Units received from preceding department	40,000	
Additional units put into process	<u>8,000</u>	<u>48,000</u>
Units transferred to finished good, storeroom	44,000	
Units still in process (1/2 materials, ¼ labor and factory overhead)	<u>4,000</u>	<u>48,000</u>

Cost charged to the department:

Cost from preceding department	<u>Total cost</u>	<u>unit cost</u>
Transferred in during the month (40,000 units)	<u>\$140,400</u>	<u>\$3.510</u>
Cost added by department		
Materials	\$17,020	\$0.370
Labor	32,400	0.720
Factory overhead	<u>19,800</u>	<u>0.440</u>
Total cost added	\$69,220	\$1.530
Adjusted unit cost of units transferred in during the month	<u>-</u>	<u>2.925</u>
Total cost to be accounted for	<u>\$209,620</u>	<u>\$4.455</u>

Cost Accounted for as Follows:

Transferred to finished goods Storeroom (44,000 x \$4.55)		\$196,020
Work in process – ending inventory		
Adjusted cost from proceeding department (4,000x\$2.925)	\$11,700	
Materials (4,000 x 1/2 x \$0.370)	740	
Labor (4,000 x ¼ x \$0.720)	720	
Factory overhead (4,000 x ¼ x \$0.440)	<u>440</u>	<u>13,600</u>
Total cost Accounted for		<u>\$209,620</u>

Additional Computations:

Equivalent production:

$$\text{Materials} = 44,000 + 4000 \div 2 = \underline{46,000} \text{ units}$$

Labor and factory overhead = $44,000 + 4,000 \div 4 = \underline{45,000}$ units

Unit costs:

Materials = $\$17,020 \div 46,000 = \underline{\$0.370}$ per unit

Labor = $\$32,400 \div 45,000 = \underline{\$0.720}$ per unit

Factory overhead = $\$19,800 \div 45,000 = \underline{\$0.440}$ per unit

Adjustment for additional units = $\$140,400 \div 48,000 = \underline{\$2.925}$ per unit

The additional 8,000 units are entered in the department's quantity schedule as "Additional Units put into process". The quantity schedule reports that 44,000 units were complete and transferred to the finished goods store room and that 4,000 units are still in process, 50% complete as to materials and 25% complete as to labor and factory overhead. Therefore, equivalent production is 46,000 units for materials and 45,000 units for labor and factory overhead. Dividing departmental materials, labor and factory overhead costs for the period by these production figures results in a new unit cost for all cost components. Total cost charged to the Terminal department as cost transferred in from preceding department must now be allocated over a greater number of units, thereby, reducing the unit cost of work done in the preceding department.

Initially the \$140,400 cost transferred to the Terminal department was absorbed by 40,000 units, resulting in a unit cost of \$3.51. Because of the 8,000 increase in units, the \$140,400 cost must now be spread over 48,000 units resulting in a unit cost for the preceding department of \$2.925. This adjusted cost is inserted in the production report as **"Adjusted cost of units transferred in during the month"** and is added to departmental unit costs to arrive at the unit cost accumulated to the end of operations in the Terminal department. When additional materials increase the number of units being processed, it is still possible to have lost units. However, should increase and lost units occur, no separate calculation is required for the lost units; only net units added are used.

Activity 4.4

- List down the additional element/s that the cost of production report of the final department has.
- Identify the possible effects of additions of materials in subsequent departments.

Summary

Process costing is used in production processes where relatively large numbers of nearly identical products are manufactured. The purpose of a process costing system is the same as that of a job-order costing system - to accumulate costs and assign these costs to units of product. Product costs are needed for planning, cost management, decision making, and reporting to various outside organizations.

The flow of costs in process costing systems and job-order costing systems is the same. Costs of direct material, direct labor and factory overhead are added to a Work-in – process Inventory account. When products are completed, the costs assigned to them are transferred either to Finished-goods inventory or to the next production departments’ work-in-process inventory account. In sequential production processes, the cost of the goods transferred from one production department to another is called transferred-in cost (costs transferred from preceding department).

There are some important differences between job-order and process-costing system. Chief among these is that job-order costing systems accumulate production costs by job or batch, whereas processes costing systems accumulate costs by department (processes). Another important difference is the focus on equivalent units in process costing. An equivalent unit is a measure of the amount of productive input that has been applied to a fully or partially completed unit of a product. In process costing, Production costs per equivalent unit are calculated for the three manufacturing cost components (Direct materials, direct labor, and Factory overhead) separately.

The key document in a process costing system is the departmental production report, rather than the job-cost record used in job-order costing. The following are the prerequisite in preparing a departmental production report (1) analyze the physical flow of units, (2) calculate the equivalent production units (3) compute the cost per equivalent production unit, and (4) analyze the total costs of the department. Job-order and process costing represent the polar extremes of product costing systems.

Self-Check Exercise**Part one: Multiple choice questions**

1. Which one of the following companies would *most likely* use process costing?
 - a. Beef Producer
 - b. Law Firm
 - c. Furniture Retailer
 - d. Printing Company
2. In a process-costing system, manufacturing costs are accumulated by:
 - a. batch.
 - b. batch and time period.
 - c. department or order
 - d. department or process.
3. Which of the following statements is false?
 - a. In job-order costing, costs are accumulated by job order.
 - b. In process costing, costs are accumulated by department.
 - c. In process costing, the cost per unit in a department is found by spreading the period's manufacturing costs over the production activity.
 - d. In process costing, the total cost of each unit is found by dividing the total factory costs by the number of units completed.
 - e. In job-order costing, the unit cost is found by dividing the job's total cost by the job's total units.
4. Companies that use a process-cost accounting system would:
 - a. establish a separate Work-in-Process Inventory account for each manufacturing department.
 - b. establish a separate Finished-Goods Inventory account for each manufacturing department.
 - c. pass completed production directly to Cost of Goods Sold.
 - d. charge goods produced with actual overhead amounts rather than applied overhead amounts.
 - e. eliminate the need for the Finished-Goods Inventory account.
5. Process costing is used to account for:
 - a. large numbers of identical products that are produced in a continuous manufacturing environment.
 - b. small numbers of products that are produced in batches.
 - c. raw materials that are converted to finished goods.
 - d. finished goods those are refined and processed further.
 - e. large numbers of products that are produced in a non repetitive process.
6. A department has no beginning work in process, has started 90,000 units and completed 60,000 units. Its ending work in process is 30,000 units, 60% complete as to conversion costs and fully complete as to materials. Its equivalent units for conversion costs are:
 - a. 60,000.
 - b. 90,000.
 - c. 78,000.
 - d. 54,000.

7. If 8,000 units were in beginning inventory, 24,000 units were started and 6,000 units were in the ending inventory, how many units were completed and transferred out?
- 30,000
 - 20,000
 - 26,000
 - 24,000
8. Which of the following is correct about abnormal lost units;
- They are unavoidable
 - Their cost is distributed over the remaining good units
 - They occurred beyond the tolerance limit of human and machine
 - Their cost is assigned to Factory overhead.

Based on the following information answer question number 9&10;

During April, 20,000 units were transferred in from department A at a cost of \$39,000. Materials cost of \$6,500 and conversion cost of \$9,000 were added in department B. On April 30, Department B had 5,000 units of Work in process, 60% complete as to conversion. Materials are added in the beginning of the process in department B. there is no lost units.

9. what would be the equivalent production units for materials and conversion costs
- 18,000 units and 20,000 units respectively
 - 20,000 units and 18,000 units respectively
 - 20,000 units and 15,000 units respectively
 - None of the above
10. The cost per equivalent units for conversion cost would be:
- \$0.200 per unit
 - \$0.361 per unit
 - \$0.500 per unit
 - Some other amount

Work out

1. For December, the production control department of Jackson chemical company reported the following production data for department 2:

➤ Transferred in from Department 1	55,000 liters
➤ Transferred out to Department 3	39,500 liters
➤ In process at the end of December(with 1/3 Labor and Factory overhead)	10,500 liters

All materials were put into process in Department 1.

The cost department collected these figures for department 2:

➤ Unit cost for units transferred in from department	\$1.80
➤ Labor cost in department 2	\$ 27,520
➤ Applied factory overhead	\$ 15,480

Required: prepare the cost of production report for department 2 for December

2. Ethio plc. Company produces a product which requires processing I in three departments. In the third department, materials are added doubling the number of units. The following data pertain to the operations of department 3 for March.

Units received from department 2	20,000	
Units transferred to finished goods storeroom	32,000	
The balance of the units are still in process-100% complete as to Materials and 50% complete as to labor and overhead		
Cost transferred from Department 2		\$30,000
Cost added by department:		
Materials	\$8,800	
Labor	9,000	
Factory overhead	<u>7,200</u>	<u>\$25,000</u>

There was no beginning Work in process inventory

Required: Prepare a cost of production report for department for March

CHAPTER FIVE

AVERAGE AND FIFO COSTING

Chapter learning objectives:

After studying this chapter, you should be able to:

- ❖ Explain the conceptual differences between the weighted average and first in, first out method of accounting for process costing.
- ❖ Compute the cost per equivalent unit under the weighted average and FIFO process costing
- ❖ Prepare a departmental production report using the weighted average and first in, first out cost flow assumptions.
- ❖ Identify the apparent difficulties that process costing has in actual practice

5.1 Introduction

This chapter is an extension of chapter four. In the last chapter the accounting procedures for the three manufacturing cost components, the flow of products in process costing, the concepts and purposes of the cost of production, the timing of lost units and the effects of additions of materials in subsequent departments were discussed in detail. The cost of production reports prepared for the month of January in chapter four lists ending work in process inventories and this ending work in process inventories became the beginning inventory for the month of February. These ending work in process inventories can accounted either in weighted average cost flow assumption or in first-in, first-out cost flow assumption. Therefore, this chapter combines historical full absorption costing with process cost accumulation and weighted average and first-in first out (FIFO) cost flow assumptions. Generally, the chapter is all about the treatment of beginning work in process inventories and the accounting procedures used accordingly.

5.2 BEGINNING WORK IN PROCESS INVENTORIES

The cost of production reports illustrated in previous chapter lists ending work in process inventories. These inventories become beginning inventories of the next period. Two of the possible methods of accounting for these beginning inventory costs are:

1. Average costing: Beginning inventory costs are added to the costs of the new period

2. First-in first-out (FIFO) costing: Beginning inventory costs are kept separate from the new costs necessary to complete the work in process inventory

5.2.1 Average costing

The average costing method of accounting for beginning work in process inventory costs involves merging these costs with the cost of the new period. To accomplish this relatively simple task, representative average unit costs must be determined.

The January cost reports of the three departments' reviewed in previous sections are used to illustrate the treatment of beginning work in process inventory and to show the relationship of costs from one period to the next. Ending inventories in January departmental cost reports become beginning work in process inventories for February and are summarized as follows:

	<u>Blending</u>	<u>Testing</u>	<u>Terminal</u>
Units	4,000	3,000	4,000
Cost from preceding Department	-	\$5,400	\$14,400
Materials in process.....	\$2,000	-	-
Labor in process	1,240	910	900
Factory overhead in Process.....	1,200	800	550

Blending Department: The February 1 work in process inventory of the Blending Department shows a \$2,000 materials cost, a \$1,240 labor cost, a \$1,200 factory overhead cost, and 4,000 units in process. During February, additional charges to the department are material, \$19,840; labor, \$24,180; and factory overhead, \$22,580. The additional materials put into process are for the production of 40,000 units. Therefore, “**units to be accounted for**” total 44,000 (4,000+40,000). Of the total units put into process, 39,000 are completed of the completed units, 38,000 units were transferred to the Testing Department and 1,000 units are awaiting transfer. At month end 3,000 units are in process, 100% complete as to materials but only 66^{2/3} % complete as to labor and overhead. During the month, 2,000 units were lost.

In the Blending Department as well as in the subsequent departments in this illustration, it is assumed that the loss applies to all good units and that the loss is within normal tolerance limits. Therefore, the effect of the losing units is an increase in the unit cost of the remaining good units.

The unit cost of work done in the Bending Department is \$1.72 consisting of \$0.52 for materials, \$0.62 for labor, and \$0.58 for factory overhead. The \$0.52 unit cost for materials is computed by adding the materials cost in the beginning work in process inventory to the materials cost for the month (\$2,000+\$19,840) and dividing the \$21,840 total by the equivalent production figure of 42,000 units. These units include the 38,000 units completed and transferred, the 1,000 units completed but still on hand, and the 3,000 units in process, which are completed as to materials. The cost of materials already in process is added to the materials cost for the month before dividing by the equivalent production figure. This method results in an **average unit cost** for work done in the current and preceding periods.

The same procedure is followed in computing unit costs for labor and factory overhead. The \$0.62 unit cost for labor is a result of dividing equivalent production of 41,000 units $[39,000 + (2/3 \times 3,000)]$ into the sum of the beginning inventory labor cost of \$1,240 and the departmental labor cost of \$24,180 for the month. The factory overhead unit cost is \$0.58 $[(\$1,200 + \$22,580)/41,000]$. Of total cost charged to the department, \$65,360 is transferred to the Testing Department when the following entry is recorded:

Work in process- Testing Department	65, 360	
Work In process- Blending Department		65,360

The cost remaining in the Testing Department, \$5,680, is assigned to the ending work in process inventory. The WIP inventory consists of \$1,720 (1,000 units x \$1.72) for units completed and on hand and of the following costs assigned to units still in process \$1,560 (3,000 units x \$0.52) for materials; \$1,240 (2,000 units x \$0.62) for labor; and \$1.160 (2,000 units x 0.58) for factory overhead. The 1,000 units completed but on hand are

listed as work in process in the Blending department as this department is still responsible for these units. The above facts are illustrated below:

ABC Company
Blending Department
Cost of production Report Average costing
For the month of February, 2007

Quantity schedule:

Units in process at beginning (all materials, ½ labor and factory overhead)	4,000	
Units started in process	<u>40,000</u>	<u>44,000</u>
Units transferred to next department	38,000	
Units completed and on hand	1,000	
Units still in process (all materials, 2/3 labor and factory overhead)	3,000	
Units lost in process	<u>2,000</u>	<u>44,000</u>

Cost charged to the department:

Cost added by department	<u>Total cost</u>	<u>unit cost</u>
Work in process-beginning inventory		
Materials	\$2,000	
Labor	1,240	
Factory overhead	1,200	
Cost added during the period		
Materials	19,840	\$0.52
Labor	24,180	0.62
Factory overhead	<u>22,580</u>	<u>0.58</u>
Total cost to be accounted for	<u>\$71,040</u>	<u>\$1.72</u>

Cost accounted for as follows:

Transferred to next department (38,000x\$1.72)		\$65,360
Work in process-ending inventory:		
Completed and on hand (1,000x\$1.72)	\$1,720	
Materials (3,000x\$0.52)	1,560	
Labor (3,000 x 2/3 x \$0.62)	1,240	
Factory overhead (3,000 x 2/3 x \$0.58)	<u>1,160</u>	<u>5,680</u>
Total cost Accounted for		<u>\$71,040</u>

Additional computations:

Equivalent production:

Materials = 38,000 + 1,000 + 3,000 = 42,000 units

Labor and factory overhead = 38,000 + 1,000 + (2/3 x 3000) = 41,000 units

Unit costs:

Materials = \$2,000 + \$19,840 = \$ 21,840, \$21,840 ÷ 42,000 = \$0.52 per unit

Labor = \$1,240 + \$24,180 = \$25,420, \$25,420 ÷ 41,000 = \$0.62 per unit

Factory overhead = \$1,200 + \$22,580 = \$23,780, \$23,780 ÷ 41,000 = \$0.58 per unit

Testing Department: Accounting for the beginning work in process inventory cost in a department other than the first requires additional analysis. When the prior period's ending work in process inventory was computed, part of the cost of this inventory came from costs added by the preceding department. Because costs assigned to the beginning work in process inventory are added to costs incurred during the period and the total is divided by equivalent production, the beginning work in process inventory of departments other than the first must be split into following two parts:

1. cost transferred from preceding departments
2. Cost added by the department itself.

The portion of the beginning work in process inventory cost from preceding department is entered in the section of the cost report entitled **“Cost from preceding department.”** It is added to the costs of transfers received from the preceding department during the current period. An average unit cost for work done in preceding departments is then computed. The other portion of the beginning inventory cost, which was added by the Testing Department, is entered as a departmental cost to be added to other departmental costs incurred during the current period. Average unit costs are then computed.

The cost of production report of the Testing Department illustrates these procedures. The analysis of the beginning work in process inventory of this department lists 3,000 units in process with cost of \$5,400 from the preceding department, a labor cost of \$910, and \$800 for factory overhead.

The following costs pertain to February: cost from the preceding department, \$65,360; labor, \$34,050; factory overhead, \$30,018. Units completed and transferred to the Terminal Department totaled 36,000; 4,000 units are in process, 50% complete as to labor and factory overhead; 1,000 units were lost in process. The \$5,400 portion of the beginning work in process inventory, which is cost from the preceding department, is entered in the current month's cost report as work in process-beginning inventory. It is added to the \$65,360 of cost transferred from the Blending Department to the Testing Department during the month. The average unit cost for work done in the preceding department is \$1.726, computed by dividing total cost received from the Cutting Department, \$70,760 (\$5,400+\$65,360), by 41,000 units. These units consist of 3,000

units in the beginning work in process inventory and 38,000 units received during the month. The unit cost is a weighted average, since it considers all units and costs received from the preceding department. It is not the average of the two unit cost, \$1.80 and \$1.72. A simple average would not be accurate, since there are more units with a unit cost of \$1.72 (38,000 units) than with a unit cost of \$1.80 (3,000 units). The \$1.726 average unit cost for work done in the Blending Department pertains to the 41,000 units transferred to the Testing Department. However, because 1,000 of these 41,000 units were lost during processing in the Testing Department, the \$1.726 unit cost figure must be adjusted. The “Adjusted cost from preceding Department” is computed on the assumption that units lost can not be identified as coming from either units in process at the beginning or from units received during the period but proportionately from both sources. The total cost of the units transferred remains at \$70,760, but 40,000 units must now absorb this total cost causing an increase of \$0.043 in the cost per unit, from \$1.726 to an adjusted cost of \$1.769 [$\$70,760 / (41,000 \text{ units} - 1,000 \text{ lost units})$].

Departmental unit costs for labor and factory overhead are computed as explained in discussing the cost report of the Blending Department. The \$910 of labor in process at the beginning is added to labor put in process during the month, \$34,050. The total of these two labor costs, \$34,960, is divided by an equivalent production figure of 38,000 unit [$36,000 + (4,000 \times 1/2)$] to arrive at unit cost of \$0.920. The factory overhead unit cost of \$0.811 is the result of dividing total factory overhead, \$30,818 ($\$800 + \$30,018$) by equivalent production of 38,000 units. The department unit cost is the sum of these two unit costs, $\$0.920 + \0.811 , or \$1.731. The departmental unit cost of \$1.731 does not have to be adjusted for units lost, since the cost of any Testing work on lost units has automatically being absorbed in the departmental unit cost by excluding lost units from the equivalent production figure. Lost units are excluded on the assumption that the loss applies to good units and that the loss is within normal tolerance limits. The \$1.769” Adjusted Cost from Preceding Department” and the \$1.731 departmental unit cost are totaled to give a cumulative unit cost figure of \$3.50.

The total cost to be accounted for is \$136,538. Of the total, \$126,000 is the cost of the 36,000 units completed and transferred. The balance is cost assigned to the ending work

in process inventory. The following entry transfers the cost of the 36,000 units to the next department:

Work in process-Terminal Department	126,000
Work in process- Testing Department	126,000

All the above facts are reflected on the following cost of production report of Testing Department

ABC Company
Testing Department
Cost of Production Report Average Costing
For the Month of February, 2007

Quantity Schedule:

Units in process at beginning (1/2 labor and Factory overhead)	3,000	
Units received from preceding department	<u>38,000</u>	<u>41,000</u>
Units transferred to next department	36,000	
Units still in process (1/2 labor and Factory overhead)	4,000	
Units lost in process	<u>1,000</u>	<u>41,000</u>

Cost charged to the Department:

Cost from preceding department	Total cost	unit cost
Work in process-beginning inventory (3,000 units)	\$5,400	\$1.800
Transferred in during this period (38,000 units)	<u>65,360</u>	<u>1.720</u>
Total (41,000 units)	<u>\$70,760</u>	<u>\$1.726</u>

Cost added by department:

 Work in process-beginning inventory:

Labor	\$910
Factory overhead	800

Cost added during period

Labor	34,050	\$0.920
Factory overhead	<u>30,018</u>	<u>0.811</u>
Total cost added	\$65,778	\$1.731
Adjustment for lost units	—	<u>0.043</u>
Total cost to be accounted for	<u>\$136,538</u>	<u>\$3.500</u>

Cost Accounted for As follows:

Transferred to next department (36,000x\$3.500)		\$126,000
Work in process-ending inventory:		
Adjusted cost from preceding department		
(4,000x (\$1.726+\$0.043))	\$7,076	
Labor (4,000 x 1/2 x \$ 0.920)	1,840	
Factory overhead (4,000x 1/2 x \$ 0.811)	<u>1,622</u>	<u>10,538</u>
Total cost accounted for		<u>\$136,538</u>

Additional Computations:

 Unit cost from preceding department = $\frac{\$70,760}{41,000}$ = \$1.726 per unit

Equivalent production:

Labor and factory overhead: $36,000 + 4000 \div 2 = \underline{38,000}$ units

Unit costs:

Labor = $\$910 + \$34,050 = \$34,960$; $\$34,960 \div 38,000 = \underline{\$0.920}$ per unit

Factory overhead = $\$800 + \$30,018 = \$30,818$; $\$30,818 \div 38,000 = \underline{\$0.811}$ per unit

Adjustment for lost units:

Method No. 1 $\underline{\$70,760} = \1.769 ; $\$1.769 - \$1.726 = \underline{\$0.043}$ per unit
40,000

Method No. 2 $1,000 \text{ units} \times \$1.726 = \$1,726$; $\$1,726 \div 40,000 = \underline{\$0.043}$ per unit

Terminal Department: The accounting treatment for the Terminal Department is similar to that of the Testing department except that the Terminal department is the last department in ABC Co. and it transfers all of its products to the Finished Goods storeroom. To complete this discussion of operation for February, the cost of production report of the Terminal department is prepared on the basis of the following data: costs added by the department during the month are labor, \$33,140 and factory overhead, \$19,430. The units are accounted for as follows: completed and transferred to finished goods, 36,000 units; 3,000 units are still in process 1/3 complete as to labor and factory overhead, 1,000 units are lost in process. The cost of production report is as follows

ABC COMPANY
Terminal Department
Cost of Production Report – Average costing
For the Month of February, 2007

Quantity Schedule:

Units in process at beginning (1/4 labor and Factory overhead)	4,000	
Units received from preceding department	<u>36,000</u>	<u>40,000</u>
Units transferred to finished goods store room	36,000	
Units still in process (1/3 labor and factory overhead)	3,000	
Units lost in process	<u>1,000</u>	<u>40,000</u>

Cost charged to the Department:

Cost from preceding department	Total cost	unit cost
Work in process-beginning inventory (4,000 units)	\$14,400	\$3.60
Transferred in during this period (36,000 units)	126,000	3.50
Total (40,000 units)	<u>\$140,400</u>	<u>\$3.51</u>

Cost added to the Department:

Work in process beginning inventory:	
Labor	\$900
Factory overhead	550
Cost added during the period	

Labor	33,140	\$0.92
Factory overhead	<u>19,430</u>	<u>0.54</u>
Total cost added	\$54,020	\$1.46
Adjustment for lost units	—	<u>0.09</u>
Total cost to be accounted for	<u>\$194,420</u>	<u>\$5.06</u>
Cost accounted for as follows:		
Transferred to finished goods store room (36,000x\$5.06)		\$182,160
Work in process-ending inventory:		
Adjusted cost from preceding department (3,000x (\$3.51+0.09))	\$10,800	
Labor (3000x 1/3 x\$0.92)	920	
Factory overhead (3,000x1/3x\$0.54)	<u>540</u>	<u>12,260</u>
Total cost accounted for		<u>\$194,420</u>

Additional Computations:

Unit cost from preceding department = $\$140,400 \div 40,000 = \3.51 per unit

Equivalent production:

Labor and factory overhead = $36,000 + 3000 \div 3 = \underline{37,000}$ units

Unit costs:

Labor = $\$900 + \$33,140 = \$34,040$; $\$34,040 \div 37,000 = \underline{\$0.92}$ per unit

Factory overhead = $\$550 + \$19,430 = \$19,980$; $\$19,980 \div 37,000 = \underline{\$0.54}$ per unit

Adjustment for lost units:

Method No.1 $\$140,400 \div 39,000 = \3.60 ; $\$3.60 - \$3.51 = \underline{\$0.09}$ per unit

Method No.2 $1000 \times \$3.51 = \$3,510$; $\$3,510 \div 39,000 = \underline{\$0.09}$ per unit

The following entry transfers the cost of the 36,000 finished units to finished goods.

Finished Good182,160
 Work in process- Terminal Department182,160

Activity 5.1

- Explain the primary differences between the cost of production report of the originating department and the second department when weighted average method is used.
- How equivalent production units are computed when weighted average method is used.

5.22 First-in First-out (FIFO) costing

The First in First Out method may be used to account for beginning work in process inventory costs in process costing. Under this method, the beginning work in process inventory costs are separated from cost incurred in the in the current period and are not averaged with the additional new costs. This procedure gives one unit cost for units completed from the beginning work in process inventory and another for units started and finished in the same period. Under the First-in, first - out costing method, the cost of completing units in process at the beginning of the period is computed first, followed by the computation of the units started and finished within the period.

To illustrate the FIFO method, the February cost of production reports for ABC Company are presented, using the same data and assumptions as were used in the average costing illustration. A comparison of these reports with those illustrated for the average costing method indicates that the two methods do not result in significantly different unit costs, since in general, manufacturing operations in process cost type industries are more or less uniform from period to period.

Blending Department: When the February cost of production report of the Blending Department using the FIFO method is compared to the average costing report, the following difference are apparent;

1. Under FIFO costing, the beginning work in process inventory cost of \$4,440 is kept separate and is not broken down into its component parts as in the average costing.
2. Under FIFO costing, the degree of completion of the beginning work in process inventory must be stated in order to compute completed unit costs. Average costing does not require the degree of completion of the beginning work in process

Under FIFO costing, the cost of completing the 4,000 units in process at the beginning of February must be computed first. No additional materials were needed; but since these

units were only 50 %complete as to labor and factory overhead, more labor and overhead cost must be added to complete these units.

To determine costs expended in completing the units in the beginning Work in process inventory and to arrive at the cost of units started and finished within the current period, unit costs are computed for materials, labor, and factory overhead added during the period. Materials added during February, costing \$19,840, were sufficient to complete an equivalent production of 38,000 units. Of these 38,000 units, 34,000 were started and completed during the period, 3,000 units are in process at month end, with all the necessary materials, and 1,000 units were complete but still on hand. Therefore, the unit cost for materials is \$0.522 ($\$19,840/38,000\text{units}$).

The labor cost for November is \$24,180, and overhead cost is \$22,580. The labor and overhead unit costs are computed after determining the number of units that could have been completed from these costs. The labor cost and the overhead cost were sufficient to complete:

1. 50 % or 2,000 of the 4,000 units in the beginning inventory;
2. 34,000 units started and completed this period,
3. 1,000 units completed and still on hand and
4. $2/3$ or 2,000 of the 3,000 units still in process. Therefore, the equivalent production for labor and factory overhead is 39,000 units. The unit cost for labor is \$0.620 ($\$24,180/39,000$), and the unit cost for factory overhead is \$0.579 ($\$22,580/39,000\text{units}$).

In average costing, the cost of the unit transferred to the next department was computed by multiplying the number of units transferred by the final unit cost. Under FIFO costing units in process at the beginning must be completed first and will usually have a completed unit cost that is different from the unit cost for work started and finished during the period. Two separate computations determine the total cost transferred to the next department.

No additional materials were needed to complete the beginning work in process inventory. The cost of labor and factory overhead used during the period in completing the beginning inventory units is added to the \$4,440, already included as a cost of these units. Labor and overhead were added at unit costs of \$0.620 for labor and \$0.79 for factory overhead to complete the equivalent of 2,000 of the 4,000 units in process. The labor cost added was \$1,240 ($2,000 \times \0.620) and factory overhead was \$1,158 ($2,000 \times \0.579). The total cost of the 4,000 units completed and transferred was \$6,838 ($\$4,440 + \$1,240 + \$1,158$). The other 34,000 units were transferred at a unit cost of \$1.721 or at a total of \$58,517. The remaining \$5,685 cost to be accounted for is in work in process at the end of the period. The above facts are illustrated below in the form of cost of production report.

ABC Company
Blending Department
Cost of Production Report – FIFO Costing
For the Month of February, 2007

Quantity Schedule:

Units in process at beginning (all materials, 1/2 Labor and factory overhead)	4,000	
Units started in process	<u>40,000</u>	<u>44,000</u>
Units transferred to next department	38,000	
Units completed and on hand	1,000	
Units still in process (all materials, 2/3 labor and Factory overhead)	3,000	
Units lost in process	<u>2,000</u>	<u>44,000</u>

Cost charged to the Department:

Work in process-beginning inventory	<u>Total cost</u>	<u>unit cost</u>
Cost added by the department	\$4,440	
Materials	\$19,840	\$0.522
Labor	24,180	0.620
Factory overhead	<u>22,580</u>	<u>0.579</u>
Total cost added	<u>\$66,600</u>	1.721
Total cost to be accounted for	<u>\$71,040</u>	

Cost accounted for as follows:

Transferred to next department:		
From beginning inventory:		
Inventory cost	\$4,440	
Labor added ($4,000 \times 1/2 \times \0.620)	1,240	
Factory overhead added ($4,000 \times 1/2 \times \0.579)	<u>1,158</u>	<u>6,838^b</u>
Form current production		

Units started and finished (34,000x\$1.721)	<u>58,517*</u>	\$65,355
Work in process-ending inventory		
Completed and on hand (1,000 x \$1.721)	\$1,721	
Materials (3,000 x \$0.522)	1,566	
Labor (3,000 x 2/3 x \$0.620)	1,240	
Factory overhead (3,000 x 2/3 x \$0.579)	<u>1,158</u>	<u>5,685^c</u>
Total cost accounted for		<u>\$71,040^a</u>

*34,000 units x \$1.721 per unit=\$58,514 to avoid a decimal discrepancy, the cost transferred from current production is computed as follows.

$$\$71,040 - (\$6,838 + \$5,685) = \underline{\underline{\$58,517}} \quad [a - (b + c)]$$

Additional computation:

<u>Equivalent production</u>	<u>Materials</u>	<u>Labor and factory over head</u>
Transferred out	38,000	38,000
Less beginning inventory (all units)	<u>4,000</u>	<u>4,000</u>
Started and finished this period	34,000	34,000
Add: beginning inventory (Worked in this period)	0	2,000
Add: ending inventory:		
Completed and on hand	1,000	1,000
Still in process (worked in this period)	<u>3,000</u>	<u>2,000</u>
	<u>38,000</u> units	<u>39,000</u> units
Unit cost: Materials = $\frac{\$19,840}{38,000} = \underline{\underline{\$0.522}}$ per unit		
Labor and factory overhead = $\frac{\$22,580}{39,000} = \underline{\underline{\$0.579}}$ per unit		

The following entry transfer the total cost of the 38,000 units sent to the next department.

Work in process –Testing Department	65,355
Work in process- Blending Department	65,355

Testing Department: Although the cost transferred out of the Blending Department was the result of two separate computations, the total cost transferred into the Testing department is shown as only one amount in its cost report. The unit cost of \$1.72 is obtained by dividing the 38,000 total units received into the total cost received of \$65,355 (\$6,838+\$58,517). This procedure seems to cancel out the apparent advantages of the FIFO method and has been criticized by some writers.

The balance of the Testing department report is consistent with the FIFO method of costing. The beginning work in process inventory, valued at \$7,110, is shown in total and

is not broken down into its component parts. Labor and factory overhead costs needed to complete the units in the beginning inventory are added to this figure to determine the completed cost of these units which are transferred to the next department. The \$0.920 units cost for labor and the \$0.811 unit cost for factory overhead are computed by dividing the equivalent production figure of 37,000 units for labor and factory overhead into the labor cost of \$34,050 and factory overhead of \$30,018. The equivalent production figure of 37,000 units consists of

1. 2,000 units of beginning inventory completed
2. 33,000 units started and finished this period and
3. 2,000 of the 4,000 units in the ending work in process inventory.

The labor cost added to the beginning work in process inventory was \$1,840 ($2,000 \times \0.920), and factory overhead added was \$1,622 ($2,000 \times \0.811). These two amounts are added to the beginning inventory cost of \$7,110 to give total cost of \$10,572. This is the completed cost of the 3,000 units in the beginning work in process inventory transferred to the Terminal Department. The departmental units cost is \$1.731. As 1,000 units were during February, computation of the cost accumulated to the end of operations in the Testing Department requires an adjustment of \$0.046 for lost unit. This adjustment is determined by dividing the previous department's total cost of \$65,355 by the good units (37,000) of the period, resulting in an adjusted unit cost of \$1.766. In FIFO costing, the lost units must be identified as all units from the beginning inventory, all new units started during the period (assumed in this illustration), or a portion in each of these two categories. The identification is needed to determine which unit cost(s) should be adjusted. Such a determination in FIFO costing is also required in cases involving the addition of materials in departments subsequent to the first, when an increase in units and a resulting change in unit cost occur.

The Testing department completed and transferred 36,000 units, of which 3,000 units came from those in process at the beginning of the period.

The following is the cost of production report of the Testing Department under FIFO costing.

ABC COMPANY
Testing Department
Cost of Production Report – FIFO Costing
For the Month of February, 2007

Quantity Schedule:

Units in process at beginning (1/3 labor and factory overhead)	3,000	
Units received from preceding department	<u>38,000</u>	<u>41,000</u>
Units transferred to next department	36,000	
Units still in process (1/2 labor and factory overhead)	4,000	
Units lost in process	<u>1,000</u>	<u>41,000</u>

Cost charged to the Department:

	Total cost	unit cost
Work in process-beginning inventory	\$7,110	
Cost from preceding department:		
Transferred in during the month (38,000 units)	<u>\$65,355</u>	<u>\$1.720</u>
Cost added by department		
Labor	\$34,050	\$0.920
Factory overhead	<u>30,018</u>	<u>0.811</u>
Total cost added	\$64,068	\$1.731
Adjustment for lost units	—	<u>0.046</u>
Total cost to be accounted for	<u>\$136,533</u>	<u>\$3.497</u>

Cost accounted for as follows:

Transferred to next department		
From beginning inventory:		
Inventory cost	\$7,110	
Labor added (3,000x2/3x\$0.920)	1,840	
Factory overhead added (3,000x2/3x\$0.811)	<u>1,622</u>	\$10,572
From current production		
Units started and finished (33,000x\$3.497)	<u>115,435*</u>	\$126,007
Work in process-ending inventory		
Adjusted cost from preceding department		
(4,000x (\$1.72+\$0.046))	\$7,064	
Labor (4,000x1/2x\$0.920)	1,840	
Factory overhead (4,000 x 1/2 x \$0.811)	<u>1,622</u>	<u>10,526</u>
Total cost accounted for		<u>\$136,533</u>

*33,000 units x \$3.497 = \$115,401. To avoid a decimal discrepancy, the cost transferred from current production is computed as follows.

$$\$136,533 - (\$10,572 + \$10,526) = \underline{\underline{\$115,435}}$$

Additional computation

<u>Equivalent production</u>	<u>Labor and factory overhead</u>
Transferred out	36,000
Less beginning inventory (all units)	<u>3,000</u>
Started and finished in this period	33,000
Add beginning inventory-(Work this period)	2,000
Add ending inventory	<u>2,000</u>

37,000 units*Units cost:*

$$\text{Labor} = \frac{\$34,050}{37,000} = \underline{\$0.920} \text{ per unit}$$

$$\text{Factory overhead} = \frac{\$30,018}{37,000} = \underline{\$0.811} \text{ per unit}$$

Adjustment for lost units:

$$\text{Method No.1} = \frac{\$65,355}{38,000 - 1,000} = \$1.766; \$1.766 - \$1.720 = \underline{\$0.046} \text{ per unit}$$

$$\text{Method No. 2.} = 1,000 \text{ units} \times \$1.720 = \$1,720; \$1,720 \div 37,000 = \underline{\$0.046} \text{ per unit}$$

The following entry transfers the cost of the 36,000 units to the next department.

Work In process-Terminal Department	126,007
Work In process- Testing Department	126,007

The remaining \$10,526 cost to be accounted for is the ending work in process inventory which is computed in the conventional manner.

Terminal Department: To complete the illustration of FIFO costing, the cost of production report of the Terminal Department is prepared and as shown below.

ABC Company
Terminal Department
Cost of Production Report – FIFO Costing
For the Month of February, 2007

Quantity Schedule:

Units in process at beginning (1/4 labor & factory overhead)	4,000	
Units received from preceding department	<u>36,000</u>	<u>40,000</u>
Units transferred to finished goods store room	36,000	
Units still in process (1/3 labor & factory overhead)	3,000	
Units lost in process	<u>1,000</u>	<u>40,000</u>

Cost charged to the Department:

	<u>Total cost</u>	<u>unit cost</u>
Work in process-beginning inventory	<u>\$15,850</u>	
Cost from preceding department:		
Transferred in during the month (36,000 units)	\$126,007	\$3.500
Cost added by department		
Labor	\$33,140	\$0.921
Factory overhead	<u>19,430</u>	<u>0.540</u>
Total cost added	\$52,570	\$1.461
Adjustment for lost units	—	<u>0.100</u>
Total cost to be accounted for	<u>\$194,427</u>	<u>\$5.061</u>

Cost accounted for as follows:

Transferred to next department		
From beginning inventory:		
Inventory cost	\$15,850	
Labor added (4,000x3/4 x \$0.921)	2,763	
Factory overhead added (4000x3/4 x\$0.540)	<u>1,620</u>	\$20,233
From current production:		
Units started and finished (32,000 x \$5.061)	<u>161,933*</u>	\$182,166
Work in process-ending inventory		
Adjusted cost from preceding department (3,000x (\$3.50X+\$0.10)	\$10,800	
Labor (3,000 x 1/3 x\$0.92)	921	
Factory over he ad (3,000x1/3x\$0.540)	<u>540</u>	<u>12,261</u>
Total cost accounted for		<u>\$194,427</u>

*32,000 x \$ 5.061 per unit = \$161,952. To avoid a decimal discrepancy, the total cost transferred from current production is computed as follows.

$$\$194,427 - (\$20,233 + \$12,261) = \underline{\$161,933}$$

Additional computation:

<u>Equivalent production</u>	<u>Labor and factory overhead</u>
Transferred out	36,000
Less beginning inventory (all units)	<u>4,000</u>
Started and finished this period	32,000
Add beginning inventory (work this period)	3,000
Add ending inventory (work this period)	<u>1,000</u>
	<u>36,000</u> units

Unit costs:

$$\text{Labor} = \frac{\$33,140}{36,000} = \underline{\$0.921} \text{ per unit}$$

$$\text{Factory overhead} = \frac{\$19,430}{36,000} = \underline{\$0.540} \text{ per unit}$$

Adjustment for lost units

$$\text{Method No. 1} = \frac{\$126,007}{36,000 - 1000} = \$3.60; \$3.60 - \$3.50 = \underline{\$0.100} \text{ per unit}$$

$$\text{Method No. 2} = 1000 \times \$3.50 = \$3,500 = \frac{\$3,500}{35,000} = \underline{\$0.100} \text{ per unit}$$

Based on this report, the entry to transfer the cost of the 36,000 finished units is:

Finished Goods	182,166	
Work in process- Terminal Department		182,166

Activity 5.2

- Identify and explain the fundamental differences between the cost of production report of the first department and subsequent departments.
- How equivalent production units are computed under FIFO cost flow assumption.
- How does the FIFO method differ from the weighted average process costing?

5.3 Average costing versus FIFO costing

Both average costing and FIFO costing have certain advantages. It would be arbitrary to state that one method is either simpler or more accurate than the other. The selection of either method depends entirely upon management's opinion regarding the most appropriate and practical cost determination procedures. Each firm should select the method which offers reliable figures for managerial guidance.

The basic difference between the two methods concerns the treatment of beginning work in process inventory. The averaging method adds beginning work in process inventory cost to the cost from the preceding department and to materials, labor, and factory overhead costs incurred during the period. Unit costs are determined by dividing these costs by equivalent production figures. Units cost are transferred to the next department as one cumulative figure.

The FIFO method retains the beginning work in process inventory cost as a separate figure costs necessary to complete the beginning inventory units are added to this cost. The sum of these two cost totals is transferred to the next department. Units started and finished during the period have their own unit cost, which is usually different from the completed unit cost of units in process at the beginning of the period. The FIFO method thus separately identifies for management the current period unit cost originating in a

department. Unfortunately, these costs are averaged out in the next department, resulting in a loss of much of the value associated with the use of the FIFO method

If the FIFO method is used, units lost during a period must be identified as to whether they came from units in process at the beginning or from units started or received during the period. Also in computing equivalent production figures in FIFO costing the degree of completion of both beginning and ending work in process inventories must be considered.

The principal disadvantage of FIFO costing is that if several unit cost figures are used at the same time, extensive detail is required within the cost of production report, which can lead to complex procedures and even inaccuracy. Whether the extra detail yields more representative unit costs than the average costing method is debatable, especially in a firm where production is continuous and more or less uniform, and appreciable fluctuation in unit costs are not expected to develop. Under such conditions, the average costing methods leads to more satisfactory cost computation.

5.4 Difficulties Encountered in Process Cost Accounting Procedures

The following difficulties in using process costing may be encountered in actual practice

1. The determination of production quantities and their stages of completion present problems. Every computation is influenced by these figures. Since the data generally come to the cost department from operating personal often working under circumstances that make a precise count difficult, a certain amount of doubtful counts and reliable estimates are bound to exist. Yet, the data submitted form the basis for the determination of inventory costs.
2. Materials cost computations frequently require careful analysis. In the illustrations, material cost is generally part of the first department's cost. In certain industries, materials costs are not even entered on production reports. When materials prices are influenced by fluctuating market questions, the

materials cost may be recorded in a separate report designs to facilitate management decisions in relation to the materials market.

3. When units are lost due shrinkage, spoilage, or evaporation, the time when the loss occurs influences the final cost calculation. Different assumptions concerning the units to which the loss pertains would result in different department unit costs which, in turn, affect inventory costs, the cost of unit transferred, and the completed unit cost. Another consideration involves the treatment of cost attributable to avoidable loss as an expense of the current period. An increase in units resulting from materials added in departments subsequent to the first requires special consideration as well.
4. Industries using process cost procedures are generally of the multiple product type. Joint processing cost must be allocated to the products resulting from the processes. Weighted unit averages or other bases are used to prorate the joint costs to the several products. If units manufactured are used as a basis for cost allocation considerable difficulties arise in determining unit costs.

Management must decide whether economy and operational cost are compatible with increased information, based on additional cost computations and procedures. Some companies use both job order and process costing procedures for various purposes in different departments. The basis for using either method should be reliable production and performance data for product costing which, when combined with output, budget, or standard cost data, will provide the foundation for effective cost control and analysis.

Activity 5.3

- Discuss the advantages and disadvantages of Average and FIFO costing method.
- Identify and describe the difficulties encountered in process costing.

Summary

Process costing systems are used in industries like chemical processing, oil refining, and breakfast cereals to cost like or similar products or services. The key feature of process costing is the averaging of costs over a quantity (often large) of those like or similar units. Part of process costing was discussed in detail in the previous chapter and the main focus of this chapter is on the accounting procedures when there are beginning Work in process inventories. The existence of beginning Work in process inventories forces us to use either the weighted average or FIFO cost flow assumption.

The weighted – Average method of process costing computes unit costs by focusing on the total costs and the total equivalent production units completed to date and assigns this average costs to units completed and to units in ending Work in process inventory. The first-in, first –out (FIFO) method of process costing assigns unit costs of the earliest equivalent units available to units completed, and the unit costs of the most recent equivalent units worked on during the period to ending Work in process inventory.

The weighted average process costing system for costs transferred in from preceding department computes weighed average costs from preceding department by merging beginning Work in process and work done in the current period. The FIFO process costing system for costs transferred in from preceding department assigns transferred in costs in the beginning Work in process to units completed, and the costs from preceding department during the current period to ending Work in process inventory.

Self-check exercises:**Part I: Multiple choices**

Based on the following data answer question 1-5

The Millwood supply Company manufactures single product on a continuous plan in three departments using average costing method. On November 1, the Work in process inventory in department 2 was:

Cost in preceding department	\$13,130
Materials - department 2	None
Labor - department 2	\$500
Factory overhead- department 2	\$50
Units in process	5,000
Costs in department 2 during November were:	
Labor	\$14,200
Factory overhead	\$3,450

During November, 70,000 units were received from Department 1 at a unit cost of \$2.64. 64,000 units were completed in department 2; of which 60,000 were transferred to department 3 and 8,000 were on hand in department 2 at the end of the month. Four thousand were still in process, estimated to be one- half complete as to labor and factory overhead. The balance was lost within the department; its cost is to be absorbed by all finished and unfinished production in the department.

1. What is the equivalent unit of production for conversion costs?
 - a. 75,000 units
 - b. 70,000 units
 - c. 60,000 units
 - d. 68,000 units
2. What are the unit costs for Labor and Factory overhead?
 - a. \$0.210 per unit and \$ 0.050 per unit respectively
 - b. \$0.55 per unit and \$0.100 per unit respectively
 - c. \$3.50 per unit and \$0.25 per unit respectively
 - d. Some other amount

3. What is the total cost to be accounted for?
 - a. \$210,000
 - b. \$230,000
 - c. \$216,000
 - d. \$216,130
4. What is the total cost of ending Work in process inventory?
 - a. \$24,072
 - b. \$10,996
 - c. \$35,588
 - d. Some other amount
5. What is the adjustment for lost unit figure?
 - a. \$0.110 per unit
 - b. \$0.43 per unit
 - c. \$0.55 per unit
 - d. Some other amount

Part II: Work out

1. Datar Company uses three departments to produce a detergent. The Finishing department is the third and last step before the product is transferred to storage. All materials needed to give the detergent its final composition are added at the beginning of the process in the Originating department. Any lost units occur only at the beginning of the process. The following data for finishing department for October has been made available.

Production data:

In process, October 1, (labor and Factory overhead, $\frac{3}{4}$ complete)	10,000 gals
Transferred in from preceding department	40,000 gals
Finished and transferred to storage	35,000 gals
In process, October 31, (labor and factory overhead, $\frac{1}{2}$ complete)	10,000 gals

Additional data:

Work in process, October1:	
Cost from preceding department	\$38,000
Cost from this department:	
Labor	39,000
Factory overhead	<u>42,000</u>
Total Work in process, October1,	<u>\$119,000</u>
Transferred in during October	\$140,000
Cost added in this department:	
Labor	\$162,000
Factory overhead	<u>130,000</u>
Total cost added	<u>\$ 292,000</u>
Total cost to be accounted for	<u>\$551,000</u>

Required: prepare the cost of production report for Finishing department for the month ended October31,

- I. Under weighted Average method.
- II. Under FIFO method method.

CHAPTER SIX

Cost Allocation

Chapter learning objectives

After you have read and studied this chapter, you should be able to:

- ❖ Discuss the various purposes of cost allocations.
- ❖ Describe how the single- rate cost allocation method differs from the dual-rate method
- ❖ Explain how the choice of budgeted versus actual allocation rates changes the risks that managers face.
- ❖ Distinguish among direct, step down and reciprocal methods of allocating support department costs.
- ❖ Distinguish between the incremental and stand-alone cost allocation methods.
- ❖ Outline the consequences of the inappropriate use of an allocation base.

6.1 Introduction

Cost allocation is an inescapable problem in nearly every organization and in nearly every facet of accounting. It is an important topic because many of the costs associated with designing, producing, and distributing products and services are not easily identified with the products and services that are created. Although an introduction to overhead cost allocation is provided in chapter 3, the overall topic is much broader than using a predetermined overhead rate. The purpose of this chapter is to extend the chapter 3 discussion to include the concepts underlying costs to the various products and services produced.

More specifically, the chapter includes three main sections. The first section provides a discussion of the conceptual foundation for cost allocations including the purposes and underlying logic associated with the various allocation methods. Section two provides a brief discussion of two (single rate versus dual rates) approaches for assigning costs to cost objects. Finally, the third section extends the discussion of departmental overhead

rates to include the allocation from support (services) departments to producing departments. This part includes three methods for accomplishing the allocation process. Allocation of common costs is also part of this section.

6.2 Purpose of cost allocations

The purpose of cost allocations are closely related to the purposes of information systems. Cost allocations are needed to value inventory for external reporting purposes, for planning and monitoring the cost of activities and processes, and for various short-term and long-term strategic decisions. Some examples include decisions to “Make or Buy” sub-components and services, how to price products and services, when to add or discontinue various products and services and when to expand or contract the size of a segment of the company. The accounting treatment for all of these aspects will be discussed in detail in cost and management accounting II.

In addition, cost allocation methods are components of the overall performance evaluation system. Therefore, system designers must also carefully consider the motivational or behavioral aspects of the alternative cost allocation methods.

Exhibit 6.1 summary of purpose of cost allocation

Purpose	Illustrations
1. To provide information for economic decisions	• To decide whether to add a new airline flight • To decide whether to manufacture a component part of a television set or to purchase it from another manufacturer • To decide on the selling price for a customized products or service.
2. To motivate managers and other employees	• To encourage the design of products that are simpler to manufacture or less costly to service • To encourage sales representatives to push high-margin products or services
3. To justify costs or compute reimbursements	• To cost products at a “fair price” often done with government defense contracts • To compute reimbursements for a consulting firm that is paid a percentage of the cost savings resulting from the implementation of its recommendations

4. To measure income and assets for meeting external regulatory and legal reporting obligations	• To cost inventories for financial reporting to stock holders, bond holders, and so on (under generally accepted accounting principles inventor able costs include manufacturing costs but exclude Research & Development, Marketing, distribution, customer service costs. • To cost inventories for reporting to tax authorities.
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6.3 The supporting logic of cost allocation methods.

While allocating cost to different units or products or some other cost objects, there should be some criteria (or supporting logical) to be followed. Accordingly, the following are some of the supporting logic (criterion) used as guidelines while allocating costs to units and /or products and services.

- a) **Cause and effect:** From the perspective of the matching concept (i.e. Matching cost and benefits) it is logical to allocate a cost to the cost object (e.g. user, activity, department, product) based on a “**cause and effect**” relationship. The idea is to allocate the cost to whatever causes or drives the cost. Using this logic or criteria, managers identify the variable or variables that cause resources to be consumed. For example, managers may use hours as the variables when allocating the costs of a quality testing area to products. Cost allocations based on the cause-and-effect criterion are likely to be the most credible to the operating personnel.
- b) **Fairness or equity:** When the driver for a cost can not be identified, or identified easily, and then an allocation scheme perceived to be “**fair and equitable**” might be used. This criterion is often cited in government contracts when cost allocations are the basis for establishing a price satisfactory to the government and its supplier. Cost allocation here is viewed as a “**reasonable**” or “fair” means of establishing a selling price in the minds of the contracting parties.
- c) **Benefit received:** using this criterion managers, identify the beneficiaries of the outputs of the cost object. The costs of the cost object are allocated among the beneficiaries in proportion to the benefits each receives. For example, consider a corporate wide advertising program that promotes the general image of the

corporation rather than any individual product. The cost of this program may be allocated on the basis of division revenues. The higher the revenues, the higher the divisions allocated cost of the advertising program. The rationale behind this allocation is the belief that divisions with higher revenues apparently benefit from the advertising more than division with lower revenues and therefore, ought to be allocated more of the advertising costs.

- d) Ability to bear:** This criterion advocates allocating costs in proportion to the cost object's ability to bear them. An example is the allocation of corporate executive salaries on the basis of divisions operating income. The presumption is that the more profitable a given division is a greater ability it has to absorb corporate head quarter's costs

Activity 6.1

- Discuss how cost allocation is an inescapable task.
- Describe the purposes of cost allocation for different activates.
- Compare and contrast the supporting logic of cost allocation methods.

6.4. Allocating costs from One department to another (interdepartmental cost allocation)

The various functional areas with in a manufacturing facility are usually separated into two types of departments. These include producing (operational) departments and service departments

Producing departments: Include those departments or units where the central purposes of the organization are carried out. These departments involve in converting raw or direct materials into finished products. Examples of such departments or units would include the surgery department in a hospital, the undergraduate and graduate programs in a university, and production departments such as milling, assembly, and painting in a manufacturing company.

Service departments: A services department is a unit in an organization that is not involved directly in producing the organization's goods or services. However, a service department does provide a service that enables the organization's production process to take place. Some examples of service departments include purchasing, receiving and storage, engineering, power maintenance, packing, shipping, inventory control inspection and quality control. For example, the maintenance department in an automobile plant does not make automobiles but if it did not exist, the production process would stop when manufacturing machine broke down.

A service department such as maintenance department must exist in order for an organization to carry out its primary function. Therefore, the cost of running a service department is part of the cost incurred by the organization in producing goods and services. In order to determine the cost of those goods or services, all service department cost must be allocated to the production department in which the goods or services are produced. For this reason, the costs incurred in an automobile plant's maintenance department are allocated to all of the production departments that have machinery. Three key issues that arise when allocating costs from one department to another are:

1. Whether to use a single rate method or dual rate method
2. Whether to use budgeted rate or actual rate
3. Whether to use budgeted quantities or actual quantities

1. Single rate cost allocation versus dual rate cost allocation methods

a) The single rate cost allocation method

This method pools all cost in one cost pool and allocates these costs to cost object using the same rate per unit of the single allocation base. There is no distinction between costs in cost pool in terms of cost behavior (such as fixed costs versus variable costs). Under some circumstances, this simple approach can results in an unfair cost allocation among the using departments and may lead to inappropriate decisions.

b) The dual-rate cost allocation method

The dual rate cost allocation refers to using separate rates or allocations for fixed and variable service costs. This method classifies costs into two sub cost pools (available cost sub pool and a fixed costs sub pool). This method by allocating fixed costs and variable costs separately avoid fairness problem that is inherent in the case of single - rate allocation method.

Illustration

Suppose that Morgan industries which has a central computer department. This department has only two users-Micro computer division and peripheral equipment Division. The following data apply to the coming budget year.

→	Fixed costs of operating the computer facility (in the relevant range of 1,000 to 15,000 hours)	\$3,000,000
→	Total capacity available	15,000 hours
→	Budgeted long-run usage (quantity) in hours	
	Micro computer Division	8,000 hours
	Peripheral equipment division	<u>4,000 hours</u>
	Total	<u>12,000 hours</u>
→	Budgeted variable costs per hour	
	In the 1,000-15,000 hours relevant range	\$200 per hour used

Central compute Department's cost allocation under single rate method

Under single-rate method, the costs of the central computer department (assuming budgeted usage is the allocation base and budgeted rates are used) would be allocated follows:

→	Total cost pool = 3,000,000 + (12,000 budgeted Hours x \$200)	\$5,400,000
→	Budgeted usage	12,000 hours
→	Budgeted total rate = $\frac{\$ 5,400,000}{12,000 \text{ hours}}$	\$450 per hour used

Here in this approach, the allocation rate for each division will be the same i.e. \$450 per hour used.

The \$450 rate includes an allocated amount of \$250 per hour (\$3000,000 ÷ 12,000 hours) for the fixed cost of operating facility. These fixed costs will be incurred whether the computer runs its 15,000 hours capacity, its 12,000 hours budgeted usage, or even, say only 6,000 hours usage.

Central computer Department's cost allocation under dual rate method

When the dual-rate method is used, allocation bases for each different sub-cost pool must be chosen. Assume that the budgeted rates are used, and the allocation quantities chosen are budgeted usage for fixed costs and actual usage for variable costs. The total budgeted usage of 12,000 hours comprises 8,000 hours for the microcomputer Division and 4,000 hours for the peripheral equipment division.

The costs allocated to the divisions would be:

→ Microcomputer Division	$\left\{ \begin{array}{l} \text{Fixed cost} = \left(\frac{8,000 \text{ hours}}{12,000 \text{ hours}} \right) \times \$3,000,000 = \$2,000,000 \\ \text{Variable cost} \quad \quad \quad \$200 \text{ per hour used} \end{array} \right.$
→ Peripheral equipment Division	$\left\{ \begin{array}{l} \text{Fixed costs} \left(\frac{4,000 \text{ hours}}{12,000 \text{ hours}} \right) \times \$3,000,000 = \$1,000,000 \\ \text{Variable cost} \quad \quad \quad \$200 \text{ per hour used} \end{array} \right.$

Assumed further that, during the coming year the actual computer hours of 9,000 and 3,000 are used by micro computer division and peripheral equipment division respectively. The costs allocated to these two divisions would be:

Under the single rate method

Micro computer division = 9,000 hours x \$450 = \$4,050,000

Peripheral equipment division = 3,000 hours X \$450 = \$350,000

Under the dual rate method:

Micro computer division = \$2,000,000 + (9,000 hours x \$200) = \$3,800,000

Peripheral equipment division = \$1,000,000 + (3,000 hours x \$200) = \$1,600,000.

One obvious benefit of using the single rate method is the low cost of implementation. It avoids the often expensive analysis necessary to classify the individual cost items of a department into fixed and variable categories. However, the allocation of these costs on the basis of single rate may lead to a decision that may adversely affect the goal of the organization. For example, the microcomputer division uses an external vendor that charges \$360 per hour, when the central computer department has excess capacity. In the short-run, Morgan industries incurs an extra \$160 per hour because this external vendor is used (\$360. external purchase price per hour minus the \$200 internal variable costs per hour) instead of its own central computer Department. The decision made by micro computer division can decrease its own cost but the overall cost to the Morgan industries will increase.

An important benefit of the dual rate method is that it signals to division managers how variable costs and fixed costs behave differently. This important information could steer division managers into making decisions that benefit the corporation as well as each division. For example, it would signal that using external vendor who charges more than \$200 per hour could result in Morgan industries being worse off than if it had used its own central computer department which has a variable cost of \$200 per hour.

2. Budgeted versus Actual rates

The decision whether to use budgeted cost rates or actual cost rates affects the level of uncertainty that user departments face.

a) Budgeted rates :

Let the user departments know the cost rates they will be charged in advance. Users are then better equipped to determine the amount of the service to request and, if the option exists, whether to use the internal department source or an external vendor. Budgeted rates also help to motivate the manager of the support department (for example, the

central Department) to improve efficiency, not the user department, bears the risk of any unfavorable cost variances. Because the user department, do not pay for any costs that exceed the budgeted rates.

The managers of support department likely would view the budgeted rates negatively, especially when unfavorable cost variances occur due to price increase of his or her control. Thus, to overcome this problems, some organizations are recognizing that it may not always be best to impose the risk of variance from budgeted amount completely on the support department (as when costs are allocated using budgeted rates) or completely on the user departments (as when costs are allocated using actual rates). For example, the two departments may agree to share the risk (through an explicit formula) of a large, uncontrollable increase in the price of materials used by the support department.

- b) Actual rates when actual rates are use the user department will not know the rates charged until the end of the period

3. Budgeted versus actual usage allocation bases:

Should a service department allocate its actual costs to operating departments or should it allocate its budgeted costs? The answer is that budgeted costs should be allocated. The problem with allocating actual costs is that it burdens the operating department with the inefficiencies of the services departments' managers. If actual costs are, then any lack of cost control on the part of the service departments' manger is simply buried in a routine allocation to other departments.

When actual usage is the allocation base, user division will not know in advance the amount of cost to be allocated to them until the end of the budget period. When budget usage is the allocation base, user divisions will know in advance their allocated costs. This information helps the user divisions with short - run and long - run planning. The main justification given for the use of budgeted usage to allocate fixed costs relates to long-run planning. Organizations commit to infrastructure costs (such as fixed costs of a support department) on the basis of long-run planning horizon; the use of budgeted usage to allocate these fixed costs is consistent with long-run horizon.

If fixed costs are allocated on the basis of budgeted long-run usage, some managers may be tempted to under estimate their planned usage. In this way, they will bear a lower percentage of the total costs (assuming all other managers do not similarly under estimate their usage). Some organizations offer rewards in the form of salary increase and promotions to managers who make accurate forecasts of long-run usage (this is the carrot approach), and or impose cost penalties for under predicting long-run usage (this is the stick approach).

Activity 6.2

- Distinguish between production departments and service departments.
- Describe the conditions and purposes of using either the single rate or the dual rate cost allocation methods.
- List down the advantages and benefits of using a budgeted rate.
- Service departments should allocate its actual costs. Do you agree or disagree? Discuss.

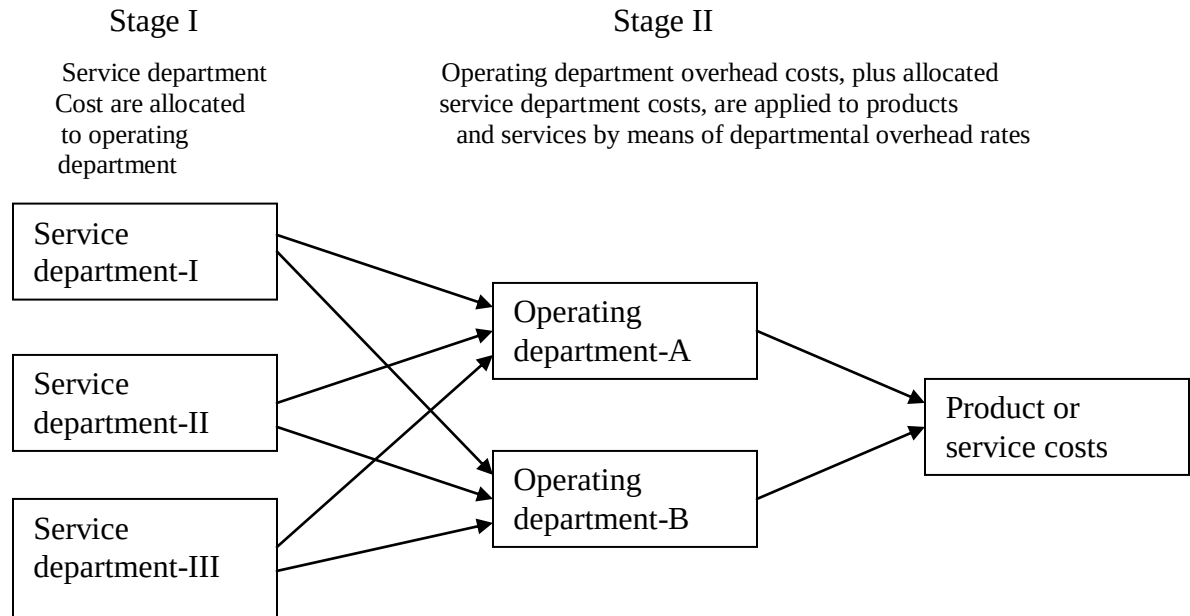
6.5 Support (service department) cost Allocation

6.5.1 Types of Relationships between Service Departments

First stage allocations may include self-services and reciprocal services between departments as well as services to producing departments. Self-services refers to situations where service departments use of their own services. For example, the power department uses some power and the maintenance department requires some maintenance. Reciprocal service refers to situations where two or more services departments provide services to each other. For example the power department provides electric power to the maintenance department and the maintenance department provides repair and maintenance to the power department. These relationships affect the choice of

allocation methods as well as the accuracy of the first stage cost allocations. The effect of service department costs on products and services is shown in the following exhibit.

Exhibit 6.1. Effect of service department costs on products and services



6.5.2 Types of support (Service) Department cost allocation

The allocation of services department costs to operating departments is very essential since the amount of costs allocated are presumed to represent each department's "fair share" of the costs of services provided for it and, the allocations are included in the performance evaluations of operating departments and also included in determining their individual profitability. There are three alternative approaches for allocating service department costs. These include :

1. The direct method
2. the step down or sequential method
3. The reciprocal method

1. Direct allocation Method

Under the direct method, each service department's costs are allocated among only operating (production) departments that consume part of the service department's output. The direct method simplifies the allocation process by ignoring self-services and reciprocal services between service departments. The benefit of direct method is its simplicity. Although this is the simplest method, it is also the least accurate method. Its disadvantage is failure to recognize reciprocal services provided among support departments. Notice that the proportion of each service department's costs to be allocated to each operating department is determined by the relative proportion of the service department's output consumed by the operating departments.

To illustrate, consider Melbreg industries which manufactures engines used in electric power generating plants. Melbreg has two support departments (plant maintenance and information system) and two operating departments (Machining and Assembly) in manufacturing facility. The two support departments at Melbreg provide reciprocal support to each other as well as to the two operating departments. The data for our example is listed in exhibit 6.2.

Exhibit 6.2. Data for allocating support department costs at Melbreg for 2001

	<u>Support Department</u>		<u>Operating Department</u>		<u>Total</u>
	<u>Plant Maintenance</u>	<u>Information system</u>	<u>Machining</u>	<u>Assembly</u>	
Budgeted factory Over head costs Before any inter Department cost					
Allocations	\$600,000	\$116,000	\$400,000	\$200,000	\$1,316,000
Support work furnished:					
By plant maintenance:					
Budgeted labor hours	—	1,600	2400	4000	8,000
Percentage	—	20%	30%	50%	100%
By information system:					
Budgeted computer time	200	-	1,600	200	2000
Percentage	10%	-	80%	10%	100%

As it can be see from the above exhibit, the plant maintenance department provides a total of 8,000 hours of support work: 20% $(1,600 \div 800)$ goes to the information systems support department, 30% $(2,400 \div 8,000)$ to the machining department and 50% $(4,000 \div 8,000)$ to the assembly department. On the basis of the above, the direct method of allocating support department costs for year 2001 at Melbreg industries is given as follows in exhibit 6.3.

Exhibit 6.3 direct method of allocating support department costs for 2001 at Melbreg.

	<u>Support Department</u>		<u>Operating Department</u>		
	Plant Maintenance	Information systems	Machining	Assembly	Total
Budgeted factory Over head costs Before any inter Departmental Cost allocation	\$600,000	\$116,000	\$400,000	\$200,000	\$1,316,000
Allocation of plant Maintenance $(3/8, 5/8)^1$	<u>$(600,000)$</u>		225,000	375,000	
	<u>\$0</u>				
Allocation of information Systems $(8/9, 1/9)^2$		<u>$(116,000)$</u>	<u>103,111</u>	<u>12,889</u>	
Total budgeted factory Over head of operating Departments		<u>\$0</u>	<u>\$ 728,111</u>	<u>\$587,889</u>	<u>\$1,316,000</u>

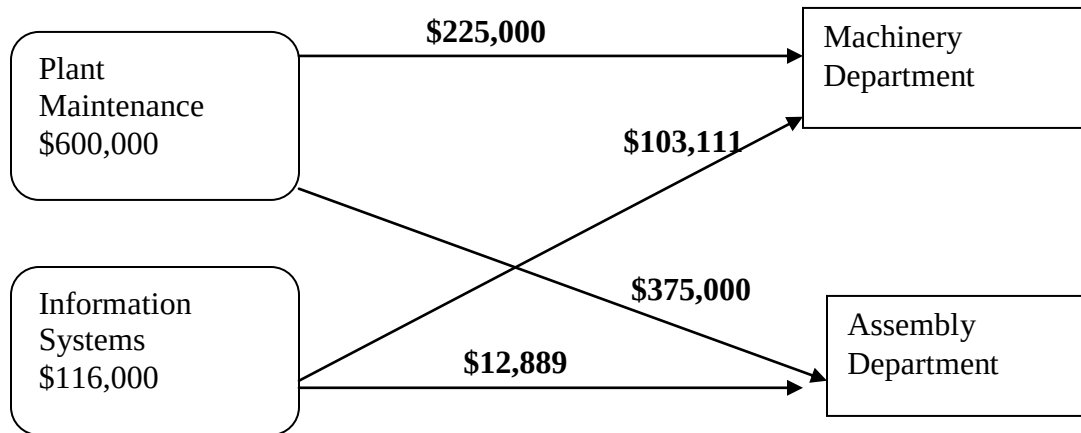
1. Base is $(2,400 + 4000)$, or 6,400 hours; $2400 \div 6,400 = 3/8$; $4000 \div 6,400 = 5/8$

2. Base is $(1,600 + 200)$, or 1,800 hours; $1,600 \div 1,800 = 8/9$; $200 \div 1,800 = 1/9$

NB → the base used to allocate plant maintenance is the budgeted total maintenance labor hours worked in operating departments.

$2,400 + 4,000 = 6,400$ hours. This excludes the 1,600 hours of support time provided by plant maintenance to information systems.

→ The based used for allocation of information systems costs is $1,600 + 200 = 1,800$ hours of computer time which excludes the 200 hours of support time provided by information systems to plant maintenance.

Summary of direct method of allocating support department costs**2. Step- down allocation method**

As Stated above, the direct method ignores the provision of services by one service department to another services department. This short coming is overcome partially by the step-dawn method of service department cost allocation. This method partially reorganizes the service rendered by service department to other services department. Under this method the managerial accountant first chooses a sequence in which to allocate the service departments costs. A common way to select the first service department in the sequence is to choose the one that serves the largest number of other service departments. The service departments are ordered in this manner, with the last service department being the one that serves the smallest number of other service departments.

Alternatively, there are two approaches for ranking service departments.

Approach 1: Rank support departments on the percentage of the support departments' total support provided to other support departments. The support department with the highest percentage is allocated first. The support department with the lowest percentage is allocated last. In our Melberg example, the chosen order would be;

	<i>Percentage of total support Service provided to other support department</i>
1. Plant maintenance	20%
2. information systems	10%

Approach 2: Rank support departments on the total dollar of service provided to other support departments. In our Melberg industries example, the chosen order would be:

***Dollar amount of total service provided
to other support departments***

1. Plant Maintenance	\$120,000
2. Information systems	\$116,000

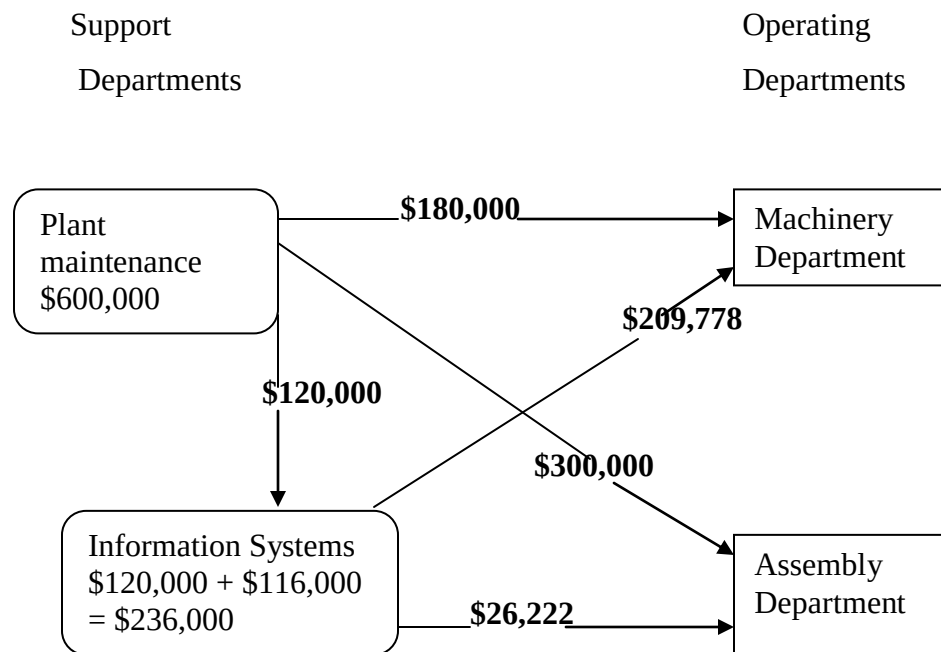
Under the step down method, once a support department's costs have been allocated no subsequent support department costs are allocated or circulated back to it. Thus, once the plant maintenance department costs are allocated they receive no further allocations from other (lower-ranked) support departments. The result is that the step down method does not recognize the total services that support departments provide to each other exhibit 6.4 reflects this fact.

Exhibit 6.4. Step down method of allocating support department costs for 2001 at Melberg industries

	<u>Support Departments</u>		<u>Operating Departments</u>		
	Plant Maintenance	Information Systems	Machining	Assembly	Total
Budgeted Factory overhead costs before any interdepartmental cost allocations	\$600,000	\$ 116,000	\$400,000	\$200,000	\$1,316,000
Allocation of plant Maintenance: (2/10, 3/10, 5/10) ¹	(600,000) <u>\$0</u>	120,000	180,000	300,000	
Allocation of Information Systems: (8/9, 1/9) ²		(236,000) <u>\$0</u>	209,778	26,222	
Total budgeted Factory Overhead of operating Departments			<u>\$789,778</u>	<u>\$526,222</u>	<u>\$1,316,000</u>

1. Base is $(1,600 + 2,400 + 4,000)$ or 8,000 hours; $1,600 \div 8,000 = 2/10$; $2,400 \div 8,000 = 3/10$; $4,000 \div 8,000 = 5/10$
2. Base is $(1,600 + 200)$ or 1,800 hours; $1,600 \div 1,800 = 8/9$; $200 \div 1,800 = 1/9$

The above exhibit (exhibit 6.4) shows the step down method where the plant maintenance cost of \$600,000 is allocated first; \$120,000 allocated to information systems (20% of \$600,000); \$180,000 to Machining (30% of \$600,000); and \$300,000 to assembly (50% of \$600,000). The cost of Information systems after allocation total \$236,000 (\$116,000 + \$120,000 from the first round allocation). This \$236,000 amount is then allocated among the two operating departments \$209,778 ($8/9 \times \$236,000$) to machining and \$26,222 ($1/9 \times \$236,000$) to assembly. Summary of step-down method of allocating support department costs.



3. Reciprocal allocation method

The reciprocal allocation method allocates costs by explicitly including the mutual services provided among all support departments. This method is more accurate than the

other two methods because it fully recognizes self-services and reciprocal services between service departments. For example, the plant maintenance department maintains all the computer equipment in the Information systems department. Similarly, Information system provides data base support for plant maintenance. The reciprocal allocation method enables us to incorporate interdepartmental relationships fully into support department cost allocation by fully accounting for the mutual provision of services. However, this method is more involved because it requires the solution to simultaneous equations. The reciprocal method requires three steps:

Step 1: Express support department costs and support department reciprocal relationships in the form of linear equations as shown below.

Let: **PM**, be the complete reciprocated costs of plant maintenance.

IS, be the complete reciprocated costs of information systems. We then express the data in Exhibit 6.2 as follows:

$$PM = \$600,000 + 0.1 IS \dots\dots\dots 1$$

$$IS = \$116,000 + 0.2 PM \dots\dots\dots 2$$

0.1 IS: the percentage of the information systems work used by plant maintenance.

0.2 PM: is the percentage of the plant maintenance work used by information systems.

The complete reciprocated costs in equations (1) and (2), means the support department's own costs plus any interdepartmental cost allocation

Step 2: Solve the system of simultaneous equations to obtain the complete reciprocated costs of each support department. Where there are two support departments the following substitution approach can be used. Thus by substituting equation (2) into equation (1) we have the following results:

$$PM = \$600,000 + [0.1 (\$116,000 + 0.2 PM)]$$

$$= \$600,000 + \$11,600 + 0.02 PM$$

$$0.98 PM = \$611,600$$

$$\therefore PM = \underline{\underline{\$624,082}}$$

By substituting the above result in to equation (2):

$$\begin{aligned} \text{IS} &= \$116,000 + 0.2 (\$624,082) \\ &= \$116,000 + \$124,816 \\ \therefore \text{Is} &= \underline{\underline{\$240,816}} \end{aligned}$$

Step 3: Allocate the complete reciprocated costs of each support department to all other departments (both support and operating departments). On the basis of the usage proportions (based on total units of services provided to all departments). For example, consider the Information systems department with a complete reciprocated cost of \$240,816. This amount would be allocated as follows.

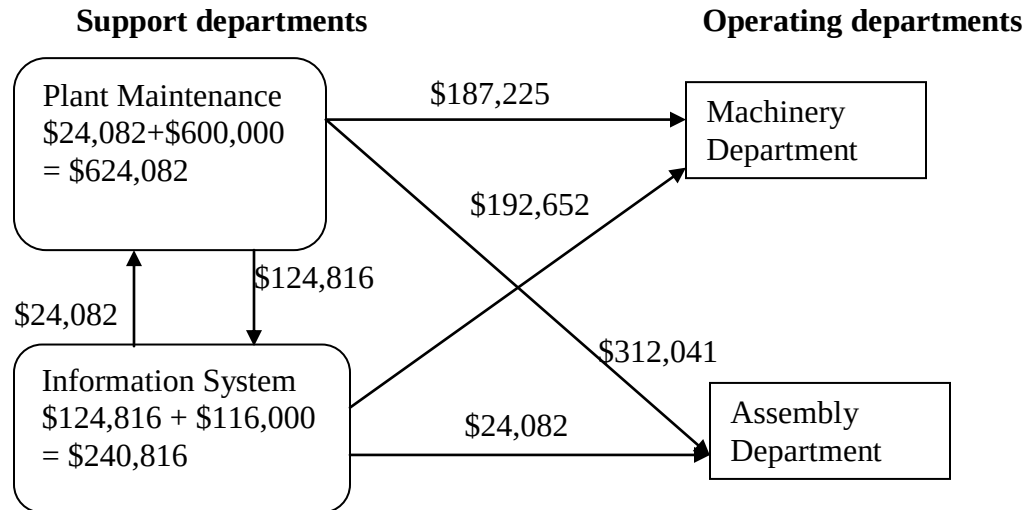
To plant maintenance (1/10 x 240, 816) =	\$24,082
To Machining (8/10 x \$240,816) =	192,653
To Assembly (1/10 x \$240,819) =	<u>24,082</u>
Total	<u><u>\$240,817</u></u>

Exhibit 6.5 Reciprocal method of allocating support Department costs for 2001 at Melberg industries:

	<u>Support Departments</u>		<u>Operating Departments</u>		
	Plant Maintenance	Information System	Machining	Assembly	Total
Budgeted Factory overhead costs before any interdepartmental cost allocations	\$600,000	\$116,000	\$400,000	\$200,000	\$1,316,000
Allocation of plant Maintenance:					
(2/10, 3/10, 5/10) ^a	(624,082)	124,816	187,225	312,041	
Allocation of Information Systems:					
(1/10, 8/10, 1/10) ^b	<u>24,082</u>	<u>(240,816)</u>	<u>192,652</u>	24,082	
	<u>\$0</u>	<u>\$ 0</u>			
Total budgeted Factory Overhead of operating Departments			<u>\$779,877</u>	<u>\$536,123</u>	<u>\$1,316,000</u>

1. Base is $(1600 + 2400 + 4,001)$ or 8000 hours; $1,600 \div 8,000 = 2/10$; $2,400 \div 8000 = 3/10$; $4,000 \div 8000 = 5/10$
2. Base is $(200 + 1600 + 200)$, or 2,000 hours; $200 \div 2,000 = 1/10$; $1,600 \div 2000 = 1/10$; $200 \div 2000 = 1/10$

Summary of reciprocal cost allocation method:



One source confusion to some managers using the reciprocal cost allocation method is, why the reciprocated cost of support departments, \$864, 898 ($\$624,082 + \$240,816$) exceeds the amount of \$ 716,000 ($600,000 + \$116,000$). This is because, the method take into account that the allocation of support costs will be to all departments using its services and not just to the operating departments. The excess of \$148,898 (\$24,082 for plant maintenance and \$124,816 for information systems) is the total costs that are allocated among support departments. The total costs allocated to the operating departments under this method are still \$716,000.

Activity 6.3

- Describe the type of relationship between the departments within an organization.
- Distinguish the three methods of allocating service department cost to users?
- How is the sequence of the service departmental cost allocation is determined in the step-down method?
- How is the service (Support) department cost allocation methods ranked in terms of product cost accuracy?

6.5.3 Allocation of common costs

A common cost is a cost of operating a facility, activities, or like cost objects that is shared by two or more users. Consider Alemu Alaka a senior student in Arba Minch University who has been invited to a job interview with an employer in Addis. After considering the costs and benefits Alemu decided to use the Ethiopian Air way. The round trip Arba Minch -Addis air fare cost is birr 1,200. A week prior to leaving, Alemu is also invited to an interview with an employer in Dire Dawa. The Arba Minch - Dire Dawa round - trip air fare costs is birr 800. Alemu Decided to combine the two recruiting steps into a Arba Minch- Addis - Dire Dawa trip that will costs birr 1,500 in air fare. The birr 1,500 is a common cost that benefits both prospective employers. Two methods for allocating this common costs between the two prospective employers are now discussed; the stand alone method and the incremental method

a Stand-alone cost allocation method

The stand alone cost-allocation method uses information pertaining to each user of a cost object as a separate operating entity to determine the costs allocation weights. For the airfare common cost of \$1,500, information about the separate (stand-alone) round - trip air fares (birr 1,200 and birr 800) is used to determine the allocation weights. The common cost allocation under standalone method would be:

$$\text{Addis employer} = \frac{\text{Birr } 1,200}{\text{Birr } 1,200 + \text{birr } 800} \times \text{Birr } 1,500 = 0.60 \times \text{birr } 1,500 = \underline{\underline{\text{Birr } 900}}$$

$$\text{Dire Dawa employer} = \frac{\text{Birr } 800}{\text{Birr } 800 + \text{birr } 1,200} \times \text{birr } 1,500 = 0.40 \times \text{Birr } 1,500 = \underline{\underline{\text{Birr } 600}}$$

b Incremental cost Allocation method

The incremental cost - allocation method ranks the individual cost objects and then uses this ranking to allocate costs among those cost objects. The first ranked cost object is termed the primary party and is allowed costs up to its costs as a standalone entity (user). The second ranked costs object is termed the incremental party and is allowed the additional cost that arise from there being two users instead of only the primary user.

Consider again Alemu and his birr 1,500 air fare cost. Assume the Addis employer is viewed as the primary party Alemu's rationale was that he had already committed to go to Addis before accepting the invitation to interview in Dire Dawa. The Addis employer is allocated the full Arba Minch – Addis air fare. The non-allocated part is then allocated to the Dire Dawa employer as follows:

Party	Cost allocated	Cost remaining to be allocated to other parties
Addis (primary)	Birr 1,200	Birr 300 (birr 1,500 –birr 1,200)
Dire Dawa (incremental)	300	0

Had the Dire Dawa employer been chosen as the primary party, the cost allocations would have been Dire Dawa birr 800 (the stand - alone Arba Minch –Dire Dawa round trip air fare) and Addis Birr 700 (Birr 1,500-birr 800).

Under the incremental method, the primary party typically receives the highest allocation of the common costs. Not surprisingly, most users in common costs situations purpose themselves as the incremental party. In some cases, the incremental party is a newly formed “organization” such as a new product line or a new sales territory.

Activity 6.4

- Distinguish between Stand-alone and Incremental cost Allocation methods.

Summary

In manufacturing industries, departments are classified into production and service department. A production department is one that is engaged in the actual manufacture of the product by changing the shape, form or nature of material worked upon or by assembling the parts into finished product. A service department on the other hand, is one which is rendering a service to production departments. Service departments are not involved directly in producing an organization's final output of goods or services, but they do provide essential services in an organization. Thus in order to determine the full cost of the organization's final services or goods, service department costs are allocated to the departments directly involved in producing the organization's final output. The supporting logic or criterion that guide cost allocation are "cause and effect" relationship between the cost and cost driver, Fairness or equity, Benefits received and Ability to bear. Three methods are used commonly in practice, the direct method, step - down method and the reciprocal method. These methods may be combined with either the single or the dual allocation approach in which variable and fixed costs are allocated separately.

Self-test Exercises:**Part I: Multiple Choices:**

Based on the following data answer question 1-5

The Solomon Company has two service departments S1 and S2, and two producing departments P1 and P2. S1 allocates cost based on the number of employees in each department and S2 allocates costs based on the cubic feet of water used in each department. Departmental data for the month are as follows:

	<u>Support Department</u>		<u>Operating Department</u>		
	S1	S2	P1	P2	Total
Budgeted factory Over head costs Before any inter Department cost					
Allocations	\$100,000	\$200,000	\$400,000	\$600,000	\$1,300,000
Support work furnished:					
By S1:					
Budgeted no. of employees hours	—	20	40	20	80
Percentage	—	25%	50%	25%	100%
By S2:					
Budgeted Cubic feet of water	10	-	40	60	110
Percentage	9.09%	-	36.36%	54.55%	100%

- Using the direct method of allocating service department costs, the service department cost allocated S1 would be;
 - Zero
 - (20/80) (\$100,000)
 - (10/110) (\$200,000)
 - None of these
- Using the direct method, the service department costs allocated to department P1 would be:
 - (40/100) (\$100,000) + (40/110) (\$200,000)
 - (40/80) (\$100,000) + (40/110) (\$200,000)
 - (40/60) (\$100,000) + (40/100) (\$200,000)
 - (40/110) (\$100,000) + (40/800) (\$200,000)

3. Using the step –down method, the costs of
 - a. Department S1 should be allocated first
 - b. Department S2 should be allocated first
 - c. Both departments should be allocated ignoring the reciprocal services
 - d. a and b
 - e. b and c
4. Now, ignore your answer to the previous questions and assume that you are asked to allocate department S2's cost first using the step- down method. The cost allocated to S1 would be:
 - a. $(10/120)$ (\$200,000)
 - b. $(10/110)$ (\$200,000)
 - c. $(10/1000)$ (\$200,000)
 - d. $(20/80)$ (\$200,000)
5. Now, ignore your answer to the previous questions and assume that you are asked to allocate department S1's cost first using the step- down method. The cost allocated to S2 would be:
 - a. Zero
 - b. $(20/100)$ (\$100,000)
 - c. $(10/120)$ (\$100,000)
 - d. $(20/80)$ (\$100,000)
 - e. $(10/110)$ (\$100,000)
6. One of the following is the correct order to put the different cost allocation methods based on their efficiency;; from least to best:
 - a. Step- down; Direct; Reciprocal
 - b. Reciprocal; Direct; Step- down
 - c. Direct; Step- down; Reciprocal
 - d. Reciprocal; Step-down; Direct
 - e. None of the above
7. One of the following is not among the purposes of cost allocation;
 - a. Provide information for decision making
 - b. Motivate workers for better production

- c. Provide information for proper income measurement
- d. Provide information for setting selling price
- e. All of the above
- f. None of the above

Based on the following information answer question 8-10

Aroon Heather, a graduating senior at a university near San Francisco, received an invitation to visit a prospective employer in New York. A few day later, she received an invitation from a prospective employer in Chicago. She decided to combine her visits, traveling from San Francisco to New York, New York to Chicago, and Chicago to San Francisco. Heather received job offers from both companies. She was puzzled about how to allocate her travel costs between the two employers.

She gathered the following data:

Regular round – trip fares with no stopovers

San Francisco to New York	\$1,400
San Francisco to Chicago	\$1,100

Heather paid \$1,800 for her three- leg flight (San Francisco to New York, New York to Chicago, Chicago to San Francisco)

- 8. The cost allocated between New York and Chicago employer using the stand alone allocation method should be
 - a. \$1,400 and \$ 1,100 respectively
 - b. \$1,008 and \$792 respectively
 - c. \$1,100 and \$ 1,400 respectively
 - d. Some other amount
- 9. Assume the New York employer is the primary party, the cost allocated should be
 - a. New York employer \$ 1,008 and Chicago employer \$792
 - b. New York employer \$1,400 and Chicago employer \$400
 - c. New York employer \$400 and Chicago employer \$ 1,400
 - d. Some other amount
- 10. Assume the Chicago employer is the primary party, the cost allocated should be
 - a. Chicago employer \$1,100 and New York employer \$700
 - b. Chicago employer \$700 and New York employer \$1,100

- c. Chicago employer \$1,008 and New York employer \$792
- d. Some other amount

Part II: Exercises

1. Midroc business group has a power plant designed and built to serve its three factories. Data for 2008 are as follows:

Factory	<u>Usage in kilowatt-hours</u>	
	Budget	Actual
Wanza	100,000	80,000
Midroc Gold	60,000	120,000
ELFORA	40,000	40,000

Actual fixed costs of the power plant were \$1,000,000 in 2008; actual variable costs, \$2,000,000

Required:

- a. Calculate the amount of power costs that would be allocated to Midroc Gold using a single- rate method.
- b. Calculate the amount of power costs that would be allocated to Midroc Gold using a dual- rate method.
2. Lee Company has a small factory with two service departments and two operating departments. The service departments, Power and Maintenance provide support to the operating departments and also use some of their own services. Budget estimates are provided in the following schedule:

	<u>Support Department</u>		<u>Operating Department</u>		Total
	Power	Maintenance	Cutting	Assembly	
Budgeted factory					
Over head costs					
Before any inter					
Department cost					
Allocations	\$100,000	\$40,000	\$200,000	\$160,000	\$500,000
Support work furnished:					
By Power:					
Budgeted kilowatt					
Hours	—	100	600	200	900
Percentage	—	11.11%	66.67%	22.22	100%
By Maintenance					
Budgeted Maintenance					
hours	20	-	150	100	270
Percentage	7.41%	-	55.55%	37.04%	100%

NB. Power costs are allocated using kilowatt hours. Maintenance costs are allocated using maintenance hours.

Required: Allocate service department costs using

- a. The direct method
- b. The step- down method

3. Phoenix Consulting provides outsourcing services and advice to both government and corporate clients. For costing purpose, Phoenix classifies its departments into two support departments (Administrative/Human Resources and Information systems) and two operating departments Government consulting and corporate consulting). For the first quarter of 2005 phoenix incurs the following costs in its four departments:

Administrative/ Human Resource (A/H)	\$600,000
Information systems (IS)	\$2,400,000
Government Consulting (GOVT)	\$8,756,000
Corporate Consulting (CORP)	\$12,542,000

The actual level of support relationship among the four departments for the first quarter of 2005 was:

		Used by			
		A/HR	IS	GOVT	CORP
Supplied	A/HR	—	25%	40%	35%
By	IS	10%	—	30%	60%

The Administrative/ Human Resource support are based on headcount. The Information Systems support percentage is based on actual hours of computer time used.

Required: Allocate the two support department costs to the two operating departments using:

- a. Direct method
- b. Step-down method (allocate Administrative/ Human Resource first)
- c. Step-down method(allocate Information systems first)
- d. Reciprocal method

Chapter 7

Spoilage, Rework and Scrap

Learning objectives

- Distinguish among spoilage, reworked, and scrap
- Illustrate costing procedures for scrap, spoiled goods and Reworked units Specific job.
- Account for spoilage in process costing using the first –in –first out (FIFO) method and weighted average method.
- Account for spoilage in job costing
- Account for rework in job costing
- Account for scrap

7.1 Introduction

This chapter discusses the topic of accounting for manufacturing defect in the manufacturing process are spoilages for and scrap companies develop accounting techniques for identifying and quantifying the cost of manufacturing defect in order to properly reload and report the value of inventory cost of goods sold, and to develop and analyze proposed or implemented cost- reduction and quality management strategies

7.2 Distinction among spoilage, rework, and scrap

Three categories of cost that result from defects in the manufacturing process are: spoilage, rework, and scrap. The term spoilage refers to unit that do not meet the required production specifications. Spoiled units would typically be recycled or sold as seconds. The term rework also refers to units that do not meet the required production specifications, but rework unit are being repaired (reworked) so that they can be sold as normal finished units. The term scrap refers to left –over or residual materials that results from the manufacturing process.

Activities 7.1 distinguish among spoilage, reworked units, and scrap.

7.3 Accounting procedures for normal and abnormal spoilage

A key element of accounting for spoilage is distinguishing between normal and abnormal spoilage. Normal spoilage is spoilage inherent in a particular production process also referred to expected spoilage, and arises even under efficient operating conditions. Cost of normal spoilage are typically treated as a component of the costs good units manufactured because good units cannot be made without the simultaneous appearance of spoiled units

In contrast abnormal spoilage is unusual and unexpected spoilage that should not occur under normal operating conditions. It is not an inherent result of the particular production process and is usually regarded as avoidable and controllable. Abnormal spoilage costs are written off as losses of the accounting period in which detection of the spoiled units occurs.

The questions to ask to account for spoilage are as follows

- 1) What is the flow of the process (FIFO or weighted average)?
- 2) Where are the quality control inspection points? It is assumed that all spoilage occurs at the inspection point.
- 3) How much spoilage is considered “normal”? All spoiled units are the same? , what makes one” normal” and one “abnormal” is a function of acceptable or expected levels of occurrence-A company may determine that from a particular production process three spoiled units should occur.

Spoiled Units are detected, then there are three normal and two abnormal’ spoiled units, if only spoiled units are detected, then both are considered to be normal spoilage

Allocation of Normal and Abnormal Spoilage Costs:

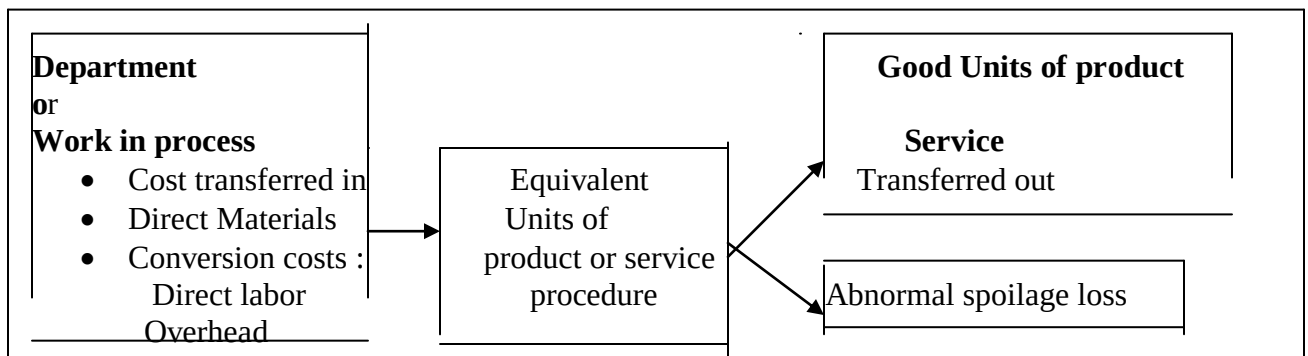


Fig. 7.1 *Allocation of Normal and Abnormal Spoilage Costs:***7.3.1 Spoiled Materials charged to total production**

The **Waa Mara** products company has monthly capacity to manufacture 125,000 three inch coil springs for use in mechanical brakes production is scheduled in response to orders received. Spoilage is caused by a variety of unpredictable factors and average \$ 0.05 per spring. During November, 100,000 springs were produced with a materials cost of \$0.40 per unit, a labor cost of \$0.50 per unit, and factory overhead charged to production at a rate of 150% of the direct labor cost. This rate is based on an estimate that includes \$ 0.05 per spring for spoilage.

The entry to record work put into production during the month is:

Work in process –materials -----	40,000
Work in process – labor - - - - -	50,000
Work in process – FOH - - - - -	75,000
Materials - - - - -	40,000
Payroll - - - - -	50,000
Applied Factory overhead - - - - -	75,000

On the last working day of the month the entire day's production of 4,000 units is spoiled due to improper heat treatment however, these units can be sold for \$ 0.50 each in the second hand market. To record this normal loss on spoiled goods and the possible resale value, the entry that charges all production during the period with a proportionate share of the spoilage is:

	Subsidiary	
Spoiled good	Record	2000
Factory overhead control - - - - -		4600
Loss on spoiled goods ----	4600	
Work in process materials - - - - -		1600
Work in process – labor - - - - -		-2000

when the inventory was recorded and the price realized at the time of sale would be a plus or minus adjustment to factory overhead control (loss on spoiled goods)

7.3.2 Spoiled materials charged to a particular job.

The **Waa Mara** products company has a contract to manufacture 10,000 heavy duty coil spring for the Turaf- state supply company. This order requires a steel wire that is harder and slightly heavier than stock normally used, but the production process, as well as labor time and overhead factors, is identical with the standard product. Material cost for each of these springs is \$0.60. This special order required exacting specification, and normal spoilage is to be charged to the order. The \$0.05 per unit spoilage factor is now eliminated from the overhead rate, and 140% of direct labor cost, or \$0.70 per unit, is the rate used on this job . the order is put into production the first day of December, and sampling during the first hour of production indicates that eleven units of production are required to secure ten good springs.

Entries to recorded costs placed into production for 11.000 units are:

Work in process – Materials -----	6,600
Work in process – labor -----	5,500
Work in process- FOH -----	7,700
Materials -- - -----	6,600
Payroll -----	5,500
Applied Factory overhead -----	7,700

One thousand units did not meet specifications and are spoiled but can be sold as second for \$0.45 per unit. The entry to record the spoilage is:

Spoiled Goods -----	450
Work in process – Materials -----	150
Work in process – labor -----	125
Work in process – Factory overhead -----	175

$$\frac{\$ 450 \text{ sales recovery}}{\$ 1,800 \text{ costs of 1,000 units}} = 25\%$$

$$25\% \left\{ \begin{array}{ll} \$ 600 \text{ materials} & = \$ 150 \\ \$ 500 \text{ Labor} & = \$ 1R5 \\ \$ 700 \text{ Factory overhead} & = 175 \end{array} \right.$$

OR

$$\frac{\$6,600 \text{ materials}}{\$19,800 \text{ total job cost}} \times \$450 \text{ sales recovery} = \$ 150$$

$$\frac{\$5,500 \text{ Labor}}{\$19,800 \text{ total job cost}} \times \$450 \text{ sales recovery} = \$125$$

$$\frac{\$7,700 \text{ Factory overhead}}{\$19,800 \text{ total Job cost}} \times \$450 \text{ sales recovery} = \$175$$

The entry transferring the completed order to finished goods would be:

Finished Goods ----- 19,350

Work process – Materials ----- 6,450

Work in process – Labor -----5,375

Work in process – FOH-----7,525

The net result of this treatment is to charge the spoilage loss of \$ 1,350 (\$ 1800-450 cost recovery) to the 10,000 good units that are delivered at the original contract price. The unit cost of completed spring is \$ 1.935 (\$19,350 ÷ 10,000 units).

Any difference between the price when the inventory was recorded and the price realized at the time of sale should be an adjustment to work in process, finished goods, or cost of goods sold, depending on the completion status of the particular job order. As an expedient, the difference might be closed to Factory overhead control.

Activity 7.2 “Normal spoilage is planned spoilage.” and “costs of abnormal spoilage are lost costs”. Discuss.

7.4 Account for Detective work (Reworked units)

Reworked units are unacceptable units of production that are subsequently reworked into good units and sold completed reworked units should be indistinguishable from non – reworked good units.

In manufacturing process, imperfection may arise because of faults in materials, labor, or machines. If the unit can be reprocessed in one or more stages and made into a standard Salable product, it is often profitable to rework the defective unit.

Two methods of accounting for the added cost to upgrade defective work are appropriate, depending upon circumstances.

- 1) If the defective work is experienced in regular manufacturing, the additional cost to correct defective units (based on previous experience) is included in the predetermined factory overhead and in the resulting factory overhead rate. Actual rework cost is charged to Factory overhead control.

To illustrate, assume that **Waa Mara** Company has an order for 500 units of a product that has direct production costs of \$5 for material and \$ 3 for labor, with factory overhead charged to production at 200 % of labor cost. Fifty units are found to be defective and are to be reworked at a total cost of \$30 for materials, \$60 for Labor, and over head at 200% of direct labor cost. The entries are:

Work in process – Materials - - - - -	2,500
Work in process – labor - - - - -	1,500
Work in process factory overhead - - - - -	3,000
Materials - - - - -	2,500
Payroll - - - - -	1,500
Applied factory overhead - - - - -	3,000
Factory overhead control - - - - -	210
Defective work - - - - -	210
Materials - - - - -	30
Payroll - - - - -	60
Applied Factory over head - - - - -	120
Finished goods - - - - -	7000
Work in process -materials - - - - -	2500
Work in process – Labor - - - - -	1,500
Work in process - FOH - - - - -	5,000

The unit cost of the completed units is \$ 14 (\$ 7000 ÷ 5000 units)

2. Suppose, however, that the same company received special order for 5000 units with the agreements stating that any defective work is chargeable to the contract. During production, so units are improperly assembled. The total cost to correct these defective units is \$ 30 for materials, \$60 for labor, and 200% of the direct labor cost for factory overhead. The entries in this case are:

i. Work in process – Materials - - - - -	2500
Work in process – Labor - - - - -	1,500
Work in process – FOH - - - - -	3,000
Materials - - - - -	2,500

	Payroll - - - - -	1,500
	Applied factory over head - - - - -	3,000
ii.	Work in process – materials - - - - -	30
	Work in process - Labor - - - - -	60
	Work in process - FOH - - - - -	120
	Materials - - - - -	30
	Payroll - - - - -	60
	Applied Factory over head - - - - -	120
iii.	Finished goods - - - - -	7,210
	Work in process – materials - - - - -	2,530
	Work in process – Labor - - - - -	1,560
	Work in process – FOH - - - - -	3,120

The unit cost in this case is \$ 14.42 instead of \$ 14. When ever the defective work cost is charged directly to the job, a slight over charge of factory overhead results because of the inclusion of rework cost in the factory over head rate. One remedy /action / to correct this discrepancy would be either to create a new independent over head rate or to separate costs for the special job.

Activity 7.3 the costs of reworking defective units are always charged to the specific jobs where the defects were originally discovered.” Do you agree? Explain.

7.5 Costing procedures for scrap and waste

In manufacturing process, waste and scrap result from:

1. the processing of materials,
2. defective and broken parts
3. obsolete stock
4. revisions or abandonment of experimental projects, and
5. worn out or obsolete machinery. This scrap should be collected and placed in storage for sale to scrap dealers. At the time of sale, the following entry may be made :

Cash (or Account Receivable) - - - - - x xx

Scrap sales /or Factory overhead control / - - - xxx

The amount realized from the sale of scrap and waste can be treated in two way with respect to the income statement:

1. The amount accumulated in scrap sales may be closed directly to Income summary and shown on the Income statement under other Income.
2. The amount may be credited to FOH control, thus reducing the total factory overhead expense and there by the cost of goods manufactured.

When Scrap is collected from a job or department, the amount realized from the sale of scrap is often treated as a reduction in the materials cost charged to the individual job or product. In this case, the entry to record the sale would be:

Cash (or Account Receivable) - - - - - xxx

Work in process - - - - -xxx

When the quantity and value of scrap material is relatively high .it should be stored in a designated place under the supervision of a storekeeper. A scrap report (as shown below) is generally prepared in duplicate to authorize transfer receipt of the scrap.

Summary of accounting for spoilage, rework, or scrap:

Type of spoilage, rework, or scrap	Accounting Treatment
Normal spoilage arising from the requirements of a specific job	Charge to the individual job
Normal spoilage occurring periodically as a regular part of all jobs	Charge to overhead
Abnormal spoilage	Charge to separate loss account
Opportunity cost of spoilage	Not measured
Rework for defect arising from the requirements of	Charge to individual job

a specific job	
Rework for defect occurring periodically during normal production	Charge to overhead
Rework for abnormal defect	Charge to separate loss account
Sale of scrap	Record at time of production or at time sold If not material: Record as other income
Scrap common to all jobs or difficult to trace to specific job	Credit to overhead
Scrap traced to individual jobs	Credit to individual job

Summary

Spoilage is unacceptable units of production that are discarded or are sold for net disposal proceeds. Reworked units are unacceptable units that are subsequently reworked and sold as acceptable finished goods. Scrap is a product that has minimal sales value compared with the a sales value of the main or joint product(s)

Accounting for spoilage in process costing systems using the weighted-average method is very similar to the weighted –average method discussed in the previous chapter. The only difference is that the normal and abnormal units are treated as additional unit groups for which cost per –equivalent-unit needs to be calculated and costs assigned.

Accounting for spoilage in process costing systems using the First-in, first-out (FIFO) method is very similar to the first-in, first-out (FIFO) method discussed in the previous chapter. The only difference is that the normal and abnormal units are treated as additional unit groups for which cost-per-equivalent-unit needs to be calculated and costs assigned.

Reworked units should be distinguishable from non-reworked good units when completed, and hence the two are assigned the same costs. Normal work can be assigned to a specific job, or if common to all jobs as part of manufacturing overhead. Abnormal rework is written off as a period cost.

Accounting for scrap is similar to the accounting for by-products discussed in chapter 7. Companies differ as to both when and how scrap is recognized in the accounting records. Scrap is sometimes reused as direct materials rather than sold as scrap. Then it should be debited to materials control as a class of direct materials and carried at its estimated net realizable value.

Self-test questions

Q1 In Night Star Shop, 5 aircraft parts out of a job lot of 500 aircraft parts are spoiled. Costs assigned up to the point of inspection are \$100 per unit. Night Star calculates these costs on the basis of its inventory costing assumptions-weighted average, FIFO, or Standard costs. We do not, however, emphasize cost flow assumptions in our presentation here. The current disposal price of the spoiled parts is estimated to be \$30 per part. When the spoilage is detected, the spoiled goods are inventoried at \$30 per unit.

Required:

- a) Pass the journal entry to recognize normal spoilage attributable to a specific job.
- b) Assume the normal spoilage is common to all jobs & Pass the necessary journal entry to recognize.
- c) If the spoilage is abnormal, the net loss is highlighted to management by charging the loss to an abnormal loss account. What is the journal entry to record this?

Q2. Consider the Night Star machine Shop data. Assume that the five spoiled parts used in our Night Star machine Shop illustration are reworked.

- a) What is the journal entry for the total costs (details of costs assumed) assigned to the five spoiled units before considering rework costs?
- b) If the rework is normal but occurs because of the requirements of a specific job, what is the journal entry to record this normal rework attributable to specific job?
- c) If the rework is normal and not attributable to any specific job, what is the journal entry to record this normal rework common to all jobs?
- d) If the rework is abnormal, what is the journal entry to charge abnormal rework to a separate loss account.

Chapter -8

Joint Products and By- Product

Learning Objectives:-

At the end of this session you should be able to:

- ❖ Define a joint product and give an example
- ❖ Differentiate the different method of allocating joint costs
- ❖ Explain why joint costs should be allocated to individual products
- ❖ Identify the split-off point(s) in a joint-cost situation
- ❖ Identify the distinction between joint products and By- products

8.1 Introduction:

In many industries, two or more products are unavoidably produced from the same process and same raw materials. These products are produced in natural proportions which cannot be changed at the will of the management. For example, in an oil refinery, when crude oil is processed, many products are simultaneously produced from the same raw materials. Examples of these products are: petrol, kerosene, diesel, grease, etc. Such products are known as joint products or by-products..

Definition:

Joint products are two or more products that are produced simultaneously from the one production process. Joint products, thus, represent two or more products separated in the course of processing, each product being in such proportion and of such a nature that no single one of them can be regarded as the main products.

Characteristics: characteristics of joint products are:

- (i) Joint products are produced from the same raw materials in natural proportion.
- (ii) They are produced simultaneously by a common process.
- (iii) They comparatively of almost equal value.
- (iv) They may require further processing after their point of separation.

Examples of entities that manufacture joint products:

Entity	Main products	By-products
Shell oil	Petrol, diesel, jet fuel	Asphalt
Flour mill	White flour, brown flour	animal feeding stuff
Cheese	Fresh cheese, butter	Butter milk

8.2 Joint products and co-products

Joint products should be distinguished from co-products. Co-products refers to more than one products being manufactured by a company but need not necessarily arise from the same raw materials and manufacturing process and the quantity of each product can be changed by the management. For example, in a bakery the various co-products are bread, cake, biscuits, etc. and the quantity of each such product may be change by the management as per needs.

8.2.1 Joint costs and subsequent costs

Joint costs /or common costs/ are all manufacturing costs incurred before that stage in manufacture at which the products get separated .i.e. they are common to all joint products and incurred prior to the split off point.

Definition: Split-off point is the stage in the production process where the joint products are identifiable as separate product.

Subsequent (or attributable) costs, on the other hand, are those costs which are incurred after the separation or split-off point. They are in ceremonial costs which can be easily traced to each specific product. They are relevant costs in sell or process further decision.

The distinction between joint costs and subsequent costs is important because when accounting for products, the joint costs are the main problem. This is because joint costs cannot be traced to individual products.

Activity 8.1 Explain why joint costs are irrelevant in a sell-or-process further decision

8.3 Accounting for joint products

Accounting for joint products means the apportionment of joint costs to each of the joint products. Such apportionment serves the following objectives:

- (a) To determine the cost per unit

- (b) To help in venture valuation
- (c) To determine the profit or loss on each line of product
- (d) To determine the price of each product.

The Various methods of apportionment of joint costs discussed below are based mainly on individual opinion and tend to produce only approximate result. This is because no perfectly logical basis exists for the apportionment of joint costs to products and most of the methods are arbitrary.

Methods of allocation /apportionment/ of joint costs are:

- (1) Relative sale value
- (2) Physical unit method
- (3) Net realizable value method
- (4) Reverse cost method
- (5) Average unit cost method

(1) Relative sales value

A method of allocating joint costs to joint products in proportion to their sales value at the split-off point i.e. the apportionment is done on the basis of weighted sales value (number of units produced and sold times selling price per unit).

The allocation of joint production does not make it a relevant cost. Because that cost remains fixed despite the decision to process it further. The method is illustrated below with assumed figures:

Illustration 8.1

Example Product	Selling price Per unit (a)	Production quantities / unit/ (b)	Sales value (a)*(b) (c)	Apportioned Joint cost. (24 :48 :28)(d)
A-----	12	200	2400	2160
B-----	8	600	4800	4320
C-----	4	700	<u>2800</u>	<u>2520</u>
			<u>10,000</u>	<u>9000</u>

Working Note

Apportioned cost ratio = individual product sales value

Total sales Value

For product	A=	$\frac{2400}{10,000}$	=24%
Fop produced	B=	$\frac{4800}{10,000}$	= 48%
Fop produced	C=	$\frac{2800}{10,000}$	= 28%

Apportioned Joint cost = Apportioned cost ratio * total joint costs

For product A Apportioned joint cost = 24% * 9000
 $= 24/100 * 9000$
 $= \$2160$

For product B Apportioned joint cost = 48% *9000
 $= 48/100 *9000$
 $= \$ 4320$

For products apportioned joint costs = 28% * 9000
 $= 28/100 * 9000$
 $= \$ 2520$

(2) Physical unit method

It is a method of allocating joint costs to joint products based on some physical characteristics of the joint products at the split-off point. This means that the joint cost is apportioned on the basis of relative weight, volume or quantity, etc of each product obtained at the point where the split-off occurs.

Illustration8.2

The following data have been extracted from the books of international chocolate company.

Joint cost	Joint product	Weight at
Allocation of	Split-off point	relative proportion
cost		joint
\$1100 {	Cocoa butter 750 kg	$\frac{3}{4}$ \$825
	Cocoa powder <u>250 kg</u>	$\frac{1}{4}$ <u>\$275</u>
	1000kg	\$1100

Working Note:

$$\begin{aligned}
 \text{Allocation of joint costs for cocoa butter} &= \text{Relative} \times \text{Joint cost proportion} \\
 &= \frac{(750)}{1000} \times 1100 \\
 &= \frac{3}{4} \times 1100 \\
 &= \underline{\underline{\$ 825}}
 \end{aligned}$$

$$\begin{aligned}
 \text{Allocation of joint cost for cocoa powder} &= \text{relative} \times \text{joint cost proportion} \\
 &= \frac{(250)}{1000} \times 1100 \\
 &= \frac{1}{4} \times 1100 \\
 &= \underline{\underline{275}}
 \end{aligned}$$

(3) Net realizable Value method.

A method of allocation joint costs to the joint products according to the relative size of the final products net realizable values.

Steps:

- Calculate the net realizable value of the individual products
- Calculate the relative proportion based on the net realizable value
- Allocate joint costs to products

Net realizable value = sales value of final product-separable cost of processing

Illustration 8.3. The following data have been extracted from the books of Coke co.

Ltd. Joint	Joint	Sales value	Separable	Net realizable	Relative
------------	-------	-------------	-----------	----------------	----------

Allocation of Joint	Cost	Products	of final	Cost of	value	Protection
\$1100		Product	processing			cost
		Tanning cream	\$3000	\$1560	\$1440*	6/11
\$600						
		Instant cocoa mix	\$2000	\$800	\$1200*	5/11
<u>500</u>						
		Total		joint		cost
allocated.....				<u>\$1100</u>		

Sales value of final product-separable cost of processing = net releasable value

$$\$3000 - \$1560 = \$1440$$

$$\$2000 - \$800 = \$1200$$

(4) Reverse cost method

In this method the joint cost is apportioned on the basis of net value of each product. The net value is calculated on the basis of net value of each product. The net value is calculated by deducting the following from the sales value.

- (a) Estimated profit margin
- (b) Selling and distribution costs, if any
- (c) After split-off processing costs.

The net value of individual products so obtained is taken as the basis for apportioning joint costs. This is known as reverse cost method because net values are calculating by working backward from sales values.

This method is particularly used when product are not sold at their stage at split-off point but require further processing. Operation of this method is illustrated below.

Illustration 8.4. Assume in processing a basic raw material, three joint products ‘x’, ‘y’ and ‘z’ are produced. The joint expenses of manufacturing are: materials \$10,000, labour \$ 8000, overhead \$ 9000 (total \$ 27,000) . Subsequent /separable/ costs are as follows.

	Products		
	X	Y	Z
Material.....	\$2000	\$1600	\$1,800
Labour.....	2500	1400	1,700
Overhead.....	<u>2500</u>	<u>1000</u>	<u>1,500</u>
Total	<u>7000</u>	<u>4000</u>	<u>5000</u>
Sales value.....	42.000	20.000	18.000
Estimated profit on sales	50%	50%	33 1/3 %

Required: Show how you would apportion the joint costs of manufacture by reverse cost method.

Solution: *statement of Apportionment of joint costs*

	<u>Products</u>		
	X	Y	Z
Sales value.....	42,000	20,000	18,000
Less: Estimated profit.....	<u>21,000</u>	<u>10,000</u>	<u>6,000</u>
Estimated total cost	21,000		10,000
12.000			
Less: Subsequent cost	<u>7,000</u>		<u>4,000</u>
<u>5,000</u>			

Joint cost(\$27,000)...	<u>14,000</u>	<u>6,000</u>
<u>7,000</u>		

(5) **Average unit cost method.**

In this method, the joint cost is apportioned by using the average unit cost which is obtained by dividing the total joint costs by the total number of units produced of the entire product. The average cost per unit of each product is the same. The procedure is illustrated as follows.

Illustration 8.5 from the following particulars, find out the cost of joint products X, Y and Z under the average unit cost method

(a) per-separation point costs \$ 30.000

(b) Other production data,

Product	Unit produced
X	1000
Y	400
Z	<u>600</u>
	<u>\$2000</u>

Solution:

$$\text{Average unit cost} = \frac{\text{Joint Cost}}{\text{Total No of unit produced}}$$

$$= \frac{\$30.000}{2000 \text{ unit}}$$

$$= \$ 15 \text{ per unit}$$

Statement of Apportionment of Joint Cost

Product	Unit	Average cost	Apportioned Cost
---------	------	--------------	------------------

	(A)	(B)	AxB = C
X	1000	\$15	15,000
Y	400	15	6000
Z	<u>600</u>	15	<u>9000</u>
Total	<u>2000</u>		<u>30,000</u>

8.4 BY-PRODUCTS

By- Products are products of relatively small value which are incidentally and unavoidably produced in the course of manufacturing the main product. For example in sugar mills, other main product is sugar. But baggasse and molasses of comparatively smaller value are incidentally produced and thus are by-products.

By-products may be:

- (a) Those sold in their original form without further processing
- (b) Those which require further processing in order to be saleable

8.5 Distinction between Joint products and By Products

There are no hard and fast rules to distinguish between joint products and by-products. A product may be treated as a by-product in one business and the same product may be treated as a joint product in another business. However, the following factors should be considered to determine if a product is a joint product or a by-product.

(a) Relative sales value

If the sales values of the entire product are more or less equal, they are treated as joint products. If however, there is wide difference in the relative sales values of products, the product with the greater sales value is treated as the main product and the products of lower value are treated as by product.

(b) Objective of manufacture

If the objective of manufacturing is product A, the unwanted product B and C be treated as by-product.

(C) Policy of management

The management may decide to treat a particular product as the main product and the other products as by products.

Summary

Joint products are two or more products that are produced simultaneously from the one production process. It consists of a main product with a high sales value as compared to other joint products. In other ways, a by-product is a product with a low sales value as compared to other joint products. The costs incurred in the production of joint products are known as joint costs. These costs are common to all the joint products and incurred prior to the split-off point. Joint costs are normally used for estimating cost of goods sold and value of inventory. Split-off point is the stage in the production process where the joint products are identifiable as separate product. The costs incurred after split-off point are called separable costs. They are incremental costs which can be easily traced to each specific product and they are relevant costs in sell or process further decision. The five most popular methods of allocating joint costs to joint products are: (1) relative sales value method, (2) physical units' method, (3) net realizable value method, (4) Reverse cost method, (5) average unit cost method.

Profit differs under each of these methods. None of these methods is accurate, because, the methods are arbitrary cost allocation methods. Allocated joint costs are unreliable for pricing as well as any other management decisions, because of their arbitrary nature.

Self Test Questions**A. True or False? Give reason in brief**

- 1) Allocation of joint cost to joint and by products does not affect overall profit or loss.
- 2) There is no difference b/n joint product and co-products.
- 3) By-product may not have any realizable value.
- 4) Sales value is a method of allocating joint costs to joint products based on some physical characteristics at the split-off point.
- 5) Some outputs of the joint production process have zero sales value and no journal entries are made in the accounting system to record the processing of such outputs with zero sales value.
- 6) The estimated NRV is the expected final sales value in the ordinary course of business minus the expected separable costs of the total production of these products during the accounting period.
- 7) Under the benefits-received criterion, the physical measure method is less preferred than the sales value at split-off method. Because it has no relationship to the revenue-producing power of the individual products.
- 8) Joint costs incurred up to the split-off point are past (sunk) costs irrelevant to the decision to sell a joint (or main) product at the split-off point or to process it further.
- 9) Joint cost allocation is important in litigation involving one or more joint products.
- 10) Cost allocation is required for cost reimbursement purposes under contracts when only a portion of a business' products or services is sold or delivered to a single customer (government agency).

B : Choose the best answer among the following alternatives.

- 1) Why is the sales value at split-off method widely used?
 - a) It is objective
 - b) It does not anticipate subsequent management decisions on further processing.
 - c) It uses a meaningful common denominator.
 - d) It is simple.
 - e) All
 - f) None
- 2) Which method of allocating joint costs should be chosen where market prices exist at split-off?
 - a) Physical measure method
 - b) Relative sales value method
 - c) Reverse cost method
 - d) average unit cost method
- 3) The _____ measure method allocates joint costs to joint products on the basis of the relative weight, volume, or other physical measure at the split-off point of the total production of these products during the accounting period.
 - a) Physical measure method
 - b) Relative sales value method
 - c) average unit cost method
 - d) Reverse cost method
 - e) None

Answer question no. 4 & 5 based on the following data. ABC Company incurred \$200,000 of joint costs to produce the following:

Product A: 10,000 units, 20,000 pounds

Product B: 10,500 units, 48,000 pounds

Product C: 11,500 units, 12,000 pounds

- 4) What are the joint costs allocated to **product A** using the number of pounds produced as the physical measure?
- a) \$120,000
 - b) 50,000
 - c) 130,000
 - d) 30,000
 - e) None
- 5) What are the joint costs allocated to **product A** and **product C** using the number of pounds produced as the physical measure?
- a) \$170,000
 - b) 150,000
 - c) 80,000
 - d) 68,000
 - e) None

C: Fill in the blank spaces

- 1) The ----- product usually has a greater sales value than by – product
- 2) The cost incurred up to the point where individual products can be identified are called - - - - - costs.
- 3) Under - - - - - method of apportionment of joint costs, the cost per unit of each product is the same.
- 4) The _____ is the juncture in the production process where one or more products in a joint-cost setting become separately identifiable.
- 5) _____ are all costs (manufacturing, marketing, distribution, etc.) incurred beyond the split-off point that are assignable to one or more individual products.

D: Theoretical questions

- 1) Define joint product and by – product
- 2) Explain briefly the method of accounting by – product
- 3) List the major characteristics of joint products.
- 4) Define subsequent costs and give an example as well.

E) Practical questions

Q1. Lubucko Company incurred \$200,000 of joint costs to produce the following:

Product A: 10,000 units, 20,000 pounds

Product B: 10,500 units, 48,000 pounds

Product C: 11,500 units, 12,000 pounds

(i) Assume the following sales values per unit: A: \$10.00, B: \$30.00, and C: \$20.00

Required:

- a) What is the sales value at split-off point?
- b) How much of the joint costs are allocated to each product?
- c) What are the joint production costs per unit?
- d) Assume all of the units produced of B and C were sold. 2,500 units of A (25%) remain in inventory. What is the gross margin percentage of each product?

(ii) Assume that Lubbock Company can process products A, B, and C further into A1, B1, and C1.

The new sales values after further processing are:

A1: $10,000 \times \$12.00 = \$120,000$

B1: $10,500 \times \$33.00 = \$346,500$

C1: $11,500 \times \$21.00 = \$241,500$

- a) Additional processing (separable) costs are as follows: A1: \$35,000; B1: \$46,500; and C1: \$51,500. What is the estimated net realizable value of each product at the split-off point?
- b) How much of the joint cost is allocated to each product?
- c) What is the production cost per unit?

Q2. In a process line of **XY** co, three joint products are produced .for the month of many 1999, the following data were available.

	Products		
	L	M	N
Sales price per Kg	\$5	\$10	\$ 20
Post – separation point cost	10,000	5,000	15,000
Out put in Kg	2500	1000	1500

Per- separation point costs amounted to \$ 20,000. The pre- separation point costs are apportioned to joint product according to their weight. Your are required to prepare a statement showing the estimated profit or loss for each product and in total.

Q3. A coke manufacturing company produces the following products by putting 5000 tones of coal & 25 per tone in to common process:

Coke	-----	3500 tones
Tar	-----	1200 “
Sulphate	-----	52 tones
Benzol	-----	48 tones

Required: apportion the joint cost amongst the products on the basis of physical unit method

(Hint: Difference b/n total input of 5000 tones and total out put of 4800 tones is assumed to be a normal loss.)

Q4. From the following information find the profit made by each product apportioning joint costs on sales value basis:

Direct material	-----	\$ 126,000
Power	-----	25,000
Petrol, oil Lubricant	-----	50,00
Labor	-----	7,500
Other charges	-----	4,100

	Product X	product Y
Selling costs - - - - -	\$ 20,000	\$ 80,000
Sales - - - - -	152,000	168,000

Q3.ArbaMinch textile has some spoiled goods that had an assigned cost of \$4,000and zero net disposal value.

Required: prepare a journal entry for each of the following conditions under both (a) process costing (Department A) and (b) job costing:

- Abnormal spoilage of \$4,000.
- Normal spoilage of \$4,000 related to general plant operations.
- Normal spoilage of \$4,000 related to specifications of a particular job.

